

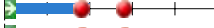
Program overview

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Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master Sustainable Energy Technology 2022							
SET3013	Renewable Energy	4					
SET3055	Economics and Regulations of Sustainable Energy Systems	4					
SET3060	Energy Systems Optimization	5					
SET3815-M	Matlab Fundamentals for Electrical Engineers	2					
SET3815-Pr	System Integration Project	7					
SET3901	Graduation Project	45					
WM0201SET	Technical Writing	2					
Profile Courses SET 2022							
Solar systems: Solar Power Storage 2022							
CH3222	Energy Storage in Batteries	4					
EE4536	DC and AC Microgrids	4					
EE4545	Electrical Power Systems of the Future	4					
ET4376	Photovoltaic Basics	4					
ET4377	Photovoltaic Technologies	4					
ET4378	Photovoltaic Systems	4					
SET3065	Intelligent Electrical Power Grids	4					
SET3080	The Necessity of Storage Technology	4					
SET3085	Hydrogen Technology	4					
SET3095	Electronic Power Conversion	4					
Bio/solar systems: Biomass Solar Storage 2022							
CH3061	Multiphase Reactor Engineering	4					
CH3222	Energy Storage in Batteries	4					
ET4376	Photovoltaic Basics	4					
ET4377	Photovoltaic Technologies	4					
ET4378	Photovoltaic Systems	4					
SET3070	Thermochemistry of Biomass Conversion	4					
SET3080	The Necessity of Storage Technology	4					
SET3085	Hydrogen Technology	4					
SET3311	Green Chemistry and Sustainable Technology	4					
Autonomous systems: Wind Storage - Solar 2022							
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4					
AE4W09	Wind Turbine Design	5					
AE4W13	Site Conditions for Wind Turbine Design	3					
CH3222	Energy Storage in Batteries	4					
ET4376	Photovoltaic Basics	4					
ET4377	Photovoltaic Technologies	4					
ET4378	Photovoltaic Systems	4					
SET3080	The Necessity of Storage Technology	4					
SET3085	Hydrogen Technology	4					
Wind & Economics: Wind Power Economics 2022							
AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4					
AE4W09	Wind Turbine Design	5					
AE4W13	Site Conditions for Wind Turbine Design	3					
EE4536	DC and AC Microgrids	4					
EE4545	Electrical Power Systems of the Future	4					
SET3065	Intelligent Electrical Power Grids	4					
SET3095	Electronic Power Conversion	4					
TPM403SET	Technology Entrepreneurship and Sustainability	4					
WM0637SET	Economic Policy for Sustainable Energy	4					
WM0931SET	Sustainable Energy Innovation and Transitions	4					
Solar & Economics: Solar Power Economics 2022							
EE4536	DC and AC Microgrids	4					
EE4545	Electrical Power Systems of the Future	4					
ET4376	Photovoltaic Basics	4					
ET4377	Photovoltaic Technologies	4					
ET4378	Photovoltaic Systems	4					
SET3065	Intelligent Electrical Power Grids	4					
SET3095	Electronic Power Conversion	4					
TPM403SET	Technology Entrepreneurship and Sustainability	4					
WM0637SET	Economic Policy for Sustainable Energy	4					
WM0931SET	Sustainable Energy Innovation and Transitions	4					

Biomass & Economics: Biomass		Storage - Economics 2022			
CH3061	Multiphase Reactor Engineering			4	
CH3222	Energy Storage in Batteries			4	
SET3070	Thermochemistry of Biomass Conversion			4	
SET3080	The Necessity of Storage Technology			4	
SET3085	Hydrogen Technology			4	
SET3311	Green Chemistry and Sustainable Technology			4	
TPM403SET	Technology Entrepreneurship and Sustainability			4	
WM0637SET	Economic Policy for Sustainable Energy			4	
WM0931SET	Sustainable Energy Innovation and Transitions			4	
Electric Mobility Systems: Electric Mobility		Power Storage 2022			
CH3222	Energy Storage in Batteries			4	
EE4536	DC and AC Microgrids			4	
EE4545	Electrical Power Systems of the Future			4	
ET4117	Electrical Machines and Drives			4	
ET4291SET	Digital modelling of electric powertrain			4	
SET3065	Intelligent Electrical Power Grids			4	
SET3080	The Necessity of Storage Technology			4	
SET3085	Hydrogen Technology			4	
SET3095	Electronic Power Conversion			4	
SET3100	Electric vehicle and charging technology			4	
Electric Mobility & Economics: Electric Mobility		Power Economics 2022			
EE4536	DC and AC Microgrids			4	
EE4545	Electrical Power Systems of the Future			4	
ET4117	Electrical Machines and Drives			4	
ET4291SET	Digital modelling of electric powertrain			4	
SET3065	Intelligent Electrical Power Grids			4	
SET3095	Electronic Power Conversion			4	
SET3100	Electric vehicle and charging technology			4	
TPM403SET	Technology Entrepreneurship and Sustainability			4	
WM0637SET	Economic Policy for Sustainable Energy			4	
WM0931SET	Sustainable Energy Innovation and Transitions			4	
Electives SET 2022					
ET4399	Extra Project (max. 15EC)			15	
ME45201	Electrochemical Energy Storage 1: Fundamentals			3	
ME45203	Electrochemical Energy Storage 2: Analytical Modelling			4	
SET3822	Internship SET			15	
SET3995	Geothermal Energy and Applications			4	
TUD4040	Joint Interdisciplinary Project			15	
WM0320TU	Ethics and Engineering			3	
Profile Electives 2022					
Biomass 2022					
AE4262	Combustion for propulsion and power technologies			4	
LM3741	Fermentation Technology & Environmental Biotechnology			6	
ME45001	Advanced Heat Transfer			4	
ME45025	Introduction to Multiphase Flow			5	
ME45070	Advanced Reaction & Separation Systems			5	
ME45135	Process and Power Plant Design			5	
ME45155	Modelling of Thermo- & Hydrodynamic Systems			5	
ME45160	Advanced Applied Thermodynamics			5	
ME45165	Equipment for Heat & Mass Transfer			5	
Power 2022					
ET4103	High Voltage Technology			4	
OE44076	Offshore Renewable Energy			6	
Economics & Society 2022					
SEN1211	Agent-based Modelling			5	
SEN1531	Design of Integrated Energy Systems			5	
TPM001A	Sociotechnology of Future Energy Systems			4	
TPM023A	Cost-Benefit Analysis: Theory and Application			4	
TPM404B	Technology Entrepreneurship and Internationalization			5	
TPM411A	Idea to Start-up IT & AI			5	
TPM413A	Idea to Start-up Energy & Sustainability			5	
TPM414A	Idea to Start-up Deep Tech			5	
Solar 2022					
EE4585	Semiconductor Device Physics			5	
EE4670	PV Materials Processing and Characterization			4	
EE4680	Photovoltaic Modelling			4	
ET4379	Photovoltaic Lab Course			4	
ME46065	Thin Film Materials			3	
Storage 2022					
AP3311	Neutrons, X-Rays and Positrons for Studying Microscopic Structures and Dynamics			6	
CH3513	Electrochemistry for Renewable Energy			4	
CH3531	Functional Ceramics			3	
Wind 2022					

AE4202	CFD for Aerospace Engineers	3	
AE4205	MDO for Aerospace Applications	4	
AE4T40	Airborne Wind Energy	3	
AE4W30	Wind Resource and Wind Farm Yield	4	
Electric Mobility 2022			
EE4515	Advanced Power Electronics	4	
EE4C04	Control System Design	5	
ET4116	Power Electronics	4	
ET4121	A.C. Machines	4	
ME41106	Intelligent Vehicles 3ME	5	
RO47017	Vehicle Dynamics and Control	5	
Recommended electives 2022			
Language courses 2022			
TPM301B	Spoken English for Academic Purposes Intermediate	2	
TPM302B	Spoken English for Academic Purposes Advanced	2	
TPM303A	Intermediate Writing in English for the University	2	
TPM304A	Advanced Writing in English for the University	2	
TPM305A	Writing a Masters Thesis in English	2	
WM1115TU	Dutch Elementary 1	3	
WM1116TU	Dutch Elementary 2	3	
WM1117TU	Dutch Intermediate 1	3	
WM1135TU	Advanced English for the University	3	
Entrepreneurship courses 2022			
TPM406A	Corporate Entrepreneurship and Start-ups	6	
TPM416A	Turning Technology into Business	6	
TPM420A	Ready to Start-up	6	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Master Sustainable Energy Technology 2022	
Program Title	Sustainable Energy Technology (SET)
In association with the Faculty of	EWI 3mE LR TNW TBM
In association with the University of	The Masters degree program in Sustainable Energy Technology is a 4TU MSc program. The program is offered at TU/e (Eindhoven University of Technology), TUD (Delft University of Technology) and UT (University of Twente). The programmes have similar learning objectives. The courses and specialisations at each university are different. Each programme has a different CROHO registration (programme registration according to the WHW). If a student is admitted to the Sustainable Energy Technology programme at one of the 3 universities, the student is also admitted to the programme at the other universities. After a student is formally enrolled in the masters programme in Sustainable Energy Technology at one of the 3 universities he or she will also obtain a secondary enrolment (neveninschrijving) at the 2 other universities. Students are allowed to choose elective subjects from the lists of elective, specialisation and core courses from each of the 3 programmes, after consultation with the mentor/supervisor from the home university and after approval by the Board of Examiners of the home university.
EC Program	120 ECTS
Program Start (Study Year)	August 31st, 2021
Program Goals	The mission is to offer internationally recognized high quality Master's programme in the field of sustainable energy technology. After finishing the MSc Programme students have a broad knowledge of the field of Sustainable Energy and can act as system integrators of the energy transition. To achieve this goal, a modern and flexible curriculum is offered, which follows the latest developments in science and technology. The programme aims: - to train students in such a way that they can work independently in an industrial R&D environment or other sector of society where they will use their knowledge and skills to bring different energy subsystems together into one system; - to stimulate interdisciplinary and multicultural cooperation in teams; - to train students to solve research and engineering problems and to make a system design taking into account fluctuating energy supply and demand, energy market dynamics and societal issues; - to provide students with the necessary skills, knowledge and insight in the field of sustainable energy technology, with emphasis on creativity and insight; - to offer high quality, student-oriented degree programmes, which fulfil the learning outcome requirements of national and international accreditation bodies in the field of sustainable energy technology; - to offer an inspiring academic environment with interesting study tracks within a programme; - to give students the opportunity to pursue part of their study abroad or to take courses in an international context.
Program Structure 1	The Masters degree programme in Sustainable Energy Technology consist of a compulsory core and a specialisation. The following specialisations are offered: - Autonomous systems - Solar systems - Bio/solar systems - Wind & economics - Solar & economics - Biomass & economics - Electric Mobility systems - Electric Mobility & Economics Students choose one of these specialisations according to the content of their prior degree programme An individual exam programme (IEP) is composed as follows: 1. a common core worth 24 credits; 2. a specialisation worth 36 credits (This year the Profile Cluster Bio/Solar Systems and Biomass & Economics have a specialisation worth 32 credits) 3. an elective space worth 15 credits and 4. a graduation project worth 45 credits. Students choose a cluster of three profiles. They then have 3x3 courses of 4 EC (total 36 EC). The clusters available are: - Solar systems: Solar Power Storage - Bio/solar systems: Biomass Solar Storage - Autonomous systems: Wind Storage - Solar - Wind & Economic: Wind Power Economics - Solar & Economics: Solar Power Economics - Biomass & Economics: Biomass Storage - Economics - Electric Mobility systems: Electric Mobility Power Storage - Electric Mobility & Economics: Electric Mobility - Power - Economics
Program Structure 2	The core courses are: - Renewable Energy - Technical Writing - Energy System Optimization - Economics and Regulations of Sustainable Energy Systems - System Integration Project
Administration by the Faculty of	EWI
Gives access to	SET graduates find jobs in industry, at the public sector and at research organisations, including universities. A relatively large fraction of graduates embark upon a PhD programme.

SET3013	Renewable Energy	4
Responsible Instructor	Prof.dr.ir. A.H.M. Smets	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	BSc in Physics, Chemistry, Mechanical Engineering, Electrical Engineering or Aerospace Engineering.	
Course Contents	A transition to sustainable energy is needed for our climate and welfare. In this course, you will learn how to assess the potential for energy reduction and the potential of renewable energy sources like hydropower, wind, solar and biomass. Assessing the energy balance of a country is important to understand the opportunities and challenges for the transition to sustainable energy. Some basic principles of energy transition models will be discussed. The physical and chemical principles that determine the impact of our energy consumption on the earth's climate is briefly discussed, as well. This course gives an overview of the most important renewable energy sources, the challenges to incorporate those in the electric network and the need for storage. You'll learn how to integrate these sources in an energy system, like an electricity network and take an engineering approach to look for solutions and design a 100% sustainable energy system.	
Study Goals	This is a core course in the Sustainable Energy Technology MSc program. After this course students are able to ... make simple estimations of the impact of planet earth's albedo, radiative forcing, greenhouse gas emissions on the climate of the planet (temperature, sea levels). Using physical principles you will be able to estimate the use of energy in transport, building and manufacturing. Using physical principles you will be able to estimate the generation of Renewable Energy. How much can you generate from all the different sources? (Hydropower, Wind Energy, Solar Energy, Biomass, Tidal Energy, Wave Energy) Using physical principles you will be able to estimate the power density and energy density of various storage technologies? You will be able to analyze large scale energy infrastructure using energy balances and identify corresponding climate effects of such energy infrastructure. You will be able to analyze the capacity factor, system efficiency, energy pay back time, circularity aspects of energy systems. You will understand how policies can stimulate the use of renewable energy and improve energy efficiency? You will be able to design a 100% sustainable energy system. You will understand the technological principles and challenges for various Renewable Energy technologies	
Education Method	The education method is the so-called flipped-class room. Lectures will be provided in small video lectures on Brightspace. The on-campus lectures are focused on problem-solving. Students will actively work on problems that cover the entire content of the course and the various engineering skills. Solutions will be discussed during the lectures. Homework for every lecture includes: watching the relevant video lectures dedicated to the upcoming course and the remaining exercises of the previous lecture. Note, for students doing the MSc program Sustainable Energy Technology: the course is aligned with the course WM0201SET (Technical writing Technology) in Q2. The content of the SET3013 course will act as the basis to write a consultancy report in the course WM0201SET in a group of students in Q2.	
Literature and Study Materials	Literature and Study Materials are provided on Brightspace: Videolectures Lectureslides. Additional materials to explain physical and chemical concepts. Additional materials to update students on the latest development like recent IPCC reports. Large set of problems & exercises including answers. Answers are posted on Brightspace after the problems have been discussed in the lectures. Large set of examples of exams and answers.	
Assessment	The course SET3013 Renewable Energy will be assessed in Q1 and in Q2 (resit) of the academic year 2022/2023 as follows: The exam is a written exam. The exam takes 3 hours. The exam consist of open questions and multiple choice questions. You can use only pen, paper and a calculator. The exam will be assessed with a mark with a resolution of 0.5. To pass the exam a score equal to or higher than 58.0 out of 100.0 is needed. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3055	Economics and Regulations of Sustainable Energy Systems	4
Responsible Instructor	Dr. S.T.H. Storm	
Instructor	Prof.dr.ir. L.J. de Vries	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	None	
Course Contents	This course is a general introduction to energy economics. In this course, you learn the essentials of how actors in the energy system make economic decisions, when markets work well and when they may fail, and what options the government has to remedy market failures. Specifically, the following topics are addressed: Introduction to energy (electricity) markets: perfect competition; oligopoly Rational (and irrational) market behaviour; ABM ? Investment decision making; LCOE Opportunity costing / discounting; NPV; IRR; discount rate Markets, (perfect) competition and market failure (esp. externalities & lock-in) Private costs, external cost and social cost; empirical evidence (related to energy) LCOE; discounting; private discount rate; IRR and NPV Social cost-benefit analysis; social discount rate	
Study Goals	After this course students are able to: evaluate and analyse the performance of energy (electricity) markets; identify cases of market failures and design remedial action to rectify these market failures; estimate the LCOE and calculate the NPV and IRR of an electricity investment project; contrast private and social costs of (renewable) electricity generation; assess technological lock-in in energy generation; and appraise the role of public policy in energy markets and in energy technology innovation.	
Education Method	This course consists of lectures. Parallel to the course a simulation game will be played by the students.	
Literature and Study Materials	To be announced.	
Assessment	The final grade is based on an individual computerexam (80%) and a report on the simulation game (20%). Students should score at least 6 for both parts. All marks will be expressed on a scale of 1 to 10. In normal circumstances, the individual exam is a closed-book (in-person, on campus) computer examination of 3 hours. In case of unforeseen circumstances or due to measures resulting from COVID-19, the exam will be given in the form of an on-line (remote) open-book & timed examination of 3 hours. Part of the exam will consist of multiple-choice questions.	
Co-Instructor	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) G.O. Ndubuisi	

SET3060	Energy Systems Optimization	5
Responsible Instructor	Dr. S.J. Pfenninger	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A basic knowledge of analysis and matrix algebra.	
Course Contents	This course is an introduction to mathematical optimization modelling applied to diverse problems in planning and operating energy systems, in particular energy systems with a high share of variable (weather-dependent) wind and solar generation. We will go from a solid theoretical grounding in formulating different kinds of optimization problems, to practical strategies to solve them under real-world conditions. The course covers linear, nonlinear, and mixed-integer optimization, as well as game theory in the context of multi-actor decision problems. We will use Matlab to formulate and solve optimization problems.	
Study Goals	This course is the first course in the SET programme. After completing the course, you should be able to: - LO1: Explain the challenges in planning and operating renewable energy systems and the role of mathematical optimisation in addressing these challenges - LO2: Formulate and solve linear, mixed-integer, and non-linear optimisation problems - LO3: Apply different methods to deal with parametric uncertainty in optimisation - LO4: Determine and justify the choice of an optimisation approach to formulate an energy system decision problem - LO5: Explain how formulating multi-actor decision problems is different from single-actor decision problems - LO6: Formulate and solve optimisation problems using Matlab	
Education Method	Weekly lectures followed by 2-hour work sessions focused on problem solving using Matlab.	
Course Relations	This course cannot be followed together with SEN1511.	
Literature and Study Materials	Slides, articles, and study material presented on Brightspace.	
Assessment	Digital exam with a mix of multiple choice and open questions. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed digital exam on campus will be modified and substituted by an individual open-book online exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr.ir. Z. Lukszo	

SET3815-M	Matlab Fundamentals for Electrical Engineers	2
Responsible Instructor	Dr. A. Lekic	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	"Learn the fundamentals of the MATLAB scripting language and model-based design using Simulink" Get acquainted with MATLAB and Simulink through a series of interactive lectures focusing on developing real-world programming skills and insight. A series of "activities" during lectures and weekly homework get you to increase your "hands-on" experience. Get to focus on common pitfalls and mistakes and how to avoid them. Get exposed to "good" programming style guidelines so that others can better understand and further work with, or modify your code and designs. You get to practice in a mid-term mini project and demonstrate your newly acquired skills in a final mini project.	
Study Goals	After having passed this course... - You will be able to install and run MATLAB/SIMULINK in either Windows, Mac OS, Linux, or Android operating systems. - You will be able to operate MATLAB/SIMULINK, at least basically, either by the graphical user interface or command-line interface. - You will be able to develop, write and evaluate simple programs that run in MATLAB/SIMULINK. - You will be able to import evaluate and export data in native formats and other common data formats. - You will be able to create visualization aids (figures, plots, graphs, charts) that comply with standards in engineering and science. - You will be able to choose and apply the right code and/or model blocks to guarantee compatibility with specific versions of MATLAB/SIMULINK. - You will be able to avoid, identify and correct common pitfalls and mistakes in writing and running code. - You will be able to follow basic design and programming guidelines that make MATLAB/SIMULINK programs easier to read and work with.	
Education Method	Interactive online lectures using Brightspace as a digital learning platform. Mini-projects carried out in groups form part of the assessment. The lectures are structured as follows: 1) Theory and concepts are introduced with live programming demonstrations. 2) After a block of theory is explained, a MATLAB/SIMULINK "activity" consisting of, e.g. a brief exercise, analyzing an example, or completing a brief questionnaire is carried on by every student, and the deliverable of the activity uploaded to its respective Brightspace folder along with possible questions or remarks on the Brightspace forum. 3) If time permits, to finalize the activity its deliverable and posted questions are analyzed and discussed. 4) These steps are repeated until the topics for the session are covered. Delivered activities by the student get a delivery notification before the beginning of the next lecture at the latest. Delivery of activities is compulsory to get a grade. The course load is spread into 8 lectures of two hours each (16 hours). Preparation time for the lectures and homework overhead (delivery of activities) is estimated at a total of 16 hours (approx 2 hours per week). A mid-term project and a final project are budgeted at 12 hours each (24 hours both) of homework time for a total study load of 2 ECTS.	
Computer Use	Having access to a PC/laptop with installation rights is mandatory in order to be able to take part in the activities.	
Assessment	This course follows a PASS or FAIL grading scheme structured as follows: Formative assessment: 1) In the form of "activities" that are introduced, analyzed, and if time permits solved and discussed during lectures. Activities can (and have to) be delivered at the latest before the beginning of the next lecture. Activities get a delivery acknowledgment. A student must have at least 80% of the activities delivered to get a final grade. 2) In the form of a "mid-term" project carried on in groups, to be delivered at around half the course schedule. A lecture session is devoted for groups to peer-review and analyze the results of this project. Delivery of this project is compulsory to get a final grade. Summative assessment: A final project carried on in groups has to be delivered at a date announced in Brightspace. Your project deliverable gets assessed by the lecturer. It has to reflect that you (each student in your group) have fulfilled the learning objectives to get a PASS mark. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Matlab	

SET3815-Pr	System Integration Project	7
Responsible Instructor	L.M. Ramirez Elizondo	
Instructor	M.B. van der Meulen	
Contact Hours / Week x/x/x/x	0/1/3/3	
Education Period	2 3 4	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc in Engineering	
Course Contents	Q2: Start System Integration Project (1 EC) Q3: System Integration Project (3 EC) with focus on distributed power conversion and oral presentation. Q4: System Integration Project (3 EC) project.	
Study Goals	By the end of Q3, the students will be able to: - Present complex (technical/academic) information to a mixed target audience consisting of both specialists and non-specialists in a clear manner. - Use voice and body language to support the content of their presentation. - Apply the structure markers offered in the course in their presentation. - Design visuals that support the content of the presentation and operate them during the presentation. - Deal professionally with questions from the audience. - Provide effective feedback on presentations by others and self-criticism on their own presentations. - Present effectively in English. - Understand basic concepts related to circuit analysis - Model components of a microgrid through mathematical equations - Understand and apply the physics of energy generation and consumption By the end of Q4, the students will be able to: - Choose components and estimate sizes of the generation devices that will take part of the microgrid, according to the power demand - Implement the required component models - Incorporate the parameters corresponding to size of the generation devices and load into the models - Design a suitable control algorithm to keep the power balance of the system - Build a 3-phase configuration of the system to connect all the individual components - Simulate different scenarios to provide insight about the operation of the microgrid - Make an economic analysis of the microgrid	
Education Method	Q2 An assignment related to the modeling of the battery will be distributed to the students at the end of the quarter. Q3 Students will complete weekly assignments based on the taught material Q4 Each week the students will be requested to complete a different task related to the learning objectives. Students will be able to model renewable energy components and integrate them in a complete system using the simulation software.	
Literature and Study Materials	Online self-study modules in Brightspace: TU Present	
Assessment	Partial Assessment in Q3: A partial mark on a scale of 0 to 10 will be based on the presentations. Students who fail, like students who have missed a lesson, will be given the opportunity to compensate for this by means of an assignment. This represents 15% of the final mark. The student should obtain at least a 5.0 in this partial assessment. Partial Assessment in Q4: The partial mark on a scale of 0 to 10 will be given in this quarter based on the progress report, peer-review reports, and the final report. This represents 85% of the final mark. The student should obtain at least a 6.0 in this partial assessment. All marks will be expressed on a scale of 0 to 10. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3901	Graduation Project	45
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	x/x/x/x Based on appointments with supervising tutor. Students may start the Master thesis project after completion of 60 EC. The student has to find a project and supervisor (usually green list supervisor). Profile coordinator has to agree.	
Education Period	None (Self Study)	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	60 EC and an approved Individual Exam Program (IEP)	
Course Contents	The graduation project (SET3901) worth 45 EC is performed under supervision of one of the research groups affiliated with the Profile Cluster chosen by the student. In the Graduation Project a student works independently on a research project. Important is that system integration is addressed in the Graduation Project. The topic of this project can be defined in several ways: - A specialized topic addressing a specific component of the energy system within one profile (Wind Energy, Solar Energy, Biomass Energy, Power, Storage, Economics & Society). In that case the relation with the energy system must also be demonstrated in the report. - A topic on system integration within one profile (Wind Energy, Solar Energy, Biomass Energy, Power, Storage, Economics & Society, Electric Mobility). - A topic on system integration across two or more profiles in a combined project. In that case one of the profiles will take the lead, but the student will have supervisors from all profiles. The research project encompasses the following activities: - Literature study, problem formulation and planning; - Practical and theoretical work in one of the research groups, including interpretation and evaluation of the results; - Oral and written presentation of the work. For procedures see the course on Brightspace => MSc SET Thesis Project EEMCS. This also holds for graduation projects partly carried out with external partners. The Graduation Project is carried out at Delft University of Technology. If needed for the successful completion of the research, it is allowed to carry out the Graduation Project at a company. This has to be approved by the profile leader of the Board of Examiners of MSc SET, and supported by a written motivation from the supervisor. The Graduation Project always needs to be carried out according to MSc SET guidelines of Delft University of Technology.	
Study Goals	After the Graduation Project the student is able: - to read and understand international literature of the topic researched; - to work independently on an academic level in a R&D environment; - to plan and execute research in a project of 45 ECTS; - to work in an interdisciplinary and multicultural team of experts and supporting technologists/analysts; - to present his/her results in English by means of a presentation to a professional audience and by means of a written scientific report.	
Education Method	This course consists of an individual project.	
Literature and Study Materials	Provided by research group and resulting from literature study	
Assessment	Exam by appointment. The final assessment consists of one mark, based on: - Theoretical knowledge and understanding; - Method and scientific approach; - Competence in doing research work; - Report; - Presentation and defense; - General competences; The mark will be expressed on a scale of 0 to 10. An extensive grading scheme with assessment criteria is published on the website: https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/afstuderen-msc/assessment The Graduation Project is subject to strict procedures. Please, check the Brightspace course MSc SET Thesis Project EEMCS. Disclaimer: information may change depending on the developments around the coronavirus.	
Co-Instructor	Prof.dr. F.M. Mulder	
Co-Instructor	Dr.ir. W.A.A.M. Bierbooms	
Co-Instructor	Dr. L.M. Kamp	
Co-Instructor	L. Botto	
Co-Instructor	Prof.dr.ir. A.H.M. Smets	

WM0201SET	Technical Writing	2
Module Manager	M.B. van der Meulen	
Instructor	M.B. van der Meulen	
Co-responsible for assignments	Drs. A.E. Kam	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Modern engineers are required to possess excellent communication skills. Not only do they need to know their business, they need to know how to communicate this business to other experts, decision makers and implementers. They need to be able to write accessible, persuasive technical reports and proposals and to use a correct tone in professional correspondence. This course focuses on both the quality of the text and on the writing process. The lessons will be used for instruction, feedback and practice. Since the course equals 2EC or 56 hours per student, students will have to put in a significant amount of time outside the classroom to pass this course. Students will be required to hand in several assignments while they are working on their main assignment. No more than one lesson may be missed.	
Study Goals	At the end of the course, students are able to: -list the required element of technical reports (from title page to appendices and everything in-between) -assess whether a technical report meets these requirements -efficiently write a clear report for a target audience that has little time -structure the report -write clearly, concisely and convincingly -write proper and appropriate language -refer correctly to sources	
Education Method	work groups	
Assessment	final assignment (writing a report as a group)	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Profile Courses SET 2022

Program Structure 1

The profile courses are:

Profile Biomass
Thermochemistry of Biomass Conversion
Green Chemistry and Sustainable Technology
Multiphase Reactor Engineering

Profile Power
Electronic Power Conversion
Intelligent Electrical Power Grids
Electric Power Systems of the Future
DC and AC Microgrids

Profile Economics and Society
Sustainable Energy Innovations and Transitions
Economic Policy for Sustainable Energy
Technology, Entrepreneurship and Sustainability

Profile Solar
Photovoltaics Basics
Photovoltaics Technologies
Photovoltaics Systems

Profile Storage
The Necessity of Storage Technology
Energy Storage in Batteries
Hydrogen Technology

Profile Wind
Introduction to Wind Turbines: Physics and Technology
Site Conditions for Wind Turbine Design
Wind Turbine Design

Profile 'Electric Mobility'
Electrical Machines and Drives
Electric Vehicles & Charging Technology
Digital Modelling of Electrical Powertrain

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Solar systems: Solar Power Storage 2022	
Introduction 1	Students choose EE4545 or EE4536 in their profile courses.

CH3222	Energy Storage in Batteries	4
Responsible Instructor	Dr.ir. M. Wagemaker	
Instructor	Dr. E.M. Kelder	
Instructor	Prof.dr. F.M. Mulder	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Efficient energy storage in batteries is of major importance for electronic equipment, electrification of cars and buses, grid stability and for the introduction of fluctuating renewable energy sources including wind and solar. In this course we will learn about the thermodynamics and charge kinetics in batteries including the electrochemical driving force, thermochemistry, phase transitions and charge transport kinetics. The principles of state of the art and future rechargeable batteries, also touching super capacitors, will be discussed. The aim is to develop understanding of the battery performance parameters and to relate these with the with underlying atomic scale material properties. Along these topics current battery developments and research directions will be discussed. After seven lectures on the fundamental properties of batteries six dedicated working lectures will focus on important battery technologies including: Redox Flow Na-ion next generation Li-ion batteries Solid State batteries Supercapacitors	
Study Goals	At the end of the battery lectures students should be able to: - List the major types of Li and other batteries and their characteristics. - Reproduce the major challenges in batteries. - Explain how a battery works and the functioning of each component. - List and explain the different reaction types in batteries. - Explain the different charge transfer processes in batteries. - List and explain the different mechanisms that limit the cycle life of batteries. - Apply Nernst Law describing redox reactions. - Calculate conductivities and overpotentials. - Apply the Tafel equation describing the charge transfer between electrode and electrolyte. - Calculate energy densities and power densities from observed properties. - Derive from a phase diagram the different phases that will occur upon lithiation. - Apply the Gibbs Phase rule to predict constant of variable output voltage. - Derive the voltage curve from the shape of the Gibbs Free energy. - Predict the basic redox reaction of two electrodes given the formation energy or the standard potentials. - Compare different battery systems and differentiate. - Discuss scientific results observed in battery performance and to relate this to material properties.	
Education Method	The course contains fourteen 2-hour lectures. The first seven 2-hour lectures focusses on building the underlying thermodynamics, material science and electrochemistry chemistry and the principles of energy storage in batteries. This knowledge will be used to develop understanding of the material properties that determine performance of current and next generation battery. In the second seven lectures will be working lectures with focus on six important battery technologies. The first half hour will be a lecture on the specific battery technology followed by half an hour assignment. In the final half hour of the working lecture the assignment will be discussed. In the final lecture example exams will be discussed.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	R.A. Huggins: Advanced Batteries. Available via the library as pdf's: http://www.springerlink.com/content/nq6404/?MUD=MP Slides Articles	
Assessment	Written exam (100% of the final grade)	
	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

EE4536	DC and AC Microgrids	4
Responsible Instructor	L.M. Ramirez Elizondo	
Responsible Instructor	Dr.ir. P. Bauer	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc Electrical Engineering or equivalent	
Course Contents	This course focuses on the modelling and control of AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Several alternatives for the provision of ancillary services by means of AC and DC microgrids will be presented. Furthermore, attention will be given to the design of energy management systems for AC and DC microgrids including concepts as demand side management.	
Study Goals	Identify the challenges introduced by distributed generation Describe different microgrids' layouts and sizing Describe the operation and control of power electronic converters in AC and DC microgrids Apply droop control in AC and DC microgrids Describe the role of energy management systems in AC and DC microgrids Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids	
Education Method	Lectures	
Literature and Study Materials	Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatzigiargyriou	
Assessment	Written exam 80%, Assignments 20%	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects) 2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTs) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	

ET4376	Photovoltaic Basics	4
Responsible Instructor	Prof.dr. M. Zeman	
Course Coordinator	Dr.ir. R. Santbergen	
Contact Hours / Week x/x/x/x	0/4/0/0 lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised.	
Study Goals	Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.	
Education Method	Lectures, short multiple-question tests, exercises, instructions and self-study using the book Solar Energy.	
Literature and Study Materials	Book Solar Energy: The physics and engineering of photovolataic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.	
Assessment	Short multiple-question tests, instructions, written exam. All marks will be expressed on a scale of 0 to 10.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

ET4377	Photovoltaic Technologies	4
Responsible Instructor	Dr. L. Mazzearella	
Responsible Instructor	Prof.dr. I.M.F. Gordon	
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/0/2/0 lectures; 0/0/2/0 instruction	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Lecture course Photovoltaic Basics (ET4 376)	
Course Contents	This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed. Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.	
Study Goals	This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required. The learning objectives (LO) are as follows: LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation. LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design. LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.	
Education Method	Flipped-classroom video lectures and on-campus instructions.	
Literature and Study Materials	Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England. ISBN: 978 1 906860 32 5 (paperback); This book is also available as a free ebook: ISBN: 978 1 906860 73 8 (ePub) ISBN: 978 1 906860 75 2 (pdf)	
Assessment	The course will be assessed by a combination of a written final examination (that is taken on campus if COVID-19 restrictions allow, or by an online test that is taken via Brightspace, followed by possible verification) and three graded assignments that are not mandatory. The final mark for this course, F, will be based on the average results for the assignments, A, and the final examination, E. All marks will be expressed on a scale of 0 to 10. For calculating A, the average of the three assignment results is used. The assignment results only will be valid within the academic year in which the assignments have been made. Assignment results obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Instructor	H. Ziar	
Instructor	P. Manganiello	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	After this course, the student will be able to - Determine optimal PV panels orientation from geographical survey - Compare between different PV systems topologies - Calculate the energy yield of a PV system - Identify different types of solar modules and other components used in PV systems - Justify the importance of bypass diodes and power conversion in PV systems - Analyze different power tracking methods - Design PV systems for different applications and estimate the cost - Defend the goodness of his/her own PV System design - Describe trends of physical quantities related to customized PV systems	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3065	Intelligent Electrical Power Grids	4
Responsible Instructor	Prof.dr. P. Palensky	
Instructor	Dr.ir. J.L. Rueda Torres	
Instructor	Dr. A.I. Stefanov	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets. Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools. You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done. This course is part of the Power track of the SET master and can also be chosen by EE master students. Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.	
Study Goals	After this course students are able to - create a demand/supply matching system, to - model and simulate a smart grid, to - specify the design requirements of a SCADA system, and to - analyse cyber security threats, threat vectors, and vulnerabilities of ITOT smart grid systems.	
Education Method	The course is organized in the following units, all units have assignments (typically a simulation): 1 Modeling Smart Grids / Automation Networks (Peter) 2 Power Flow Analysis (Jose) 3 Power System Dynamics (Jose) 4 Lab Session: Modeling of Generator Controllers (TAs) 5 Lab Session: Modeling of Wind Farms in Modelica (TAs) 6 AI and ML in Energy (Pedro/Jochen) 7 SCADA/WAMS (Peter) 8 Cyber Security (Alex)	
Literature and Study Materials	On Brightspace	
Assessment	The grade is formed based on the following elements: 50% Assignment reports (in time!) 50% multiple choice Moebius Computer Exam In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3080	The Necessity of Storage Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic knowledge on chemistry and physics, as well as knowledge on Matlab. Level as of MSc SET students.	
Course Contents	In a renewable energy future energy storage is an additional necessary means to match the supply and demand of energy. - Week 1: Future energy generation systems, the combination of sources on a large scale. Group work 1: program in Matlab a numerical model for energy generation, with flexible input of PV, solar heat, wind, hydro, biofuel - Week 2: Future energy use possibilities: look into current typical use patterns of the different sectors, and look into possible changes: electrification, different sources, etc. Group work 2: program in Matlab typical use patterns for the different sectors - Week 3: Estimate the possibilities for 'demand response' in the different sectors and program this in Matlab. Deduce the resulting demand for short term and long term storage. Determine the demand for transport of energy and estimate possible transport losses. Group work 3: program in Matlab the different magnitudes for demand response, short and long storage throughout the year. - Week 4: Storage methods introduction, pumped hydro, mechanical, compressed air, deployment possibilities. Group work 4: calculate different factors for energy density, capacity, timescale of storage. Investigate approximate efficiency. - Week 5: Thermal storage, high T and low T. Combined heat and power possibilities. Group work 5: Investigate energy densities, efficiencies, capacities, exergies. - Week 6: Biomass and biofuels, future prospects for storage. Introduction to scalable chemical energy storage methods: batteries and fuels. What would electrification of the chemical industry mean. Group work: investigate biofuel resources, calculate efficiencies and compare to solar PV and electrolysis numbers. - Week 7: Hand in report and model. Group presentations of the model calculations (where each group has different boundary conditions). Discuss how the boundary conditions influence the outcome of the storage demand.	
Study Goals	At the end of the lectures and group work the students should be able to: - Reproduce and reflect on the role of storage in a renewable energy based society. - Explain how different timescales for energy storage are relevant for the future system. - List and explain the different storage solutions in terms of typical energy density, cost and efficiency. - Tell about the different drivers for energy conversion including demand for storage, and demand for transport of energy. - Reflect on the energy efficiency of intermediate storage.	
Education Method	The course contains seven weeks with 4-hour contact time. The 4 hours will be divided in lectures, group work applying the lecture content and a question time. The goal is to come to a finished assignment after six weeks that can be presented in class the seventh week. The structure of the course is as follows: each week there is first a class in which a background is handed out, then there is a working group session where the supplied information is used to start modelling aspects of a future energy system. The students will be divided over groups with each group an individual model question. The lesson is concluded with a question session. This will continue 6 weeks and the 7th week the different models will be presented by the students.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	Slides Articles, reports and reviews.	
Assessment	The final grade consists of the following parts: Individual exam (70% of the final grade). Group work (30% of the final grade, based on a report produced by the group). The groupwork cannot be redone individually. For one part may be scored 5,0 as long as the other part compensates this. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Prof.dr. J.J.C. Geerlings	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	Written exam	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

SET3095	Electronic Power Conversion	4
Responsible Instructor	A. Shekhar	
Instructor	H.J.M. Olsthoorn	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.	
Study Goals	To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.	
Education Method	Lectures Problem solving sessions Practicum Lectures	
Assessment	Written exam (closed book)	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Bio/solar systems: Biomass Solar Storage 2022

CH3061	Multiphase Reactor Engineering	4
Responsible Instructor	Dr.ir. J.R. van Ommen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student is expected to: 1. have followed a course on mass and energy balances (such as Molecular Transport Phenomena) 2. have followed a basic course in reactor engineering (such as Reactors and Kinetics) 3. is able to work with Python (or similar software; e.g., has followed a course on numerical techniques)	
Course Contents	Multiphase reactor types: Fixed beds, trickle beds, fluidized beds, bubble columns, slurry reactors, microreactors, structured (monolith) reactors, electrocatalytic reactors. Engineering aspects: Flow regimes, mass transfer, conversion models. Hatta number, reaction enhancement. Residence time distribution. Catalytic reactors structured in time and space. Unifying concepts. Design aspects: reactor selection, decoupling of kinetics and transport phenomena, combination of reaction and separation, reaction coupling, separating catalytic steps, controlled energy input, circularity of materials.	
Study Goals	1. The student is able to describe for each of the archetypes of multiphase reactors, the various embodiments with practical relevance. 2. The student is able to use the scientific literature to find relevant information for the various reactor types. 3. The student is able to analyse a multiphase reactor and to describe the physical and chemical process steps in words. 4. The student is able to explain the basic measurement techniques used in studying multiphase reactors. 5. The student is able to write down the mass, energy, and momentum balances for the overall system and the steps identified under objective 3. 6. The student is able to implement customary models in the balances mentioned under 5, such that a system of equations describing the reactor is obtained. 7. The student is able to simplify the system of equations based on a quantitative analysis and identification the rate-determining steps. 8. The student is able to use current software packages to solve the systems of equations numerically for the basic types of reactors. 9. The student is able to use the developed models to analyse defined cases. 10. The student is able to make a quantitative design of a multiphase reactor for a given application. 11. The student is able to present, both orally and in writing, the outcomes of modeling, calculations, and evaluation.	
Education Method	Each lecture will give a short overview of a specific multiphase reactor type (only 45 minutes). The slides available on Brightspace. You will get references to papers and/or chapters from textbooks that give more information about the reactor type. These will be made available via Brightspace. After the lecture, an assignment (about 8 hours work) will be available via Brightspace for the specific reactor type treated. These assignments are made in teams of four students; this number may be changed depending on the number of participants. You are free to choose which software package you want to use for making the assignments, but Python is recommended. After two weeks, each team hands in a detailed solution of the problem on paper the day before the discussion before the announced deadline. Moreover, each team prepares a summary of the solution for presentation in PowerPoint. Please bring this presentation on a memory stick. Each week, a few teams present their solution; the instructors choose the student that is presenting. The other teams challenge this solution. At the end, each team will make a larger final project.	
Computer Use	You are free to choose which software package you want to use for making the assignments, but Python is recommended.	
Literature and Study Materials	All material is provided via Brightspace.	
Assessment	The final grade consists of three parts: * the solution to the minor assignments * an individual part; the nature of this part will depend on the number of participants * the final assignment Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

CH3222	Energy Storage in Batteries	4
Responsible Instructor	Dr.ir. M. Wagemaker	
Instructor	Dr. E.M. Kelder	
Instructor	Prof.dr. F.M. Mulder	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Efficient energy storage in batteries is of major importance for electronic equipment, electrification of cars and buses, grid stability and for the introduction of fluctuating renewable energy sources including wind and solar. In this course we will learn about the thermodynamics and charge kinetics in batteries including the electrochemical driving force, thermochemistry, phase transitions and charge transport kinetics. The principles of state of the art and future rechargeable batteries, also touching super capacitors, will be discussed. The aim is to develop understanding of the battery performance parameters and to relate these with the with underlying atomic scale material properties. Along these topics current battery developments and research directions will be discussed. After seven lectures on the fundamental properties of batteries six dedicated working lectures will focus on important battery technologies including: Redox Flow Na-ion next generation Li-ion batteries Solid State batteries Supercapacitors	
Study Goals	At the end of the battery lectures students should be able to: - List the major types of Li and other batteries and their characteristics. - Reproduce the major challenges in batteries. - Explain how a battery works and the functioning of each component. - List and explain the different reaction types in batteries. - Explain the different charge transfer processes in batteries. - List and explain the different mechanisms that limit the cycle life of batteries. - Apply Nernst Law describing redox reactions. - Calculate conductivities and overpotentials. - Apply the Tafel equation describing the charge transfer between electrode and electrolyte. - Calculate energy densities and power densities from observed properties. - Derive from a phase diagram the different phases that will occur upon lithiation. - Apply the Gibbs Phase rule to predict constant of variable output voltage. - Derive the voltage curve from the shape of the Gibbs Free energy. - Predict the basic redox reaction of two electrodes given the formation energy or the standard potentials. - Compare different battery systems and differentiate. - Discuss scientific results observed in battery performance and to relate this to material properties.	
Education Method	The course contains fourteen 2-hour lectures. The first seven 2-hour lectures focusses on building the underlying thermodynamics, material science and electrochemistry chemistry and the principles of energy storage in batteries. This knowledge will be used to develop understanding of the material properties that determine performance of current and next generation battery. In the second seven lectures will be working lectures with focus on six important battery technologies. The first half hour will be a lecture on the specific battery technology followed by half an hour assignment. In the final half hour of the working lecture the assignment will be discussed. In the final lecture example exams will be discussed.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	R.A. Huggins: Advanced Batteries. Available via the library as pdf's: http://www.springerlink.com/content/nq6404/?MUD=MP Slides Articles	
Assessment	Written exam (100% of the final grade)	
	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

ET4376	Photovoltaic Basics	4
Responsible Instructor	Prof.dr. M. Zeman	
Course Coordinator	Dr.ir. R. Santbergen	
Contact Hours / Week x/x/x/x	0/4/0/0 lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised.	
Study Goals	Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.	
Education Method	Lectures, short multiple-question tests, exercises, instructions and self-study using the book Solar Energy.	
Literature and Study Materials	Book Solar Energy: The physics and engineering of photovolataic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.	
Assessment	Short multiple-question tests, instructions, written exam. All marks will be expressed on a scale of 0 to 10.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

ET4377	Photovoltaic Technologies	4
Responsible Instructor	Dr. L. Mazzearella	
Responsible Instructor	Prof.dr. I.M.F. Gordon	
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/0/2/0 lectures; 0/0/2/0 instruction	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Lecture course Photovoltaic Basics (ET4 376)	
Course Contents	This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed. Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.	
Study Goals	This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required. The learning objectives (LO) are as follows: LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation. LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design. LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.	
Education Method	Flipped-classroom video lectures and on-campus instructions.	
Literature and Study Materials	Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England. ISBN: 978 1 906860 32 5 (paperback); This book is also available as a free ebook: ISBN: 978 1 906860 73 8 (ePub) ISBN: 978 1 906860 75 2 (pdf)	
Assessment	The course will be assessed by a combination of a written final examination (that is taken on campus if COVID-19 restrictions allow, or by an online test that is taken via Brightspace, followed by possible verification) and three graded assignments that are not mandatory. The final mark for this course, F, will be based on the average results for the assignments, A, and the final examination, E. All marks will be expressed on a scale of 0 to 10. For calculating A, the average of the three assignment results is used. The assignment results only will be valid within the academic year in which the assignments have been made. Assignment results obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Instructor	H. Ziar	
Instructor	P. Manganiello	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	After this course, the student will be able to - Determine optimal PV panels orientation from geographical survey - Compare between different PV systems topologies - Calculate the energy yield of a PV system - Identify different types of solar modules and other components used in PV systems - Justify the importance of bypass diodes and power conversion in PV systems - Analyze different power tracking methods - Design PV systems for different applications and estimate the cost - Defend the goodness of his/her own PV System design - Describe trends of physical quantities related to customized PV systems	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3070	Thermochemistry of Biomass Conversion	4
Responsible Instructor	Prof.dr.ir. W. de Jong	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	To start this course you need to have a good knowledge of BSc. level thermodynamics, chemistry, kinetics, basic reactor modelling (batch, plug flow, continuously well stirred reactor), basic process engineering (mass, energy, momentum balances), and physical transport phenomena. In case you think you miss some of this pre-existing knowledge, please perform selfstudy of the first six chapters of the book 'Biomass as a Sustainable Energy Source for the Future', Wiley (2015), eds. De Jong and Van Ommen	
Summary	Biomass is available in abundance on a global level; it is naturally stored solar energy in a wide diversity of chemical structures. Among the renewable energy sources, biomass can be stored most easily, of course depending once again on the resources characteristics. Due to the heterogeneous nature of biomass also many different conversion technologies have been developed in the context of energy conversion chains with combustion being the oldest performed by mankind. This course deals with the background and design and implementation of different technologies to convert biomass thermochemically.	
Course Contents	In this course you will learn about the following aspects of biomass conversion based on thermo-chemical conversion technologies. - Biomass Characterisation (biomass types, properties, analyses) - Pretreatment (harvesting, size reduction, drying, torrefaction) - Thermochemical conversion (combustion, gasification and pyrolysis; reactor types) - Transportation fuels / biorefinery (syngas conversion, economics) - Process design (structured design approach)	
Study Goals	During this course, the students will obtain knowledge of and insight into energy conversion technologies based on biomass thermo-chemical conversion processes for a wide variety of applications, like supply of power, heat and fuels. They will be offered practical problems to solve and projects to perform in this area. These main learning objectives can be divided into several smaller learning objectives: 1) The student has knowledge on the typical biomass composition, related properties (like e.g. heating value, ash melting behaviour) and can critically identify and select fuel characterization techniques. 2) The student is able to identify and perform basic capacity calculations concerning thermo-chemical biomass conversion systems for small and larger scale (combined) heat and power generation as well as systems for transportation fuel production. 3) The student is able to set up reaction equations (stoichiometry) for a wide range of energy supply related fuel conversion processes. 4) The student can set up and perform heat and mass balance calculations regarding single phase and simple multi-phase biomass thermo-chemical conversion reactors and extended (downstream) systems for power, heat and fuels generation. 5) The student knows how to perform basic ideal gas phase chemical reaction equilibrium calculations to calculate product compositions and extents of reactions given initial reaction mixture composition, pressure and temperature. 6) The student can understand and describe reaction kinetic expressions and also apply them in order to solve (design) problems related to the conversion of fuels and the formation of emission components. 7) The student demonstrates good knowledge of mass and heat transfer mechanisms and phenomena applied to the chemical conversion of both liquid and solid biomass (derived) fuels. 8) The student can derive equations for idealised model reactors (well-stirred reactor and plug flow reactor concepts) and apply these simplified reactor concepts to solve engineering problems related to the thermo-chemical conversion of biomass fuels, in particular combustion, gasification and pyrolysis. 9) The student is able to perform calculations concerning the conversion of small particles; important in this respect is the identification of the regime of conversion. Related dimensionless numbers such as the Thiele Modulus and the Biot number are key indicators in this respect. 10) The student is able to understand the basics of thermo-chemical conversion processes and to translate this into designing and evaluating such (sometimes hybrid) systems, including their economic evaluation based on characteristic parameters like NPV, POT, and ROI. 11) The student demonstrates good communication skills, both oral and written, concerning presentation of (applied) scientific results.	
Education Method	This course consists of six 2-hour lectures, also using blended learning methods. Each lecture is followed by a 2-hour work session or excursion(s)/lab visit(s) in the same week -depending on the actual COVID situation- focused on working out minor assignments including literature surveys, problem solving and work on the major project. In the week of the exams you will present the major assignment project results.	
Course Relations	This course is a core course in the Sustainable Energy Biomass Energy profile.	
Literature and Study Materials	Book: "Biomass, a sustainable energy source for the future", de Jong and van Ommen, Wiley., 1st edition, Wiley (2015). Lecture slides	
Assessment	The final grade consists of three parts: * the solution to the (three) assignments (team work); 40% of the grade * a larger final assignment (team work); 50% of the grade * evaluation of the final assignment's presentation; 10% of the grade The assessment method may be adapted to actual student population. You can get a chance to resit the larger final assignment part of the course in case of having obtained an insufficient grade for this part by means of a repair additional work for the same assignment to obtain a sufficient grade. If you pass only part of the course, all parts need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3080	The Necessity of Storage Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic knowledge on chemistry and physics, as well as knowledge on Matlab. Level as of MSc SET students.	
Course Contents	In a renewable energy future energy storage is an additional necessary means to match the supply and demand of energy. - Week 1: Future energy generation systems, the combination of sources on a large scale. Group work 1: program in Matlab a numerical model for energy generation, with flexible input of PV, solar heat, wind, hydro, biofuel - Week 2: Future energy use possibilities: look into current typical use patterns of the different sectors, and look into possible changes: electrification, different sources, etc. Group work 2: program in Matlab typical use patterns for the different sectors - Week 3: Estimate the possibilities for 'demand response' in the different sectors and program this in Matlab. Deduce the resulting demand for short term and long term storage. Determine the demand for transport of energy and estimate possible transport losses. Group work 3: program in Matlab the different magnitudes for demand response, short and long storage throughout the year. - Week 4: Storage methods introduction, pumped hydro, mechanical, compressed air, deployment possibilities. Group work 4: calculate different factors for energy density, capacity, timescale of storage. Investigate approximate efficiency. - Week 5: Thermal storage, high T and low T. Combined heat and power possibilities. Group work 5: Investigate energy densities, efficiencies, capacities, exergies. - Week 6: Biomass and biofuels, future prospects for storage. Introduction to scalable chemical energy storage methods: batteries and fuels. What would electrification of the chemical industry mean. Group work: investigate biofuel resources, calculate efficiencies and compare to solar PV and electrolysis numbers. - Week 7: Hand in report and model. Group presentations of the model calculations (where each group has different boundary conditions). Discuss how the boundary conditions influence the outcome of the storage demand.	
Study Goals	At the end of the lectures and group work the students should be able to: - Reproduce and reflect on the role of storage in a renewable energy based society. - Explain how different timescales for energy storage are relevant for the future system. - List and explain the different storage solutions in terms of typical energy density, cost and efficiency. - Tell about the different drivers for energy conversion including demand for storage, and demand for transport of energy. - Reflect on the energy efficiency of intermediate storage.	
Education Method	The course contains seven weeks with 4-hour contact time. The 4 hours will be divided in lectures, group work applying the lecture content and a question time. The goal is to come to a finished assignment after six weeks that can be presented in class the seventh week. The structure of the course is as follows: each week there is first a class in which a background is handed out, then there is a working group session where the supplied information is used to start modelling aspects of a future energy system. The students will be divided over groups with each group an individual model question. The lesson is concluded with a question session. This will continue 6 weeks and the 7th week the different models will be presented by the students.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	Slides Articles, reports and reviews.	
Assessment	The final grade consists of the following parts: Individual exam (70% of the final grade). Group work (30% of the final grade, based on a report produced by the group). The groupwork cannot be redone individually. For one part may be scored 5,0 as long as the other part compensates this. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Prof.dr. J.J.C. Geerlings	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	Written exam	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3311	Green Chemistry and Sustainable Technology	4
Responsible Instructor	Dr. F. Hollmann	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Organic Chemistry (LB1045 or 4051OCSTRY) Chemical Biology (LB2621) OR Catalysis (4052KATALY)	
Course Contents	Depleting fossil fuels and environmental concerns force us to change the way we produce our goods. The transition of the chemical industry to becoming sustainable represents one of the major challenges to be faced in the 21st century. Catalytic reactions will increasingly substitute the existing energy- and resource-intensive production procedures; hence reducing or preventing toxic pollutants. Biotechnology will play a major role in this transition and complement advanced chemical production technologies. This course will highlight the emerging role of catalysis in the chemical industry. A major focus will lie on biotechnological production of chemicals (fermentation processes, isolated enzymes). These methods will be related to their chemocatalytic counterparts. Thus, ample examples including the underlying mechanistic principles - of bio- and chemocatalytic processes will be discussed. Finally, the emerging field of renewable feedstocks will be discussed not only from a chemical standpoint, but also on socioeconomical and ecological grounds. The 12 Principles of Green Chemistry will serve as guideline to assess the scope and limitations of the examples discussed.	
Study Goals	After attending this course the student will: (1) have detailed knowledge of sustainable catalysts (enzyme- and chemo-catalysis) (2) know industrial examples of sustainable manufacturing practice (3) be able to use the 12 Principles of Green Chemistry to assess the sustainability of a given process (4) be able to critically assess the purport of green claims	
Education Method	Lecture Sheets, specialized books, articles and patents	
Assessment	(1) Written exam (digital off-campus (open book) or on-campus, depending on the COVID-19 measures), 75% of the overall grade (2a) only for SET students: written assignment about a topic of Green Chemistry (groups of 3-4 students), 25% of the overall grade (2b) for non-SET students: group presentation about a topic of Green Chemistry (groups of 4-5 students), 25% of the overall grade Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Autonomous systems: Wind Storage - Solar 2022

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: - describe the components and configurations of wind turbines - describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine - describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design - explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. - identify the drivers of dynamic behaviour and calculate their typical values - describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure - describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Special Information	Assessment may differ due to COVID-19 measures. Changes to the assessment method will be communicated via Brightspace.	
Remarks	This course is an elective course for students from various faculties (AE, EEMCS, CEG,...). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AE4W09	Wind Turbine Design	5
Responsible Instructor	Dr. D.A. von Terzi	
Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.	
Parts	In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4. The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.	
Course Contents	Overview of the course and the group assignment System design and scaling rules Aerodynamic and structural rotor design and analysis Drive train and electrical system Wind turbine control Blade materials and assessment of fatigue damage Wind data and description The use of standards for load calculations	
Study Goals	In this course you learn various aspects of designing a wind turbine, including: - Selecting approaches for design and analysis - Collect information and data - Use (simulation) programs, models and calculations - Analyse and interpret results and draw conclusions about the design - Make design decisions Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.	
Education Method	lectures + software tutorials + group assignment + discussion meetings	
Computer Use	Spreadsheet-type calculations are used to support design decisions. The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.	
Literature and Study Materials	Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.	
Assessment	A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.	
Remarks	This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master. Supplementary courses are (these treat similar subjects as this course, but more in-depth): AE4W13 Site conditions for wind turbine design AE4135 Rotor / wake aerodynamics AE4W21-14 Wind turbine aeroelasticity ET4117 Electrical machines and drives OE44135 Offshore wind support structures	

AE4W13	Site Conditions for Wind Turbine Design	3
Responsible Instructor	Dr.ir. W.A.A.M. Bierbooms	
Instructor	Dr.ir. D.J.N. Allaerts	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	It is assumed that the student has a basic understanding of fluid mechanics and stochastics (on Brightspace self tests are available). A basic skill of Matlab is assumed: you should be able to read from and write to a text file, write and/or modify simple functions (m-files). Furthermore, it is recommended that you already have a basic knowledge of wind turbines; e.g. AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Course Contents	Introduction Atmospheric Boundary Layer Meteorology Fluid Dynamics Turbulence Wind power meteorology Measurement instruments Statistics Methods in turbulence Offshore wind climate External conditions in standards Wave and current climate and loading Example analysis site data Turbulence spectra Satellite remote sensing techniques Simulation of turbulent wind fields and gusts	
Study Goals	After successfully completing the course, students will be able to: Discuss the relevance of particular site data for wind turbine design, operation and siting. Analyse simple problems related to wind power meteorology Interpret scientific dissemination by experts on main points Apply analysis methods to convert basic site data to wind turbine design parameters	
Education Method	Lecture + assignments + compulsory practical	
Course Relations	Follow up courses AE4-W09, AE4135	
Literature and Study Materials	www.eos.ubc.ca/books/Practical_Meteorology Chapters 12 to 14; Pijush K. Kundu, Ira M. Cohen, Fluid Mechanics, Elsevier. Lecture notes and reader	
Prerequisites	AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Assessment	The written exam consists out of two parts, one for each period (scheduled in the exam period). The final grade is a weighted average of these 2 exams and the grade for one of the assignments (which one will be specified during the course). In order to take part in Exam2 one should join the compulsory practical The resit is an oral exam (to be scheduled with the teacher) on basis of pass/fail (i.e. in case of a pass the grade 6 will be put in Osiris).	
Set-up	The course combines lectures and student assignments/practical. For several topics guest speakers will be invited.	

CH3222	Energy Storage in Batteries	4
Responsible Instructor	Dr.ir. M. Wagemaker	
Instructor	Dr. E.M. Kelder	
Instructor	Prof.dr. F.M. Mulder	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Efficient energy storage in batteries is of major importance for electronic equipment, electrification of cars and buses, grid stability and for the introduction of fluctuating renewable energy sources including wind and solar. In this course we will learn about the thermodynamics and charge kinetics in batteries including the electrochemical driving force, thermochemistry, phase transitions and charge transport kinetics. The principles of state of the art and future rechargeable batteries, also touching super capacitors, will be discussed. The aim is to develop understanding of the battery performance parameters and to relate these with the with underlying atomic scale material properties. Along these topics current battery developments and research directions will be discussed. After seven lectures on the fundamental properties of batteries six dedicated working lectures will focus on important battery technologies including: Redox Flow Na-ion next generation Li-ion batteries Solid State batteries Supercapacitors	
Study Goals	At the end of the battery lectures students should be able to: - List the major types of Li and other batteries and their characteristics. - Reproduce the major challenges in batteries. - Explain how a battery works and the functioning of each component. - List and explain the different reaction types in batteries. - Explain the different charge transfer processes in batteries. - List and explain the different mechanisms that limit the cycle life of batteries. - Apply Nernst Law describing redox reactions. - Calculate conductivities and overpotentials. - Apply the Tafel equation describing the charge transfer between electrode and electrolyte. - Calculate energy densities and power densities from observed properties. - Derive from a phase diagram the different phases that will occur upon lithiation. - Apply the Gibbs Phase rule to predict constant of variable output voltage. - Derive the voltage curve from the shape of the Gibbs Free energy. - Predict the basic redox reaction of two electrodes given the formation energy or the standard potentials. - Compare different battery systems and differentiate. - Discuss scientific results observed in battery performance and to relate this to material properties.	
Education Method	The course contains fourteen 2-hour lectures. The first seven 2-hour lectures focusses on building the underlying thermodynamics, material science and electrochemistry chemistry and the principles of energy storage in batteries. This knowledge will be used to develop understanding of the material properties that determine performance of current and next generation battery. In the second seven lectures will be working lectures with focus on six important battery technologies. The first half hour will be a lecture on the specific battery technology followed by half an hour assignment. In the final half hour of the working lecture the assignment will be discussed. In the final lecture example exams will be discussed.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	R.A. Huggins: Advanced Batteries. Available via the library as pdf's: http://www.springerlink.com/content/nq6404/?MUD=MP Slides Articles	
Assessment	Written exam (100% of the final grade)	
	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

ET4376	Photovoltaic Basics	4
Responsible Instructor	Prof.dr. M. Zeman	
Course Coordinator	Dr.ir. R. Santbergen	
Contact Hours / Week x/x/x/x	0/4/0/0 lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised.	
Study Goals	Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.	
Education Method	Lectures, short multiple-question tests, exercises, instructions and self-study using the book Solar Energy.	
Literature and Study Materials	Book Solar Energy: The physics and engineering of photovoltaic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.	
Assessment	Short multiple-question tests, instructions, written exam. All marks will be expressed on a scale of 0 to 10.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4377	Photovoltaic Technologies	4
Responsible Instructor	Dr. L. Mazzearella	
Responsible Instructor	Prof.dr. I.M.F. Gordon	
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/0/2/0 lectures; 0/0/2/0 instruction	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Lecture course Photovoltaic Basics (ET4 376)	
Course Contents	This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed. Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.	
Study Goals	This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required. The learning objectives (LO) are as follows: LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation. LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design. LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.	
Education Method	Flipped-classroom video lectures and on-campus instructions.	
Literature and Study Materials	Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England. ISBN: 978 1 906860 32 5 (paperback); This book is also available as a free ebook: ISBN: 978 1 906860 73 8 (ePub) ISBN: 978 1 906860 75 2 (pdf)	
Assessment	The course will be assessed by a combination of a written final examination (that is taken on campus if COVID-19 restrictions allow, or by an online test that is taken via Brightspace, followed by possible verification) and three graded assignments that are not mandatory. The final mark for this course, F, will be based on the average results for the assignments, A, and the final examination, E. All marks will be expressed on a scale of 0 to 10. For calculating A, the average of the three assignment results is used. The assignment results only will be valid within the academic year in which the assignments have been made. Assignment results obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Instructor	H. Ziar	
Instructor	P. Manganiello	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	After this course, the student will be able to - Determine optimal PV panels orientation from geographical survey - Compare between different PV systems topologies - Calculate the energy yield of a PV system - Identify different types of solar modules and other components used in PV systems - Justify the importance of bypass diodes and power conversion in PV systems - Analyze different power tracking methods - Design PV systems for different applications and estimate the cost - Defend the goodness of his/her own PV System design - Describe trends of physical quantities related to customized PV systems	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3080	The Necessity of Storage Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic knowledge on chemistry and physics, as well as knowledge on Matlab. Level as of MSc SET students.	
Course Contents	In a renewable energy future energy storage is an additional necessary means to match the supply and demand of energy. - Week 1: Future energy generation systems, the combination of sources on a large scale. Group work 1: program in Matlab a numerical model for energy generation, with flexible input of PV, solar heat, wind, hydro, biofuel - Week 2: Future energy use possibilities: look into current typical use patterns of the different sectors, and look into possible changes: electrification, different sources, etc. Group work 2: program in Matlab typical use patterns for the different sectors - Week 3: Estimate the possibilities for 'demand response' in the different sectors and program this in Matlab. Deduce the resulting demand for short term and long term storage. Determine the demand for transport of energy and estimate possible transport losses. Group work 3: program in Matlab the different magnitudes for demand response, short and long storage throughout the year. - Week 4: Storage methods introduction, pumped hydro, mechanical, compressed air, deployment possibilities. Group work 4: calculate different factors for energy density, capacity, timescale of storage. Investigate approximate efficiency. - Week 5: Thermal storage, high T and low T. Combined heat and power possibilities. Group work 5: Investigate energy densities, efficiencies, capacities, exergies. - Week 6: Biomass and biofuels, future prospects for storage. Introduction to scalable chemical energy storage methods: batteries and fuels. What would electrification of the chemical industry mean. Group work: investigate biofuel resources, calculate efficiencies and compare to solar PV and electrolysis numbers. - Week 7: Hand in report and model. Group presentations of the model calculations (where each group has different boundary conditions). Discuss how the boundary conditions influence the outcome of the storage demand.	
Study Goals	At the end of the lectures and group work the students should be able to: - Reproduce and reflect on the role of storage in a renewable energy based society. - Explain how different timescales for energy storage are relevant for the future system. - List and explain the different storage solutions in terms of typical energy density, cost and efficiency. - Tell about the different drivers for energy conversion including demand for storage, and demand for transport of energy. - Reflect on the energy efficiency of intermediate storage.	
Education Method	The course contains seven weeks with 4-hour contact time. The 4 hours will be divided in lectures, group work applying the lecture content and a question time. The goal is to come to a finished assignment after six weeks that can be presented in class the seventh week. The structure of the course is as follows: each week there is first a class in which a background is handed out, then there is a working group session where the supplied information is used to start modelling aspects of a future energy system. The students will be divided over groups with each group an individual model question. The lesson is concluded with a question session. This will continue 6 weeks and the 7th week the different models will be presented by the students.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	Slides Articles, reports and reviews.	
Assessment	The final grade consists of the following parts: Individual exam (70% of the final grade). Group work (30% of the final grade, based on a report produced by the group). The groupwork cannot be redone individually. For one part may be scored 5,0 as long as the other part compensates this. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Prof.dr. J.J.C. Geerlings	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	Written exam	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Wind & Economics: Wind Power Economics 2022	
Introduction 1	Students choose EE4545 or EE4536 in their profile courses.

AE4W02TU	Introduction to Wind Turbines: Physics and Technology	4
Responsible Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AE4W09 (Wind Turbine Design).	
Expected prior knowledge	A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.	
Course Contents	Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.	
Study Goals	By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to: - describe the components and configurations of wind turbines - describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine - describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design - explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc. - identify the drivers of dynamic behaviour and calculate their typical values - describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure - describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes	
Education Method	Self-study of online material, with interactive lectures.	
Literature and Study Materials	Handouts, videos and additional course material on Brightspace.	
Assessment	Written exam at the end of period 2, with a resit at the end of period 3.	
Special Information	Assessment may differ due to COVID-19 measures. Changes to the assessment method will be communicated via Brightspace.	
Remarks	This course is an elective course for students from various faculties (AE, EEMCS, CEG,...). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.	
Set-up	Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.	

AE4W09	Wind Turbine Design	5
Responsible Instructor	Dr. D.A. von Terzi	
Instructor	Dr. M.B. Zaayer	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.	
Parts	In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4. The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.	
Course Contents	Overview of the course and the group assignment System design and scaling rules Aerodynamic and structural rotor design and analysis Drive train and electrical system Wind turbine control Blade materials and assessment of fatigue damage Wind data and description The use of standards for load calculations	
Study Goals	In this course you learn various aspects of designing a wind turbine, including: - Selecting approaches for design and analysis - Collect information and data - Use (simulation) programs, models and calculations - Analyse and interpret results and draw conclusions about the design - Make design decisions Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.	
Education Method	lectures + software tutorials + group assignment + discussion meetings	
Computer Use	Spreadsheet-type calculations are used to support design decisions. The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.	
Literature and Study Materials	Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.	
Assessment	A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.	
Remarks	This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master. Supplementary courses are (these treat similar subjects as this course, but more in-depth): AE4W13 Site conditions for wind turbine design AE4135 Rotor / wake aerodynamics AE4W21-14 Wind turbine aeroelasticity ET4117 Electrical machines and drives OE44135 Offshore wind support structures	

AE4W13	Site Conditions for Wind Turbine Design	3
Responsible Instructor	Dr.ir. W.A.A.M. Bierbooms	
Instructor	Dr.ir. D.J.N. Allaerts	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	It is assumed that the student has a basic understanding of fluid mechanics and stochastics (on Brightspace self tests are available). A basic skill of Matlab is assumed: you should be able to read from and write to a text file, write and/or modify simple functions (m-files). Furthermore, it is recommended that you already have a basic knowledge of wind turbines; e.g. AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Course Contents	Introduction Atmospheric Boundary Layer Meteorology Fluid Dynamics Turbulence Wind power meteorology Measurement instruments Statistics Methods in turbulence Offshore wind climate External conditions in standards Wave and current climate and loading Example analysis site data Turbulence spectra Satellite remote sensing techniques Simulation of turbulent wind fields and gusts	
Study Goals	After successfully completing the course, students will be able to: Discuss the relevance of particular site data for wind turbine design, operation and siting. Analyse simple problems related to wind power meteorology Interpret scientific dissemination by experts on main points Apply analysis methods to convert basic site data to wind turbine design parameters	
Education Method	Lecture + assignments + compulsory practical	
Course Relations	Follow up courses AE4-W09, AE4135	
Literature and Study Materials	www.eos.ubc.ca/books/Practical_Meteorology Chapters 12 to 14; Pijush K. Kundu, Ira M. Cohen, Fluid Mechanics, Elsevier. Lecture notes and reader	
Prerequisites	AE4W02TU, Introduction to Wind Turbines: Physics and Technology.	
Assessment	The written exam consists out of two parts, one for each period (scheduled in the exam period). The final grade is a weighted average of these 2 exams and the grade for one of the assignments (which one will be specified during the course). In order to take part in Exam2 one should join the compulsory practical The resit is an oral exam (to be scheduled with the teacher) on basis of pass/fail (i.e. in case of a pass the grade 6 will be put in Osiris).	
Set-up	The course combines lectures and student assignments/practical. For several topics guest speakers will be invited.	

EE4536	DC and AC Microgrids	4
Responsible Instructor	L.M. Ramirez Elizondo	
Responsible Instructor	Dr.ir. P. Bauer	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc Electrical Engineering or equivalent	
Course Contents	This course focuses on the modelling and control of AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Several alternatives for the provision of ancillary services by means of AC and DC microgrids will be presented. Furthermore, attention will be given to the design of energy management systems for AC and DC microgrids including concepts as demand side management.	
Study Goals	Identify the challenges introduced by distributed generation Describe different microgrids' layouts and sizing Describe the operation and control of power electronic converters in AC and DC microgrids Apply droop control in AC and DC microgrids Describe the role of energy management systems in AC and DC microgrids Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids	
Education Method	Lectures	
Literature and Study Materials	Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatzigiargyriou	
Assessment	Written exam 80%, Assignments 20%	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects) 2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTs) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	

SET3065	Intelligent Electrical Power Grids	4
Responsible Instructor	Prof.dr. P. Palensky	
Instructor	Dr.ir. J.L. Rueda Torres	
Instructor	Dr. A.I. Stefanov	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets. Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools. You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done. This course is part of the Power track of the SET master and can also be chosen by EE master students. Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.	
Study Goals	After this course students are able to - create a demand/supply matching system, to - model and simulate a smart grid, to - specify the design requirements of a SCADA system, and to - analyse cyber security threats, threat vectors, and vulnerabilities of ITOT smart grid systems.	
Education Method	The course is organized in the following units, all units have assignments (typically a simulation): 1 Modeling Smart Grids / Automation Networks (Peter) 2 Power Flow Analysis (Jose) 3 Power System Dynamics (Jose) 4 Lab Session: Modeling of Generator Controllers (TAs) 5 Lab Session: Modeling of Wind Farms in Modelica (TAs) 6 AI and ML in Energy (Pedro/Jochen) 7 SCADA/WAMS (Peter) 8 Cyber Security (Alex)	
Literature and Study Materials	On Brightspace	
Assessment	The grade is formed based on the following elements: 50% Assignment reports (in time!) 50% multiple choice Moebius Computer Exam In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3095	Electronic Power Conversion	4
Responsible Instructor	A. Shekhar	
Instructor	H.J.M. Olsthoorn	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.	
Study Goals	To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.	
Education Method	Lectures Problem solving sessions Practicum Lectures	
Assessment	Written exam (closed book) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

TPM403SET	Technology Entrepreneurship and Sustainability	4
Module Manager	H. Khodaei	
Instructor	Dr.ing. V.E. Scholten	
Instructor	H. Khodaei	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	No specific knowledge is required prior to entering the course.	
Course Contents	This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course. This course is the third course in the SET Economics & Society profile.	
Study Goals	Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to: 1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing; 2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems; 3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products; 4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones; 5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular; 6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.	
Literature and Study Materials	Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted on Brightspace-site. Slides presented in class: Will be posted on Brightspace-site.	
Assessment	The assessment is composed of several elements: 1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade. 2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 20% of the final grade. 3. Individual: Open questions (30%) and theory quizzes (20%) All marks will be expressed on a scale of 1-10.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	

WM0637SET	Economic Policy for Sustainable Energy	4
Module Manager	Dr. E. Schröder	
Responsible for assignments	Dr. E. Schröder	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Prior knowledge of algebra and calculus is a must. The typical master student in a STEM subject has mathematical competencies that exceed those required for this course.	
Course Contents	Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis? This is an introductory course in environmental and ecological economics designed for master students with little or no background in economics. It introduces students to key concepts and theories with respect to: - consumption and production - efficiency and welfare - inter-temporal resource allocation and sustainability - happiness and economic growth - the choice of environmental policy targets and instruments - integrated assessment models	
Study Goals	After completing the course, students should be able to: - explain the main theoretical concepts used in environmental economics and how they relate to one another - explain how a market economy undervalues the environment - explain how environmental economists value the environment - apply economic theory to environmental problems such as climate change - evaluate economic policies for solving environmental problems such as climate change	
Education Method	Classes, quizzes, and exercises.	
Literature and Study Materials	The main text is: - Harris & Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor & Francis. The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.	
Assessment	The final grade is based exclusively on the performance in the final exam. In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.	

WM0931SET		Sustainable Energy Innovation and Transitions	4
Module Manager	Dr. L.M. Kamp		
Responsible for assignments	Dr. L.M. Kamp		
Co-responsible for assignments	Dr.ir. J.N. Quist		
Contact Hours / Week x/x/x/x	0/3/0/0		
Education Period	2		
Start Education	2		
Exam Period	none		
Course Language	English		
Expected prior knowledge	none		
Course Contents	This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition. This course is the first course in the SET Economics & Society profile.		
Study Goals	After this course, the student: 1. has knowledge of: * the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective. * the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability. 2. has insight in: * the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts. 3. has experience and basic skills in: * exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition.		
Education Method	This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.		
Literature and Study Materials	Scientific articles which are made available via the TU library or via Brightspace.		
Assessment	Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.		

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Solar & Economics: Solar Power Economics 2022	
Introduction 1	Students choose EE4545 or EE4536 in their profile courses.

EE4536	DC and AC Microgrids	4
Responsible Instructor	L.M. Ramirez Elizondo	
Responsible Instructor	Dr.ir. P. Bauer	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc Electrical Engineering or equivalent	
Course Contents	This course focuses on the modelling and control of AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Several alternatives for the provision of ancillary services by means of AC and DC microgrids will be presented. Furthermore, attention will be given to the design of energy management systems for AC and DC microgrids including concepts as demand side management.	
Study Goals	Identify the challenges introduced by distributed generation Describe different microgrids' layouts and sizing Describe the operation and control of power electronic converters in AC and DC microgrids Apply droop control in AC and DC microgrids Describe the role of energy management systems in AC and DC microgrids Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids	
Education Method	Lectures	
Literature and Study Materials	Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatziaargyriou	
Assessment	Written exam 80%, Assignments 20%	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects) 2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTs) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	

ET4376	Photovoltaic Basics	4
Responsible Instructor	Prof.dr. M. Zeman	
Course Coordinator	Dr.ir. R. Santbergen	
Contact Hours / Week x/x/x/x	0/4/0/0 lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised.	
Study Goals	Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.	
Education Method	Lectures, short multiple-question tests, exercises, instructions and self-study using the book Solar Energy.	
Literature and Study Materials	Book Solar Energy: The physics and engineering of photovolataic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.	
Assessment	Short multiple-question tests, instructions, written exam. All marks will be expressed on a scale of 0 to 10.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

ET4377	Photovoltaic Technologies	4
Responsible Instructor	Dr. L. Mazzearella	
Responsible Instructor	Prof.dr. I.M.F. Gordon	
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/0/2/0 lectures; 0/0/2/0 instruction	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Lecture course Photovoltaic Basics (ET4 376)	
Course Contents	This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed. Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.	
Study Goals	This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required. The learning objectives (LO) are as follows: LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation. LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design. LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.	
Education Method	Flipped-classroom video lectures and on-campus instructions.	
Literature and Study Materials	Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England. ISBN: 978 1 906860 32 5 (paperback); This book is also available as a free ebook: ISBN: 978 1 906860 73 8 (ePub) ISBN: 978 1 906860 75 2 (pdf)	
Assessment	The course will be assessed by a combination of a written final examination (that is taken on campus if COVID-19 restrictions allow, or by an online test that is taken via Brightspace, followed by possible verification) and three graded assignments that are not mandatory. The final mark for this course, F, will be based on the average results for the assignments, A, and the final examination, E. All marks will be expressed on a scale of 0 to 10. For calculating A, the average of the three assignment results is used. The assignment results only will be valid within the academic year in which the assignments have been made. Assignment results obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4378	Photovoltaic Systems	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Instructor	H. Ziar	
Instructor	P. Manganiello	
Contact Hours / Week x/x/x/x	0/0/0/2 lectures and 0/0/0/2 lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376) - Good command of Matlab scripting language (or equivalent)	
Course Contents	This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.	
Study Goals	After this course, the student will be able to - Determine optimal PV panels orientation from geographical survey - Compare between different PV systems topologies - Calculate the energy yield of a PV system - Identify different types of solar modules and other components used in PV systems - Justify the importance of bypass diodes and power conversion in PV systems - Analyze different power tracking methods - Design PV systems for different applications and estimate the cost - Defend the goodness of his/her own PV System design - Describe trends of physical quantities related to customized PV systems	
Education Method	Lectures, self study and teaming up for carrying out the design of an actual PV System.	
Literature and Study Materials	- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available) - Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library) - Handouts - Scientific articles - PV Systems course in "Solar Engineering MicroMasters" (https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/) - PV Resources (http://www.pvresources.com/) - PVGIS (http://re.jrc.ec.europa.eu/pvgis/) - Sandia National Laboratories PVPerformance (https://pvpmc.sandia.gov/) - Dutch PV Portal 3.0 (https://www.tudelft.nl/dutchpvportal)	
Assessment	Final project report (25% of the final mark) and written test (75% of the final mark). Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual. To access to the written test, the grade of the final project report must be equal to or greater than 5.75. To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3065	Intelligent Electrical Power Grids	4
Responsible Instructor	Prof.dr. P. Palensky	
Instructor	Dr.ir. J.L. Rueda Torres	
Instructor	Dr. A.I. Stefanov	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets. Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools. You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done. This course is part of the Power track of the SET master and can also be chosen by EE master students. Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.	
Study Goals	After this course students are able to - create a demand/supply matching system, to - model and simulate a smart grid, to - specify the design requirements of a SCADA system, and to - analyse cyber security threats, threat vectors, and vulnerabilities of ITOT smart grid systems.	
Education Method	The course is organized in the following units, all units have assignments (typically a simulation): 1 Modeling Smart Grids / Automation Networks (Peter) 2 Power Flow Analysis (Jose) 3 Power System Dynamics (Jose) 4 Lab Session: Modeling of Generator Controllers (TAs) 5 Lab Session: Modeling of Wind Farms in Modelica (TAs) 6 AI and ML in Energy (Pedro/Jochen) 7 SCADA/WAMS (Peter) 8 Cyber Security (Alex)	
Literature and Study Materials	On Brightspace	
Assessment	The grade is formed based on the following elements: 50% Assignment reports (in time!) 50% multiple choice Moebius Computer Exam In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3095	Electronic Power Conversion	4
Responsible Instructor	A. Shekhar	
Instructor	H.J.M. Olsthoorn	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.	
Study Goals	To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.	
Education Method	Lectures Problem solving sessions Practicum Lectures	
Assessment	Written exam (closed book) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

TPM403SET	Technology Entrepreneurship and Sustainability	4
Module Manager	H. Khodaei	
Instructor	Dr.ing. V.E. Scholten	
Instructor	H. Khodaei	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	No specific knowledge is required prior to entering the course.	
Course Contents	This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course. This course is the third course in the SET Economics & Society profile.	
Study Goals	Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to: 1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing; 2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems; 3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products; 4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones; 5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular; 6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.	
Literature and Study Materials	Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted on Brightspace-site. Slides presented in class: Will be posted on Brightspace-site.	
Assessment	The assessment is composed of several elements: 1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade. 2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 20% of the final grade. 3. Individual: Open questions (30%) and theory quizzes (20%) All marks will be expressed on a scale of 1-10.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	

WM0637SET	Economic Policy for Sustainable Energy	4
Module Manager	Dr. E. Schröder	
Responsible for assignments	Dr. E. Schröder	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Prior knowledge of algebra and calculus is a must. The typical master student in a STEM subject has mathematical competencies that exceed those required for this course.	
Course Contents	Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis? This is an introductory course in environmental and ecological economics designed for master students with little or no background in economics. It introduces students to key concepts and theories with respect to: - consumption and production - efficiency and welfare - inter-temporal resource allocation and sustainability - happiness and economic growth - the choice of environmental policy targets and instruments - integrated assessment models	
Study Goals	After completing the course, students should be able to: - explain the main theoretical concepts used in environmental economics and how they relate to one another - explain how a market economy undervalues the environment - explain how environmental economists value the environment - apply economic theory to environmental problems such as climate change - evaluate economic policies for solving environmental problems such as climate change	
Education Method	Classes, quizzes, and exercises.	
Literature and Study Materials	The main text is: - Harris & Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor & Francis. The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.	
Assessment	The final grade is based exclusively on the performance in the final exam. In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.	

WM0931SET	Sustainable Energy Innovation and Transitions	4
Module Manager	Dr. L.M. Kamp	
Responsible for assignments	Dr. L.M. Kamp	
Co-responsible for assignments	Dr.ir. J.N. Quist	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	none	
Course Contents	This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition. This course is the first course in the SET Economics & Society profile.	
Study Goals	After this course, the student: 1. has knowledge of: * the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective. * the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability. 2. has insight in: * the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts. 3. has experience and basic skills in: * exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition.	
Education Method	This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.	
Literature and Study Materials	Scientific articles which are made available via the TU library or via Brightspace.	
Assessment	Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Biomass & Economics: Biomass Storage - Economics 2022
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CH3061	Multiphase Reactor Engineering	4
Responsible Instructor	Dr.ir. J.R. van Ommen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student is expected to: 1. have followed a course on mass and energy balances (such as Molecular Transport Phenomena) 2. have followed a basic course in reactor engineering (such as Reactors and Kinetics) 3. is able to work with Python (or similar software; e.g., has followed a course on numerical techniques)	
Course Contents	Multiphase reactor types: Fixed beds, trickle beds, fluidized beds, bubble columns, slurry reactors, microreactors, structured (monolith) reactors, electrocatalytic reactors. Engineering aspects: Flow regimes, mass transfer, conversion models. Hatta number, reaction enhancement. Residence time distribution. Catalytic reactors structured in time and space. Unifying concepts. Design aspects: reactor selection, decoupling of kinetics and transport phenomena, combination of reaction and separation, reaction coupling, separating catalytic steps, controlled energy input, circularity of materials.	
Study Goals	1. The student is able to describe for each of the archetypes of multiphase reactors, the various embodiments with practical relevance. 2. The student is able to use the scientific literature to find relevant information for the various reactor types. 3. The student is able to analyse a multiphase reactor and to describe the physical and chemical process steps in words. 4. The student is able to explain the basic measurement techniques used in studying multiphase reactors. 5. The student is able to write down the mass, energy, and momentum balances for the overall system and the steps identified under objective 3. 6. The student is able to implement customary models in the balances mentioned under 5, such that a system of equations describing the reactor is obtained. 7. The student is able to simplify the system of equations based on a quantitative analysis and identification the rate-determining steps. 8. The student is able to use current software packages to solve the systems of equations numerically for the basic types of reactors. 9. The student is able to use the developed models to analyse defined cases. 10. The student is able to make a quantitative design of a multiphase reactor for a given application. 11. The student is able to present, both orally and in writing, the outcomes of modeling, calculations, and evaluation.	
Education Method	Each lecture will give a short overview of a specific multiphase reactor type (only 45 minutes). The slides available on Brightspace. You will get references to papers and/or chapters from textbooks that give more information about the reactor type. These will be made available via Brightspace. After the lecture, an assignment (about 8 hours work) will be available via Brightspace for the specific reactor type treated. These assignments are made in teams of four students; this number may be changed depending on the number of participants. You are free to choose which software package you want to use for making the assignments, but Python is recommended. After two weeks, each team hands in a detailed solution of the problem on paper the day before the discussion before the announced deadline. Moreover, each team prepares a summary of the solution for presentation in PowerPoint. Please bring this presentation on a memory stick. Each week, a few teams present their solution; the instructors choose the student that is presenting. The other teams challenge this solution. At the end, each team will make a larger final project.	
Computer Use	You are free to choose which software package you want to use for making the assignments, but Python is recommended.	
Literature and Study Materials	All material is provided via Brightspace.	
Assessment	The final grade consists of three parts: * the solution to the minor assignments * an individual part; the nature of this part will depend on the number of participants * the final assignment Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

CH3222	Energy Storage in Batteries	4
Responsible Instructor	Dr.ir. M. Wagemaker	
Instructor	Dr. E.M. Kelder	
Instructor	Prof.dr. F.M. Mulder	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Efficient energy storage in batteries is of major importance for electronic equipment, electrification of cars and buses, grid stability and for the introduction of fluctuating renewable energy sources including wind and solar. In this course we will learn about the thermodynamics and charge kinetics in batteries including the electrochemical driving force, thermochemistry, phase transitions and charge transport kinetics. The principles of state of the art and future rechargeable batteries, also touching super capacitors, will be discussed. The aim is to develop understanding of the battery performance parameters and to relate these with the with underlying atomic scale material properties. Along these topics current battery developments and research directions will be discussed. After seven lectures on the fundamental properties of batteries six dedicated working lectures will focus on important battery technologies including: Redox Flow Na-ion next generation Li-ion batteries Solid State batteries Supercapacitors	
Study Goals	At the end of the battery lectures students should be able to: - List the major types of Li and other batteries and their characteristics. - Reproduce the major challenges in batteries. - Explain how a battery works and the functioning of each component. - List and explain the different reaction types in batteries. - Explain the different charge transfer processes in batteries. - List and explain the different mechanisms that limit the cycle life of batteries. - Apply Nernst Law describing redox reactions. - Calculate conductivities and overpotentials. - Apply the Tafel equation describing the charge transfer between electrode and electrolyte. - Calculate energy densities and power densities from observed properties. - Derive from a phase diagram the different phases that will occur upon lithiation. - Apply the Gibbs Phase rule to predict constant of variable output voltage. - Derive the voltage curve from the shape of the Gibbs Free energy. - Predict the basic redox reaction of two electrodes given the formation energy or the standard potentials. - Compare different battery systems and differentiate. - Discuss scientific results observed in battery performance and to relate this to material properties.	
Education Method	The course contains fourteen 2-hour lectures. The first seven 2-hour lectures focusses on building the underlying thermodynamics, material science and electrochemistry chemistry and the principles of energy storage in batteries. This knowledge will be used to develop understanding of the material properties that determine performance of current and next generation battery. In the second seven lectures will be working lectures with focus on six important battery technologies. The first half hour will be a lecture on the specific battery technology followed by half an hour assignment. In the final half hour of the working lecture the assignment will be discussed. In the final lecture example exams will be discussed.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	R.A. Huggins: Advanced Batteries. Available via the library as pdf's: http://www.springerlink.com/content/nq6404/?MUD=MP Slides Articles	
Assessment	Written exam (100% of the final grade)	
	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

SET3070	Thermochemistry of Biomass Conversion	4
Responsible Instructor	Prof.dr.ir. W. de Jong	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	To start this course you need to have a good knowledge of BSc. level thermodynamics, chemistry, kinetics, basic reactor modelling (batch, plug flow, continuously well stirred reactor), basic process engineering (mass, energy, momentum balances), and physical transport phenomena. In case you think you miss some of this pre-existing knowledge, please perform selfstudy of the first six chapters of the book 'Biomass as a Sustainable Energy Source for the Future', Wiley (2015), eds. De Jong and Van Ommen	
Summary	Biomass is available in abundance on a global level; it is naturally stored solar energy in a wide diversity of chemical structures. Among the renewable energy sources, biomass can be stored most easily, of course depending once again on the resources characteristics. Due to the heterogeneous nature of biomass also many different conversion technologies have been developed in the context of energy conversion chains with combustion being the oldest performed by mankind. This course deals with the background and design and implementation of different technologies to convert biomass thermochemically.	
Course Contents	In this course you will learn about the following aspects of biomass conversion based on thermo-chemical conversion technologies. - Biomass Characterisation (biomass types, properties, analyses) - Pretreatment (harvesting, size reduction, drying, torrefaction) - Thermochemical conversion (combustion, gasification and pyrolysis; reactor types) - Transportation fuels / biorefinery (syngas conversion, economics) - Process design (structured design approach)	
Study Goals	During this course, the students will obtain knowledge of and insight into energy conversion technologies based on biomass thermo-chemical conversion processes for a wide variety of applications, like supply of power, heat and fuels. They will be offered practical problems to solve and projects to perform in this area. These main learning objectives can be divided into several smaller learning objectives: 1) The student has knowledge on the typical biomass composition, related properties (like e.g. heating value, ash melting behaviour) and can critically identify and select fuel characterization techniques. 2) The student is able to identify and perform basic capacity calculations concerning thermo-chemical biomass conversion systems for small and larger scale (combined) heat and power generation as well as systems for transportation fuel production. 3) The student is able to set up reaction equations (stoichiometry) for a wide range of energy supply related fuel conversion processes. 4) The student can set up and perform heat and mass balance calculations regarding single phase and simple multi-phase biomass thermo-chemical conversion reactors and extended (downstream) systems for power, heat and fuels generation. 5) The student knows how to perform basic ideal gas phase chemical reaction equilibrium calculations to calculate product compositions and extents of reactions given initial reaction mixture composition, pressure and temperature. 6) The student can understand and describe reaction kinetic expressions and also apply them in order to solve (design) problems related to the conversion of fuels and the formation of emission components. 7) The student demonstrates good knowledge of mass and heat transfer mechanisms and phenomena applied to the chemical conversion of both liquid and solid biomass (derived) fuels. 8) The student can derive equations for idealised model reactors (well-stirred reactor and plug flow reactor concepts) and apply these simplified reactor concepts to solve engineering problems related to the thermo-chemical conversion of biomass fuels, in particular combustion, gasification and pyrolysis. 9) The student is able to perform calculations concerning the conversion of small particles; important in this respect is the identification of the regime of conversion. Related dimensionless numbers such as the Thiele Modulus and the Biot number are key indicators in this respect. 10) The student is able to understand the basics of thermo-chemical conversion processes and to translate this into designing and evaluating such (sometimes hybrid) systems, including their economic evaluation based on characteristic parameters like NPV, POT, and ROI. 11) The student demonstrates good communication skills, both oral and written, concerning presentation of (applied) scientific results.	
Education Method	This course consists of six 2-hour lectures, also using blended learning methods. Each lecture is followed by a 2-hour work session or excursion(s)/lab visit(s) in the same week -depending on the actual COVID situation- focused on working out minor assignments including literature surveys, problem solving and work on the major project. In the week of the exams you will present the major assignment project results.	
Course Relations	This course is a core course in the Sustainable Energy Biomass Energy profile.	
Literature and Study Materials	Book: "Biomass, a sustainable energy source for the future", de Jong and van Ommen, Wiley., 1st edition, Wiley (2015). Lecture slides	
Assessment	The final grade consists of three parts: * the solution to the (three) assignments (team work); 40% of the grade * a larger final assignment (team work); 50% of the grade * evaluation of the final assignment's presentation; 10% of the grade The assessment method may be adapted to actual student population. You can get a chance to resit the larger final assignment part of the course in case of having obtained an insufficient grade for this part by means of a repair additional work for the same assignment to obtain a sufficient grade. If you pass only part of the course, all parts need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3080	The Necessity of Storage Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic knowledge on chemistry and physics, as well as knowledge on Matlab. Level as of MSc SET students.	
Course Contents	In a renewable energy future energy storage is an additional necessary means to match the supply and demand of energy. - Week 1: Future energy generation systems, the combination of sources on a large scale. Group work 1: program in Matlab a numerical model for energy generation, with flexible input of PV, solar heat, wind, hydro, biofuel - Week 2: Future energy use possibilities: look into current typical use patterns of the different sectors, and look into possible changes: electrification, different sources, etc. Group work 2: program in Matlab typical use patterns for the different sectors - Week 3: Estimate the possibilities for 'demand response' in the different sectors and program this in Matlab. Deduce the resulting demand for short term and long term storage. Determine the demand for transport of energy and estimate possible transport losses. Group work 3: program in Matlab the different magnitudes for demand response, short and long storage throughout the year. - Week 4: Storage methods introduction, pumped hydro, mechanical, compressed air, deployment possibilities. Group work 4: calculate different factors for energy density, capacity, timescale of storage. Investigate approximate efficiency. - Week 5: Thermal storage, high T and low T. Combined heat and power possibilities. Group work 5: Investigate energy densities, efficiencies, capacities, exergies. - Week 6: Biomass and biofuels, future prospects for storage. Introduction to scalable chemical energy storage methods: batteries and fuels. What would electrification of the chemical industry mean. Group work: investigate biofuel resources, calculate efficiencies and compare to solar PV and electrolysis numbers. - Week 7: Hand in report and model. Group presentations of the model calculations (where each group has different boundary conditions). Discuss how the boundary conditions influence the outcome of the storage demand.	
Study Goals	At the end of the lectures and group work the students should be able to: - Reproduce and reflect on the role of storage in a renewable energy based society. - Explain how different timescales for energy storage are relevant for the future system. - List and explain the different storage solutions in terms of typical energy density, cost and efficiency. - Tell about the different drivers for energy conversion including demand for storage, and demand for transport of energy. - Reflect on the energy efficiency of intermediate storage.	
Education Method	The course contains seven weeks with 4-hour contact time. The 4 hours will be divided in lectures, group work applying the lecture content and a question time. The goal is to come to a finished assignment after six weeks that can be presented in class the seventh week. The structure of the course is as follows: each week there is first a class in which a background is handed out, then there is a working group session where the supplied information is used to start modelling aspects of a future energy system. The students will be divided over groups with each group an individual model question. The lesson is concluded with a question session. This will continue 6 weeks and the 7th week the different models will be presented by the students.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	Slides Articles, reports and reviews.	
Assessment	The final grade consists of the following parts: Individual exam (70% of the final grade). Group work (30% of the final grade, based on a report produced by the group). The groupwork cannot be redone individually. For one part may be scored 5,0 as long as the other part compensates this. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Prof.dr. J.J.C. Geerlings	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	Written exam	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3311	Green Chemistry and Sustainable Technology	4
Responsible Instructor	Dr. F. Hollmann	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Organic Chemistry (LB1045 or 4051OCSTRY) Chemical Biology (LB2621) OR Catalysis (4052KATALY)	
Course Contents	Depleting fossil fuels and environmental concerns force us to change the way we produce our goods. The transition of the chemical industry to becoming sustainable represents one of the major challenges to be faced in the 21st century. Catalytic reactions will increasingly substitute the existing energy- and resource-intensive production procedures; hence reducing or preventing toxic pollutants. Biotechnology will play a major role in this transition and complement advanced chemical production technologies. This course will highlight the emerging role of catalysis in the chemical industry. A major focus will lie on biotechnological production of chemicals (fermentation processes, isolated enzymes). These methods will be related to their chemocatalytic counterparts. Thus, ample examples including the underlying mechanistic principles - of bio- and chemocatalytic processes will be discussed. Finally, the emerging field of renewable feedstocks will be discussed not only from a chemical standpoint, but also on socioeconomical and ecological grounds. The 12 Principles of Green Chemistry will serve as guideline to assess the scope and limitations of the examples discussed.	
Study Goals	After attending this course the student will: (1) have detailed knowledge of sustainable catalysts (enzyme- and chemo-catalysis) (2) know industrial examples of sustainable manufacturing practice (3) be able to use the 12 Principles of Green Chemistry to assess the sustainability of a given process (4) be able to critically assess the purport of green claims	
Education Method	Lecture Sheets, specialized books, articles and patents	
Assessment	(1) Written exam (digital off-campus (open book) or on-campus, depending on the COVID-19 measures), 75% of the overall grade (2a) only for SET students: written assignment about a topic of Green Chemistry (groups of 3-4 students), 25% of the overall grade (2b) for non-SET students: group presentation about a topic of Green Chemistry (groups of 4-5 students), 25% of the overall grade Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

TPM403SET	Technology Entrepreneurship and Sustainability	4
Module Manager	H. Khodaei	
Instructor	Dr.ing. V.E. Scholten	
Instructor	H. Khodaei	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	No specific knowledge is required prior to entering the course.	
Course Contents	This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course. This course is the third course in the SET Economics & Society profile.	
Study Goals	Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to: 1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing; 2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems; 3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products; 4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones; 5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular; 6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.	
Literature and Study Materials	Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted on Brightspace-site. Slides presented in class: Will be posted on Brightspace-site.	
Assessment	The assessment is composed of several elements: 1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade. 2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 20% of the final grade. 3. Individual: Open questions (30%) and theory quizzes (20%) All marks will be expressed on a scale of 1-10.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	

WM0637SET	Economic Policy for Sustainable Energy	4
Module Manager	Dr. E. Schröder	
Responsible for assignments	Dr. E. Schröder	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Prior knowledge of algebra and calculus is a must. The typical master student in a STEM subject has mathematical competencies that exceed those required for this course.	
Course Contents	Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis? This is an introductory course in environmental and ecological economics designed for master students with little or no background in economics. It introduces students to key concepts and theories with respect to: - consumption and production - efficiency and welfare - inter-temporal resource allocation and sustainability - happiness and economic growth - the choice of environmental policy targets and instruments - integrated assessment models	
Study Goals	After completing the course, students should be able to: - explain the main theoretical concepts used in environmental economics and how they relate to one another - explain how a market economy undervalues the environment - explain how environmental economists value the environment - apply economic theory to environmental problems such as climate change - evaluate economic policies for solving environmental problems such as climate change	
Education Method	Classes, quizzes, and exercises.	
Literature and Study Materials	The main text is: - Harris & Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor & Francis. The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.	
Assessment	The final grade is based exclusively on the performance in the final exam. In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.	

WM0931SET	Sustainable Energy Innovation and Transitions	4
Module Manager	Dr. L.M. Kamp	
Responsible for assignments	Dr. L.M. Kamp	
Co-responsible for assignments	Dr.ir. J.N. Quist	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	none	
Course Contents	This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition. This course is the first course in the SET Economics & Society profile.	
Study Goals	After this course, the student: 1. has knowledge of: * the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective. * the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability. 2. has insight in: * the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts. 3. has experience and basic skills in: * exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition.	
Education Method	This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.	
Literature and Study Materials	Scientific articles which are made available via the TU library or via Brightspace.	
Assessment	Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Electric Mobility Systems: Electric Mobility Power Storage 2022
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CH3222	Energy Storage in Batteries	4
Responsible Instructor	Dr.ir. M. Wagemaker	
Instructor	Dr. E.M. Kelder	
Instructor	Prof.dr. F.M. Mulder	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Efficient energy storage in batteries is of major importance for electronic equipment, electrification of cars and buses, grid stability and for the introduction of fluctuating renewable energy sources including wind and solar. In this course we will learn about the thermodynamics and charge kinetics in batteries including the electrochemical driving force, thermochemistry, phase transitions and charge transport kinetics. The principles of state of the art and future rechargeable batteries, also touching super capacitors, will be discussed. The aim is to develop understanding of the battery performance parameters and to relate these with the with underlying atomic scale material properties. Along these topics current battery developments and research directions will be discussed. After seven lectures on the fundamental properties of batteries six dedicated working lectures will focus on important battery technologies including: Redox Flow Na-ion next generation Li-ion batteries Solid State batteries Supercapacitors	
Study Goals	At the end of the battery lectures students should be able to: - List the major types of Li and other batteries and their characteristics. - Reproduce the major challenges in batteries. - Explain how a battery works and the functioning of each component. - List and explain the different reaction types in batteries. - Explain the different charge transfer processes in batteries. - List and explain the different mechanisms that limit the cycle life of batteries. - Apply Nernst Law describing redox reactions. - Calculate conductivities and overpotentials. - Apply the Tafel equation describing the charge transfer between electrode and electrolyte. - Calculate energy densities and power densities from observed properties. - Derive from a phase diagram the different phases that will occur upon lithiation. - Apply the Gibbs Phase rule to predict constant of variable output voltage. - Derive the voltage curve from the shape of the Gibbs Free energy. - Predict the basic redox reaction of two electrodes given the formation energy or the standard potentials. - Compare different battery systems and differentiate. - Discuss scientific results observed in battery performance and to relate this to material properties.	
Education Method	The course contains fourteen 2-hour lectures. The first seven 2-hour lectures focusses on building the underlying thermodynamics, material science and electrochemistry chemistry and the principles of energy storage in batteries. This knowledge will be used to develop understanding of the material properties that determine performance of current and next generation battery. In the second seven lectures will be working lectures with focus on six important battery technologies. The first half hour will be a lecture on the specific battery technology followed by half an hour assignment. In the final half hour of the working lecture the assignment will be discussed. In the final lecture example exams will be discussed.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	R.A. Huggins: Advanced Batteries. Available via the library as pdf's: http://www.springerlink.com/content/nq6404/?MUD=MP Slides Articles	
Assessment	Written exam (100% of the final grade)	
	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

EE4536	DC and AC Microgrids	4
Responsible Instructor	L.M. Ramirez Elizondo	
Responsible Instructor	Dr.ir. P. Bauer	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc Electrical Engineering or equivalent	
Course Contents	This course focuses on the modelling and control of AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Several alternatives for the provision of ancillary services by means of AC and DC microgrids will be presented. Furthermore, attention will be given to the design of energy management systems for AC and DC microgrids including concepts as demand side management.	
Study Goals	Identify the challenges introduced by distributed generation Describe different microgrids' layouts and sizing Describe the operation and control of power electronic converters in AC and DC microgrids Apply droop control in AC and DC microgrids Describe the role of energy management systems in AC and DC microgrids Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids	
Education Method	Lectures	
Literature and Study Materials	Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatzigiargyriou	
Assessment	Written exam 80%, Assignments 20%	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects) 2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTs) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	

ET4117	Electrical Machines and Drives	4
Responsible Instructor	G.R. Chandra Mouli	
Instructor	J. Dong	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Bachelor program, preferably electrical engineering	
Course Contents	The course gives an overview of different types of electrical machines and drives. Different types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5	
Assessment	Written exam (100% Weightage), Closed book, Scientific (non-graphic) calculator can be used, No formula sheets Disclaimer: Information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	In order to participate in the written exam, students need to complete the three half day lab practicals. During corona period, three remote practicals are offered that are done individually. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ET4291SET	Digital modelling of electric powertrain	4
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	Lab 0,0,0,4 and 0,0,0,1 instruction	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Digital modelling of the power train including the converter, motor and the control, including the following contents: - Dynamic modelling of DC and AC Machines supplied with power electronic converters. - Modelling of mechanical systems and thermal effects in drives. - Modelling and simulation of digital control of electrical drives. - Controller design for cascaded motor controller. - Scalar and vector control, field weakening control.	
Study Goals	Students should be able to: - Model the dynamic behaviour of DC and AC machines using state space representation. - Integrate ideal power electronic converter models and cascade control systems in the drive model. - Model simple mechanical systems (wheel, gears, pulley&belt) in the electrical drive. - Calculate the steady state performance of DC and AC drives. - Implement scalar control system and vector control system for induction machines. - Implement vector control system for synchronous machines. - Implement field weakening control for DC and AC drives. - Implement the dynamic simulation models in MATLAB/Simulink. - Explain the practical design of electrical drives. - Evaluate the performance of an electrical drive system.	
Education Method	Lecture / Self-study / Simulation practical / Discussion	
Books	- Rik De Doncker, Duco W.J. Pulle, Andre Veltman, Advanced Electrical Drives Analysis, Modeling, Control, Springer, available: https://www.springer.com/gp/book/9789400701816 - Self-study materials on Brightspace.	
Assessment	Individual assignment and oral exam	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

SET3065	Intelligent Electrical Power Grids	4
Responsible Instructor	Prof.dr. P. Palensky	
Instructor	Dr.ir. J.L. Rueda Torres	
Instructor	Dr. A.I. Stefanov	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets. Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools. You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done. This course is part of the Power track of the SET master and can also be chosen by EE master students. Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.	
Study Goals	After this course students are able to - create a demand/supply matching system, to - model and simulate a smart grid, to - specify the design requirements of a SCADA system, and to - analyse cyber security threats, threat vectors, and vulnerabilities of ITOT smart grid systems.	
Education Method	The course is organized in the following units, all units have assignments (typically a simulation): 1 Modeling Smart Grids / Automation Networks (Peter) 2 Power Flow Analysis (Jose) 3 Power System Dynamics (Jose) 4 Lab Session: Modeling of Generator Controllers (TAs) 5 Lab Session: Modeling of Wind Farms in Modelica (TAs) 6 AI and ML in Energy (Pedro/Jochen) 7 SCADA/WAMS (Peter) 8 Cyber Security (Alex)	
Literature and Study Materials	On Brightspace	
Assessment	The grade is formed based on the following elements: 50% Assignment reports (in time!) 50% multiple choice Moebius Computer Exam In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3080	The Necessity of Storage Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic knowledge on chemistry and physics, as well as knowledge on Matlab. Level as of MSc SET students.	
Course Contents	In a renewable energy future energy storage is an additional necessary means to match the supply and demand of energy. - Week 1: Future energy generation systems, the combination of sources on a large scale. Group work 1: program in Matlab a numerical model for energy generation, with flexible input of PV, solar heat, wind, hydro, biofuel - Week 2: Future energy use possibilities: look into current typical use patterns of the different sectors, and look into possible changes: electrification, different sources, etc. Group work 2: program in Matlab typical use patterns for the different sectors - Week 3: Estimate the possibilities for 'demand response' in the different sectors and program this in Matlab. Deduce the resulting demand for short term and long term storage. Determine the demand for transport of energy and estimate possible transport losses. Group work 3: program in Matlab the different magnitudes for demand response, short and long storage throughout the year. - Week 4: Storage methods introduction, pumped hydro, mechanical, compressed air, deployment possibilities. Group work 4: calculate different factors for energy density, capacity, timescale of storage. Investigate approximate efficiency. - Week 5: Thermal storage, high T and low T. Combined heat and power possibilities. Group work 5: Investigate energy densities, efficiencies, capacities, exergies. - Week 6: Biomass and biofuels, future prospects for storage. Introduction to scalable chemical energy storage methods: batteries and fuels. What would electrification of the chemical industry mean. Group work: investigate biofuel resources, calculate efficiencies and compare to solar PV and electrolysis numbers. - Week 7: Hand in report and model. Group presentations of the model calculations (where each group has different boundary conditions). Discuss how the boundary conditions influence the outcome of the storage demand.	
Study Goals	At the end of the lectures and group work the students should be able to: - Reproduce and reflect on the role of storage in a renewable energy based society. - Explain how different timescales for energy storage are relevant for the future system. - List and explain the different storage solutions in terms of typical energy density, cost and efficiency. - Tell about the different drivers for energy conversion including demand for storage, and demand for transport of energy. - Reflect on the energy efficiency of intermediate storage.	
Education Method	The course contains seven weeks with 4-hour contact time. The 4 hours will be divided in lectures, group work applying the lecture content and a question time. The goal is to come to a finished assignment after six weeks that can be presented in class the seventh week. The structure of the course is as follows: each week there is first a class in which a background is handed out, then there is a working group session where the supplied information is used to start modelling aspects of a future energy system. The students will be divided over groups with each group an individual model question. The lesson is concluded with a question session. This will continue 6 weeks and the 7th week the different models will be presented by the students.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	Slides Articles, reports and reviews.	
Assessment	The final grade consists of the following parts: Individual exam (70% of the final grade). Group work (30% of the final grade, based on a report produced by the group). The groupwork cannot be redone individually. For one part may be scored 5,0 as long as the other part compensates this. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Prof.dr. J.J.C. Geerlings	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	Written exam	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

SET3095	Electronic Power Conversion	4
Responsible Instructor	A. Shekhar	
Instructor	H.J.M. Olsthoorn	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.	
Study Goals	To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.	
Education Method	Lectures Problem solving sessions Practicum Lectures	
Assessment	Written exam (closed book)	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3100	Electric vehicle and charging technology	4
Responsible Instructor	G.R. Chandra Mouli	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	SET3095/ET4119 Electronic Power Conversion ET4117 Electric machines & drives	
Course Contents	This course focusses on the technology of (hybrid) electric vehicles and its charging technology Tank to Wheel, Well to Tank, Well to Wheel efficiencies and emission estimation for conventional, hybrid and fully electric vehicles based on energy source for propulsion. Types, operational principles, advantages and disadvantages of vehicle architectures: internal combustion engine; micro, mild and full hybrids (series, parallel, series-parallel), plug-in hybrids, full electrics, fuel cell vehicles; Vehicle load forces & dynamics, mechanical transmission components, power and energy consumption, drive cycles, battery sizing and electric vehicle range, MPGe, planetary gears as a power-split, regenerative braking. Principles of battery operation, battery types, the key performance parameters of batteries, from cell to battery pack, charge-discharge curves, battery and thermal management systems. Fundamentals and types of electric machine drives used in electric cars: its characteristics, efficiency, basics of motor control. Basics of power electronic converters used in electric cars and chargers, topologies and control. Charging methods of EVs: AC charging, DC charging, vehicle-to-grid technology, wireless charging, battery swap; charging time estimation, effects of fast charging on battery; Smart charging of EVs and vehicle-to-grid, communication protocols for smart charging. Modelling and investigating the performance of the electric vehicle drivetrain using quasi-static simulation in MATLAB/Simulink. Case study of EV: Toyota Prius HEV, Tesla Model S, Porsche Taycan, Audi e-tron, Hyundai Ioniq 5 The course will have a design project where students have to form teams to design their own EV by making choices for the vehicle drivetrain and corresponding charging technology. Emphasis will be on justifying and quantifying the technological design choices including drivetrain architecture, well-to-wheel efficiency and emissions, energy storage, power electronics, type of motor and charging method.	
Study Goals	The goal of this course is to educate students to design and evaluate various (hybrid) electric vehicles with respect to their performance, efficiency, sustainability, charging methodologies and potential for grid support. At the end of the course, the students should be able to: Evaluate the performance of hybrids (series, parallel, series-parallel), plug-in hybrid and fully electric drivetrain architectures based on power and energy consumption, efficiency, driving range, well-to-wheel emissions and charging time. Explain the design trade-offs in the energy sources (battery, fuel cell, super capacitors) and motors used in electric vehicles based on the key performance indicators like efficiency, energy and/or power density, lifetime, and control. Analyse various charging technologies for electric vehicles (conductive, inductive or battery swap) based on power electronics converters used, type of the charger (AC Vs DC), connector types, power levels, and modes. Investigate the performance of the (hybrid) electric vehicle drivetrain using quasi-static modelling and simulation in MATLAB/Simulink. Design a (hybrid) electric drivetrain by sizing and making the relevant choice of the electric machine, internal combustion engine, energy source (batteries, fuel cells, etc.), power electronic converters, transmissions components, and charging technology for a given vehicle specification.	
Education Method	Classroom lectures and problem solving workshops, Discussion and case study, Demonstrations using tools and animation in the classroom, Online exercises with self-evaluation, Design project	
Literature and Study Materials	- Lecture slides & handouts - Primary reading - Modern Electric, Hybrid Electric, and Fuel Cell Vehicles, Third Edition, 2018, Mehrdad Ehsani, Yimin Gao, Ali Emadi, CRC Press, ISBN 9781138330498 - Primary reading - Chapter 1 & 14 of Electric Powertrain: Energy Systems, Power Electronics and Drives for Hybrid, Electric and Fuel Cell Vehicles, John G. Hayes, G. Abas Goodarzi, ISBN: 978-1-119-06364-3, 2018 - Supplementary reading Electric and Hybrid Vehicles: Design Fundamentals, Second Edition, 2010, Iqbal Husain, CRC Press, ISBN 9781439811757 The enlisted books are available via the TU Delft library (free e-books), via the ETV desk and ETV bookstore https://etv.itdepartment.nl/ and also via https://www.studystore.nl/ and the publisher website	
Assessment	Written exam (70% weightage), Closed book, Scientific (non-graphic) calculator can be used Design assignment with Written report + oral presentation (30% weightage) Students must register in groups for the design assignment within the first two weeks of the course. If student does not pass either the Written exam or the Design assessment, he/she will get a chance to resit that part of the course. Disclaimer: Information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Follow up course: ET4291SET Digital modelling of electric powertrain/ ET4291 Control of Electrical Drives	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Electric Mobility & Economics: Electric Mobility Power Economics 2022

EE4536	DC and AC Microgrids	4
Responsible Instructor	L.M. Ramirez Elizondo	
Responsible Instructor	Dr.ir. P. Bauer	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc Electrical Engineering or equivalent	
Course Contents	This course focuses on the modelling and control of AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Several alternatives for the provision of ancillary services by means of AC and DC microgrids will be presented. Furthermore, attention will be given to the design of energy management systems for AC and DC microgrids including concepts as demand side management.	
Study Goals	Identify the challenges introduced by distributed generation Describe different microgrids' layouts and sizing Describe the operation and control of power electronic converters in AC and DC microgrids Apply droop control in AC and DC microgrids Describe the role of energy management systems in AC and DC microgrids Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids	
Education Method	Lectures	
Literature and Study Materials	Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatzigiargyriou	
Assessment	Written exam 80%, Assignments 20%	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

EE4545	Electrical Power Systems of the Future	4
Responsible Instructor	Dr.ir. J.L. Rueda Torres	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Pre-requisites (entry levels acquired in Q1 and Q2): Master SET / Master EE / Master EWEM SET3065 Intelligent Electrical Power Grids ET4107 Power System Analyses II Matlab and Simulink skills (self study tutorial is available) Optional EE4665 Uncertainty modelling and risk assessment in electrical power systems	
Course Contents	The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.	
Study Goals	Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system. Learning objectives(LOs): LO1: Explain the impact of high RES penetration on the resiliency of future power systems. LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration. LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply. LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.	
Education Method	Lectures (conceptual aspects) 2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site Estimated study load: 112 hours (4 ECTs) Lectures 20 hours Practical 24 hours Self study 68 hours	
Assessment	Practicum assignments and digital exam/re-exam The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.	
Remarks	Company visit in combination with lectures and practicum @ Mariëndaal Centre of Excellence, TenneT Arnhem Dont forget identity card or passport! 8:00 Departure from EWI, Travel by touring bus to Arnhem. Lectures on Operational challenges by RES for system operations. Lessons learnt from large disturbances. Visit TenneT Control Room (including lunch). Practical Exercises Operational challenges and large disturbances. Travel by touring bus to Arnhem 16:30 Arrival at EWI	
maximum aantal deelnemers	80	

ET4117	Electrical Machines and Drives	4
Responsible Instructor	G.R. Chandra Mouli	
Instructor	J. Dong	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Bachelor program, preferably electrical engineering	
Course Contents	The course gives an overview of different types of electrical machines and drives. Different types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5	
Assessment	Written exam (100% Weightage), Closed book, Scientific (non-graphic) calculator can be used, No formula sheets Disclaimer: Information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	In order to participate in the written exam, students need to complete the three half day lab practicals. During corona period, three remote practicals are offered that are done individually. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ET4291SET	Digital modelling of electric powertrain	4
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	Lab 0,0,0,4 and 0,0,0,1 instruction	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Digital modelling of the power train including the converter, motor and the control, including the following contents: - Dynamic modelling of DC and AC Machines supplied with power electronic converters. - Modelling of mechanical systems and thermal effects in drives. - Modelling and simulation of digital control of electrical drives. - Controller design for cascaded motor controller. - Scalar and vector control, field weakening control.	
Study Goals	Students should be able to: - Model the dynamic behaviour of DC and AC machines using state space representation. - Integrate ideal power electronic converter models and cascade control systems in the drive model. - Model simple mechanical systems (wheel, gears, pulley&belt) in the electrical drive. - Calculate the steady state performance of DC and AC drives. - Implement scalar control system and vector control system for induction machines. - Implement vector control system for synchronous machines. - Implement field weakening control for DC and AC drives. - Implement the dynamic simulation models in MATLAB/Simulink. - Explain the practical design of electrical drives. - Evaluate the performance of an electrical drive system.	
Education Method	Lecture / Self-study / Simulation practical / Discussion	
Books	- Rik De Doncker, Duco W.J. Pulle, Andre Veltman, Advanced Electrical Drives Analysis, Modeling, Control, Springer, available: https://www.springer.com/gp/book/9789400701816 - Self-study materials on Brightspace.	
Assessment	Individual assignment and oral exam	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

SET3065	Intelligent Electrical Power Grids	4
Responsible Instructor	Prof.dr. P. Palensky	
Instructor	Dr.ir. J.L. Rueda Torres	
Instructor	Dr. A.I. Stefanov	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets. Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools. You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done. This course is part of the Power track of the SET master and can also be chosen by EE master students. Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.	
Study Goals	After this course students are able to - create a demand/supply matching system, to - model and simulate a smart grid, to - specify the design requirements of a SCADA system, and to - analyse cyber security threats, threat vectors, and vulnerabilities of ITOT smart grid systems.	
Education Method	The course is organized in the following units, all units have assignments (typically a simulation): 1 Modeling Smart Grids / Automation Networks (Peter) 2 Power Flow Analysis (Jose) 3 Power System Dynamics (Jose) 4 Lab Session: Modeling of Generator Controllers (TAs) 5 Lab Session: Modeling of Wind Farms in Modelica (TAs) 6 AI and ML in Energy (Pedro/Jochen) 7 SCADA/WAMS (Peter) 8 Cyber Security (Alex)	
Literature and Study Materials	On Brightspace	
Assessment	The grade is formed based on the following elements: 50% Assignment reports (in time!) 50% multiple choice Moebius Computer Exam In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3095	Electronic Power Conversion	4
Responsible Instructor	A. Shekhar	
Instructor	H.J.M. Olsthoorn	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.	
Study Goals	To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.	
Education Method	Lectures Problem solving sessions Practicum Lectures	
Assessment	Written exam (closed book) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3100	Electric vehicle and charging technology	4
Responsible Instructor	G.R. Chandra Mouli	
Instructor	Dr.ir. P. Bauer	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	SET3095/ET4119 Electronic Power Conversion ET4117 Electric machines & drives	
Course Contents	This course focusses on the technology of (hybrid) electric vehicles and its charging technology Tank to Wheel, Well to Tank, Well to Wheel efficiencies and emission estimation for conventional, hybrid and fully electric vehicles based on energy source for propulsion. Types, operational principles, advantages and disadvantages of vehicle architectures: internal combustion engine; micro, mild and full hybrids (series, parallel, series-parallel), plug-in hybrids, full electrics, fuel cell vehicles; Vehicle load forces & dynamics, mechanical transmission components, power and energy consumption, drive cycles, battery sizing and electric vehicle range, MPGe, planetary gears as a power-split, regenerative braking. Principles of battery operation, battery types, the key performance parameters of batteries, from cell to battery pack, charge-discharge curves, battery and thermal management systems. Fundamentals and types of electric machine drives used in electric cars: its characteristics, efficiency, basics of motor control. Basics of power electronic converters used in electric cars and chargers, topologies and control. Charging methods of EVs: AC charging, DC charging, vehicle-to-grid technology, wireless charging, battery swap; charging time estimation, effects of fast charging on battery; Smart charging of EVs and vehicle-to-grid, communication protocols for smart charging. Modelling and investigating the performance of the electric vehicle drivetrain using quasi-static simulation in MATLAB/Simulink. Case study of EV: Toyota Prius HEV, Tesla Model S, Porsche Taycan, Audi e-tron, Hyundai Ioniq 5 The course will have a design project where students have to form teams to design their own EV by making choices for the vehicle drivetrain and corresponding charging technology. Emphasis will be on justifying and quantifying the technological design choices including drivetrain architecture, well-to-wheel efficiency and emissions, energy storage, power electronics, type of motor and charging method.	
Study Goals	The goal of this course is to educate students to design and evaluate various (hybrid) electric vehicles with respect to their performance, efficiency, sustainability, charging methodologies and potential for grid support. At the end of the course, the students should be able to: Evaluate the performance of hybrids (series, parallel, series-parallel), plug-in hybrid and fully electric drivetrain architectures based on power and energy consumption, efficiency, driving range, well-to-wheel emissions and charging time. Explain the design trade-offs in the energy sources (battery, fuel cell, super capacitors) and motors used in electric vehicles based on the key performance indicators like efficiency, energy and/or power density, lifetime, and control. Analyse various charging technologies for electric vehicles (conductive, inductive or battery swap) based on power electronics converters used, type of the charger (AC Vs DC), connector types, power levels, and modes. Investigate the performance of the (hybrid) electric vehicle drivetrain using quasi-static modelling and simulation in MATLAB/Simulink. Design a (hybrid) electric drivetrain by sizing and making the relevant choice of the electric machine, internal combustion engine, energy source (batteries, fuel cells, etc.), power electronic converters, transmissions components, and charging technology for a given vehicle specification.	
Education Method	Classroom lectures and problem solving workshops, Discussion and case study, Demonstrations using tools and animation in the classroom, Online exercises with self-evaluation, Design project	
Literature and Study Materials	- Lecture slides & handouts - Primary reading - Modern Electric, Hybrid Electric, and Fuel Cell Vehicles, Third Edition, 2018, Mehrdad Ehsani, Yimin Gao, Ali Emadi, CRC Press, ISBN 9781138330498 - Primary reading - Chapter 1 & 14 of Electric Powertrain: Energy Systems, Power Electronics and Drives for Hybrid, Electric and Fuel Cell Vehicles, John G. Hayes, G. Abas Goodarzi, ISBN: 978-1-119-06364-3, 2018 - Supplementary reading Electric and Hybrid Vehicles: Design Fundamentals, Second Edition, 2010, Iqbal Husain, CRC Press, ISBN 9781439811757 The enlisted books are available via the TU Delft library (free e-books), via the ETV desk and ETV bookstore https://etv.itdepartment.nl/ and also via https://www.studystore.nl/ and the publisher website	
Assessment	Written exam (70% weightage), Closed book, Scientific (non-graphic) calculator can be used Design assignment with Written report + oral presentation (30% weightage) Students must register in groups for the design assignment within the first two weeks of the course. If student does not pass either the Written exam or the Design assessment, he/she will get a chance to resit that part of the course. Disclaimer: Information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Follow up course: ET4291SET Digital modelling of electric powertrain/ ET4291 Control of Electrical Drives	

TPM403SET	Technology Entrepreneurship and Sustainability	4
Module Manager	H. Khodaei	
Instructor	Dr.ing. V.E. Scholten	
Instructor	H. Khodaei	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	No specific knowledge is required prior to entering the course.	
Course Contents	This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course. This course is the third course in the SET Economics & Society profile.	
Study Goals	Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to: 1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing; 2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems; 3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products; 4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones; 5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular; 6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.	
Literature and Study Materials	Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted on Brightspace-site. Slides presented in class: Will be posted on Brightspace-site.	
Assessment	The assessment is composed of several elements: 1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade. 2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 20% of the final grade. 3. Individual: Open questions (30%) and theory quizzes (20%) All marks will be expressed on a scale of 1-10.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	

WM0637SET	Economic Policy for Sustainable Energy	4
Module Manager	Dr. E. Schröder	
Responsible for assignments	Dr. E. Schröder	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Prior knowledge of algebra and calculus is a must. The typical master student in a STEM subject has mathematical competencies that exceed those required for this course.	
Course Contents	Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis? This is an introductory course in environmental and ecological economics designed for master students with little or no background in economics. It introduces students to key concepts and theories with respect to: - consumption and production - efficiency and welfare - inter-temporal resource allocation and sustainability - happiness and economic growth - the choice of environmental policy targets and instruments - integrated assessment models	
Study Goals	After completing the course, students should be able to: - explain the main theoretical concepts used in environmental economics and how they relate to one another - explain how a market economy undervalues the environment - explain how environmental economists value the environment - apply economic theory to environmental problems such as climate change - evaluate economic policies for solving environmental problems such as climate change	
Education Method	Classes, quizzes, and exercises.	
Literature and Study Materials	The main text is: - Harris & Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor & Francis. The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.	
Assessment	The final grade is based exclusively on the performance in the final exam. In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.	

WM0931SET	Sustainable Energy Innovation and Transitions	4
Module Manager	Dr. L.M. Kamp	
Responsible for assignments	Dr. L.M. Kamp	
Co-responsible for assignments	Dr.ir. J.N. Quist	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	none	
Course Contents	This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition. This course is the first course in the SET Economics & Society profile.	
Study Goals	After this course, the student: 1. has knowledge of: * the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective. * the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability. 2. has insight in: * the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts. 3. has experience and basic skills in: * exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition.	
Education Method	This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.	
Literature and Study Materials	Scientific articles which are made available via the TU library or via Brightspace.	
Assessment	Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Electives SET 2022

Program Structure 1	For more information about the List of Electives, see the Brightspace Organisation of the Master Sustainable Energy Technology. The elective space should be used for one of more of the following purposes: - To gain work experience (SET3822 Internship MSc SET, 15 credits), - To gain research experience (ET4399 Extra Project, max. 15 credits), - To broaden their knowledge of the SET field, - To deepen knowledge of the SET profiles (profile electives) - To gain entrepreneurship skills (max. 6 credits) - Do To further develop language skills (max. 3 credits) For students who wish to deepen their knowledge of their SET profile, each profile offers an advanced course. Please note that the thesis supervisor may make one of these courses mandatory.
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ET4399	Extra Project (max. 15EC)	15
Responsible Instructor	Dr.ir. N.P. van der Meijs	
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Course Coordinator	Dr.ir. A.J. van Genderen	
Course Coordinator	Dr. D. Cavallo	
Course Coordinator	S.M. Alavi	
Course Coordinator	Dr.ir. R.C. Hendriks	
Course Coordinator	M. Babaie	
Course Coordinator	J. Dong	
Course Coordinator	Dr. A. Lekic	
Course Coordinator	M.A. Kitsak	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	This Extra Project offers students the opportunity to enhance lab skills (such as a specific training), academic skills (such as writing a paper for publishing or participating in a conference with a poster presentation or paper) or for studying special topics (such as capita selecta with (guest) lecturers). Note: the Extra Project can only be taken as part of the free electives. The supervisor needs to be part of the scientific staff within the Faculty of Electrical Engineering, Mathematics and Computer Science, this means they need to be a Full Professor, Assistant Professor or Associate Professor. For the MSc SET programme they should be on the Green list.	
Study Goals	To be formulated by the student in the Extra Project proposal (see education method) and to be approved beforehand by the supervisor and the MSc coordinator.	
Education Method	Before the start of the Extra Project, a proposal of the project needs to be provided according to a specified format; the format of this Extra Project proposal can be downloaded from the Brightspace page of the Extra Project. The following information is required: name of the TUD supervisor, proposed number of credits, a motivation for the extra project, learning goals and a description of the planned activities and output. This project proposal needs to be approved by the TUD supervisor, and for EE and CE the MSc coordinator, and for SET the director of studies. When submitting the Individual Exam Programme to the Board of Examiners, the Extra Project proposal needs to be attached. After the assignment, a final report needs to be produced. The Board of Examiners may ask to see the report to check the study load and quality at the time of the final MSc exam. At most one Extra Project may be performed and it can only be taken in the free elective space of the MSc programme.	
Assessment	The Extra Project and the resulting report is evaluated and graded by the supervisor. The grade is Pass/Fail. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ME45201	Electrochemical Energy Storage 1: Fundamentals	3
Responsible Instructor	Dr. R. Kortlever	
Instructor	J.W. Haverkort	
Contact Hours / Week x/x/x/x	0/0/4/0 (2 lecture hours and 2 seminar hours)	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic thermodynamics. Basic chemistry and knowledge of kinetics is not required, but will be helpful.	
Course Contents	One of the main challenges that mankind faces in the 21st century is the energy transition from fossil fuels and feedstocks to renewable sources. Most renewable energy sources, such as solar and wind energy, are intermittent. Therefore, large scale energy storage is required to successfully increase the share of these renewable energy sources. One way of achieving this is by storing energy using electrochemical systems and devices. This course introduces the principals of electrochemical energy storage and conversion to students new to field. Examples of topics discussed during this course are the thermodynamics and kinetics of electrochemical systems, transport phenomena, the electrochemical double layer, electrocatalysis and an introduction to electrochemical methods and measurements. We will discuss these topics in the light of electrochemical applications such as batteries, electrolyzers and fuel cells.	
Study Goals	After successful completion of this course, the student is able to independently: 1. Apply electrochemical nomenclature to describe the electrochemical cell and electrochemical measurements; 2. Explain the concept of the electrochemical double layer and mass transport in the electrolyte; 3. Apply basic principles of electrochemical thermodynamics, such as the Nernst equation, overpotentials and (standard) equilibrium potentials; 4. Apply basic principles of electrochemical kinetics, such as the Tafel equation and the Butler-Volmer model; 5. Explain the principles that determine the electrocatalytic performance of materials; 6. Interpret scientific literature reports on electrochemical energy storage and conversion systems and devices.	
Education Method	Combination of two (online, dependent on the situation) lecture hours per week and self-study. Homework assignments will be given after every lecture and will be discussed in class.	
Assessment	In case there is an opportunity for examinations on campus, the method of assessment will be a written exam. If, due to measures resulting from COVID-19, on campus examination is not possible the prescribed assessment will be as follows: oral exam and presentation on a literature report.	
Department	3mE Department Process & Energy	

ME45203	Electrochemical Energy Storage 2: Analytical Modelling	4
Responsible Instructor	J.W. Haverkort	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Familiarity with transport phenomena, basic fluid dynamics, and elementary differential equations. A working knowledge of electrochemistry and in particular the Butler-Volmer equation is also assumed. Ideal preparation include the courses ME45201, CH3513, or CH3222. A previous course on fuel cells, batteries, or similar, or additional self-study may also be sufficient.	
Course Contents	Electrochemistry plays a pivotal role in the transition towards a sustainable energy future. Renewable electrical energy can be stored in batteries or used to synthesize chemicals or fuels in electrochemical reactors, like an electrolyzer producing hydrogen. This course will introduce to a student, the knowledge and skills required to understand, model and optimize such electrochemical devices. Content overview: Porous electrode theory, multiphase flow, and transport models will be used to derive analytical current-voltage relations of various electrochemical devices including fuel cells, batteries, redox flow batteries, and electrolyzers. Some of the topics that will be discussed in this course include: - The Nernst-Planck equations for ionic transport in binary electrolytes. - Porous electrode theory using the concept of an electrochemical Thiele modulus. - The flooded agglomerate and relative permeability models for gas diffusion electrodes. - The single particle and Newman's P2D battery models. - Tobias model for gas-liquid electrolyzers.	
Study Goals	After successful completion of this course, the student is able to independently: 1. explain and apply porous electrode theory. 2. derive 0D, 1D, and 2D mathematical models to describe the current-voltage relationship of various electrochemical cells including flow, multiple phases, and porous electrodes. 3. optimize an electrochemical cell design with respect to, for example, energy efficiency.	
Education Method	Two lecture hours + two exercise hours per week.	
Course Relations	This course is a natural sequel to ME45201, CH3513, and CH3222 for students interested in analytical modelling of electrochemical systems.	
Literature and Study Materials	Reader, lectures slides, exercises.	
Assessment	Written exam.	

SET3822	Internship SET	15
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Course Coordinator	M.J.W. Smits	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	NA	
Course Contents	Internship SET is a three-month (15 ECTS) project on a sustainable energy engineering topic performed at a company or institute, either in the Netherlands or abroad. It is not allowed to do an internship which is longer than three months. The final report needs to be written and handed in within three months after the start of the internship. General guidelines are that it should concern a design, model, system integration or methodology implementation, or an experimental study relevant for the company involved and within the Profile Cluster. Literature study can only be a minor part of the internship. When carrying out Internship SET (SET3822) at a company or institute, the following conditions apply: The student joins in with the company or institute culture in general, including sharing the company-established balance between on-site and online work. The student joins in with the day-to-day running of the company or institute similar to regular employees. This includes having regular consultation and interaction with colleagues and manager(s), and use of and access to the company-specific resources and tools applicable to the internship. Such resources and tools can be physical/on-site (e.g. measurement equipment, company machines and infrastructure) as well as virtual (online, software tools, databases, ...). NB: Internship SET (SET3822) can never be part of a Graduation Project (SET3901) and it is not allowed to carry out both your internship and your thesis at the same company. Detailed information on projects, procedures and application is available on the Brightspace course SET3822.	
Study Goals	Learn about sustainable energy projects in a professional environment, and apply sustainable energy related knowledge on a (new or persistent) problem relevant for the company or institute. Produce a project report that gives useful answers to the initial questions of the company and that gives the TU Delft supervisor sufficiently detailed insight in the project and its results. You will also be required to write an experience report. Guidelines for writing your reports can be found on Brightspace course SET3822. There are three main educational goals for the Internship SET (SET3822): I. Profession related aspects: Insight into the profession of an engineer in Sustainable Energy Technology. Insight into the duties and the responsibilities of an engineer in a company. II. Social-psychological aspects: Learning to move in another culture (both within and outside the company). Dealing with another language for a longer period. Gaining insight into the position of the company with respect to comparable companies. III. Professional knowledge and skills related aspects: Learning to apply the gained knowledge and skills in a situation different from the (own) university situation. Learning to acquire new knowledge and skills that are necessary to meet the requirements for the Internship SET (SET3822).	
Education Method	In order to ensure that students start the projects with a solid skill set, students can apply for Internship SET after the third quarter of the first year of the MSc SET programme. In order to start Internship SET the following requirements should be fulfilled: 1. The Individual Exam Programme (IEP) must have been submitted to and approved by the Board of Examiners. 2. The examination of courses of the first three quarters of the first year of the MSc SET programme must have been completed or the student must have obtained at least 45 ECTS. 3. The courses of the fourth quarter of the first year of the MSc SET programme must have been finished, including the exams, though the results of the exams do not have to be published yet. For MSc SET students detailed information on projects, procedures and application is available on the Brightspace course SET3822. The internship application form should be submitted to the Internship Office prior (!) to starting Internship SET. This application form contains a project description that the student writes with input from the company supervisor and a TU Delft supervisor. The TU Delft supervisor is an assistant, associate, or full professor who is on the Green List of the MSc SET programme. The profile leader in the Board of Examiners of MSc SET has to approve the internship and co-sign the application form. Before you can start the internship, you need to meet the entrance requirements, which are checked by the Internship Office.	
Literature and Study Materials	Enrol in the Brightspace course SET3822 for more information about the Internship SET. You can find the step-by-step guide explaining all procedures, forms, internship offers, experiences from other students, and additional information e.g. about grants and funding for internships abroad.	
Assessment	You need to write a scientific report during the internship and send it to your company supervisor and the TU Delft supervisor on the final day of the internship. You need to write your evaluation report and send it, together with the scientific report, to the internship office, on the final day of your internship. Your company supervisor needs to send the evaluation within two weeks after the end of the internship to the TU Delft supervisor. The TU Delft supervisor needs to send the TU evaluation and the company evaluation to the internship office, within two weeks after receiving the evaluation from the company supervisor. When the internship office has received the two reports and the two assessment forms, your grade will be processed within a week. The Internship Grading Scheme will be used as a guideline to determine the mark. Please, consult SET3822 on Brightspace for the detailed grading scheme.	

	When the nature of the project requires so, it is possible to censor parts of the report. Details like numbers and names may be left out of the report (or strikethrough), for example, as long as the report represents the achievements of the project and the TU Delft supervisor is able to assess the internship. The TU Delft supervisor is responsible for the final grading. The internship will be graded with a pass or a fail (not numerically). Disclaimer: information may change depending on the developments around the coronavirus.
Co-Instructor	Prof.dr. F.M. Mulder
Co-Instructor	Prof.dr.ir. A.H.M. Smets
Co-Instructor	Dr.ir. W.A.A.M. Bierbooms
Co-Instructor	Dr. L.M. Kamp
Co-Instructor	L. Botto

SET3995	Geothermal Energy and Applications	4
Responsible Instructor	Prof.dr. D.F. Bruhn	
Responsible Instructor	A. Daniilidis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Scope: Due to the growing interest in sustainable energy production, the Department of Geoscience and Engineering offers a course on geothermal energy. This introductory course presents the fundamentals of geothermal resource exploitation. Students will gain a general understanding of what a geothermal resource is and of the basic steps in their economic exploitation, including lifetime analysis, characterization and design of a geothermal reservoir, reservoir engineering, plant design and construction, basics of the operation and maintenance of geothermal systems, and geothermal field management, as well as some environmental aspects. Contents: - Introduction to geothermal energy - Geology and geomechanics - Reservoir characterization - Design of a geothermal system - Geothermal reservoir engineering - Numerical modeling of geothermal doublets - Techno-economic assessment of geothermal systems	
Study Goals	After successfully completing this course you will be able to: - Classify geothermal systems based on their different geological settings - Describe and apply geomechanical principles related to geothermal systems - Describe solute and heat transfer in porous media by mathematical equations and explain their role in different geothermal systems (Darcy flow, conductive and convective heat transfer, diffusion and advection equations) - Develop a 2D and 3D Finite Element (FE) model in COMSOL describing a direct use geothermal system by implementing real world data - Design a prototype geothermal system: -Choose the appropriate location and well configuration -Assess the system lifetime, power, generated energy and stochastic economic output -Justify your geothermal development choices from an energy, engineering and economic point of view"	
Education Method	- Lectures - Computer practicals - excursion (half day) - own study - group work	
Assessment	- Homework (formative) - Written exam (summative, 20% of final grade) - Group project presentation and report (summative, 80% of final grade)	
Permitted Materials during Tests	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Basic calculator	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

WM0320TU	Ethics and Engineering	3
Module Manager	Dr. F. Santoni De Sio	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course is identical to the initial part of the course WM0329TU. You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and interactive work sessions with partial assessment, concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	Written exam (60%), presentation and active participation during the working group sessions (40%). In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. Please enrol as soon as possible and check Brightspace for more information on the enrolment on the enrolment procedure and timeline.	
Remarks	The course is run twice each year in the first and third quarter. The course is identical to the initial part of the course wm0329tu (6 erts).	
Elective	Yes	
Category	MSc niveau	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Profile Electives 2022

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Biomass 2022

AE4262	Combustion for propulsion and power technologies	4
Responsible Instructor	Dr. I. Langella	
Responsible Instructor	A. Gangoli Rao	
Instructor	Prof.dr. D.J.E.M. Roekaerts	
Contact Hours / Week x/x/x/x	0/0/0/4+2lab	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	WI1421LR-I Calculus I, WI1402LR Calculus II, AE1240-I Thermodynamics, AE4238 Aero Engine Technology	
Summary	This course is designed to provide the student with a fundamental understanding of combustion processes and their application within advanced propulsion and power technologies. The goal is to reach a knowledge on how to improve the efficiency and mitigate pollutant formation of practical combustion systems. Since currently about 90% of the worldwide energy usage (e.g. transportation, electricity generation, heating) is provided by combustion, it is of the utmost importance to continue developing highly efficient combustion technologies with the lowest possible level of emissions. This is especially true for aero engines, whereby the use of combustion engines will continue in the foreseeable future.	
Course Contents	1. Thermodynamics of combustion 2. Chemical kinetics 3. Radiation and transport processes 4. Premixed and diffusion flames 5. Turbulent reactive flows 6. Combustion experiments and diagnostics 7. Laboratory Tutorial & Exercise 8. Practical combustion systems for aerospace applications	
Study Goals	Write chemical equations and perform stoichiometry calculations. Compute heating values of fuels, enthalpies of formation- and adiabatic flame temperature. Apply principles of thermodynamics to analyze reacting mixtures at chemical equilibrium. Describe the fundamentals of the physics involved in premixed- and diffusion flames. Derive equations related to turbulent reacting flows and chemistry interaction. Explain laser diagnostics and their role in understanding combustion. Discuss critical mechanisms for improving the efficiency and reducing the emissions of air-breathing combustion systems. Provide phenomenological descriptions of pollutant formation of conventional combustion systems.	
Education Method	The course consists of weekly lectures and exercise sessions. In addition, the student is required to complete project assignments and perform a laboratory exercise. The project assignments and associated numerical problems are relevant to practical combustion systems for aerospace applications. The laboratory exercise deals with laminar flame speed measurements of the simplest hydrocarbon fuel, methane. The student will gain hands-on experience in assembling the experimental setup, acquiring data, performing data analysis and writing a technical report.	
Course Relations	Aero Engine Technology	
Literature and Study Materials	Slides and selected material from several textbooks are made available (e.g., J. Warnatz, U. Maas, and R.W. Dibble COMBUSTION);	
Books	J. Warnatz · U. Maas · R.W. Dibble "Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation"	
Assessment	To pass the course, the student must pass the oral examination, the evaluation of the laboratory exercise, and successfully complete the assignments given during the course. The work on the laboratory exercise and assignments should be summarized in reports, to be written by a single student or in a group of two. The grade is dictated by the result of the oral examination (50%) and the grades obtained in the assignments and lab exercise (50%).	

LM3741	Fermentation Technology & Environmental Biotechnology	6
Responsible Instructor	Dr. W.M. van Gulik	
Responsible Instructor	Dr.ir. R. Kleerebezem	
Instructor	Dr. W.M. van Gulik	
Instructor	Prof.dr.ir. M.C.M. van Loosdrecht	
Instructor	Dr.ir. R. Kleerebezem	
Instructor	Prof.dr.ir. H.J. Noorman	
Contact Hours / Week x/x/x/x	0/5/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Microbial diversity, Black box kinetics of growth/ product formation, mass balances, transport phenomena.	
Course Contents	The FTEB course is divided into two sections: Microorganism characterization (i), and Bioreactor design (ii). Microorganism characterization deals with stoichiometry, kinetics, and thermodynamics of microbial processes in the field of FT and EB. Specific attention is paid to identification of parameters and parameter sensitivity. The bioreactor design section includes batch, fed-batch and continuous bioreactor operation. Specific attention is paid to mixing, gas-liquid mass transfer, biofilm and other biomass retention based bioprocesses, as well as integration of different microorganisms in a single process configuration. All lectures are supplemented with practical examples. Students will elaborate in groups of 3-4 students a case study that should result eventually in the design of a bioprocess. As in the lecture series, students will first elaborate the properties of the microorganisms required and subsequently on the implementation in the bioreactor. Both parts of the case study will be discussed in classical sessions.	
Study Goals	After this course the students are able to analyse and design fermentation and environmental bioprocesses. Most relevant transferable skills: - Problem solving - Teamwork - Communication - Responsibility - Time management - Presentation skills	
Education Method	Lectures and assignments	
Literature and Study Materials	Lecture notes, lecture slides and relevant scientific papers, which will be provided via Brightspace.	
Assessment	Case study (group work) and exam (individual). The scores for the case study and the exam determine for 50% the final grade. The minimum exam score to pass is 5.	
Enrolment / Application	On Brightspace. Not later than one week before the first lecture	
Schedule	Part 1: Microorganism - Introduction - Stoichiometry - Thermodynamics - Basic kinetics - Including product formation - Determining parameters from experimental data - Fed-batch, batch, chemostat - Case study part 1: microorganisms Part 2: Bioreactor - Mass transfer modelling - Design transfer processes in bioreactors - Biofilms structure,function, flux analysis, and process design - Integrating mass transfer, biofilms, and biomass retention - FT-examples - EB-examples - Case study part 2: Bioreactor	
Location	Delft	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in 3ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications. Indicate implications when more detailed distributions of convective heat transfer are involved.	
Education Method	- Lectures (2x2 hours per week) including experiments, computer demonstrations as well as instruction. In 2021, all lectures will be online. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction/office hours, during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. Weekly instruction is likely to be on campus.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Office Hour if it's not standard. Lecture slides, short videos, Matlab scripts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Department	3mE Department Process & Energy	

ME45025	Introduction to Multiphase Flow	5
Responsible Instructor	Dr.ir. W.P. Breugem	
Instructor	Prof.dr.ir. R.A.W.M. Henkes	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	A multiphase flow is the simultaneous flow of two or more immiscible phases of matter like gasses, liquids and/or solids. Such flows are important in many environmental and industrial processes and equipment. Examples are gas and oil transport, dredging, steel making, refineries and chemical plants, sediment transport in rivers, cloud formation, etc. This course gives a broad introduction to the fundamental fluid mechanics of such flows. Topics include: classification of multiphase flows into dispersed and separated multiphase flows, relevant dimensionless numbers, stability of gas/liquid two-phase flows, spatial/statistical averaging of the Navier-Stokes equations and continuum modelling, flow pattern maps and mechanistic modelling of multiphase pipe flows, physics of particles (droplets, bubbles, solids), dynamics of particle-laden flow, effects of turbulence, experimental techniques and computational methods for multiphase flows. Examples will be given of the use of multiphase flow theory in industrial applications.	
Study Goals	At the end of this course the student is able to: 1.a classify multiphase flows into separated and dispersed multiphase flows. 1.b formulate appropriate interfacial conditions for a fluid/fluid and a fluid/solid interface. 1.c formulate relevant dimensionless numbers for multiphase flows. 1.d explain the concept of dynamic similarity of multiphase flows. 1.e reflect on scaling choices when designing a laboratory experiment or setting up a numerical simulation. 2.a apply a linear stability analysis to two-phase flows such as for instance a liquid jet in a gas (Rayleigh-Plateau instability) and two horizontal fluid layers in relative motion with each other (Kelvin-Helmholtz instability). 2.b give physical mechanisms for the transition of stratified to non-stratified (bubble or slug) multiphase flows in both (tilted) horizontal and vertical pipes. 3.a apply averaging methods (such as statistical, spatial and temporal averaging methods) to the Navier-Stokes equations for describing the macroscopic behavior of multiphase flows. 3.b formulate appropriate closure models needed to solve the averaged Navier-Stokes equations. 4.a classify gas/liquid pipe flows into stratified, slug and bubble flows, dependent on the superficial gas and liquid velocities. 4.b apply simplified 1D (cross-section averaged) models to each of the flow regimes. 4.c formulate appropriate closure models in order to compute the streamwise pressure drop and liquid holdup in two-phase (gas/liquid) pipe flow. 5.a classify dispersed multiphase flows into flows with solid particles, gas bubbles and/or liquid droplets. 5.b describe similarities and differences between the dynamics of solid particles, gas bubbles and liquid droplets. 5.c apply the Newton-Euler equations of motion to an analysis of the dynamics of solid particles. 5.d describe the settling of a solid sphere under gravity in a stagnant fluid in terms of the Archimedes number and the fluid/solid mass density ratio. 5.e describe possible interactions between solid particles and turbulence of the carrier phase. 6.a explain changes in the shape of a rising gas bubble in a stagnant fluid dependent on the Reynolds number, Eotvos number and Haberman-Morton number. 6.b estimate the terminal rise velocity of a bubble in a stagnant liquid from the equation of motion and an appropriate model for the drag. 6.c give physical mechanisms for the breakup of bubbles/droplets in turbulence. 7. describe the dynamics of some particle-laden flows such as sediment transport through a pipe and/or sediment transport in rivers. 8. describe the merits and shortcomings of different experimental methods for multiphase flows. 9. describe the merits and shortcomings of different numerical methods for simulating multiphase flows. 10.a give examples of multiphase flow in industrial processes. 10.b describe the types of models used to design multiphase industrial processes.	
Education Method	Oral lecture 0/0/2/2; Plenary discussion of homework assignments 0/0/2/2	
Assessment	Written exam. The student can earn bonus points by making homework assignments. The final grade is the sum of the grade for the written exam and the bonus points from the homework.	
Department	3mE Department Process & Energy	

ME45070	Advanced Reaction & Separation Systems	5
Responsible Instructor	Dr. A.A. Kiss	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic chemical & process engineering (e.g. BSc Process Technology course); heat and mass transfer	
Course Contents	1. Introduction to Process Intensification(PI): - sustainability-related issues in process industry; - definitions of Process Intensification; - fundamental principles and approaches of PI. 2. How to design a sustainable, inherently safer processing plant - presentation of PI case study assignments. 3. PI Approaches: - STRUCTURE - PI approach in spatial domain (incl. "FOCUS ON" guest lecture) - ENERGY - PI approach in thermodynamic domain (incl. "FOCUS ON" guest lecture) - SYNERGY - PI approach in functional domain (incl. "FOCUS ON" guest lecture) - TIME - PI approach in temporal domain (incl. "FOCUS ON" guest lecture) The series of lectures given in Q3 is followed in Q4 by a case-study project, in which groups of students develop conceptual design of an intensified chemical plant.	
Study Goals	Upon completion of the course the student will have basic knowledge of process intensification principles, equipment and processing methods, and will be able to apply them to a conceptual design of intensified, inherently safer chemical plants.	
Education Method	Lectures, case-study project in groups	
Literature and Study Materials	Basic study material: 1.Lecture handouts 2. Book "Fundamentals of Process Intensification" by A. Stankiewicz, T. van Gerven and G. Stefanidis, Wiley-VCH, 2019. Additional materials will be distributed at the start of the case study project.	
Assessment	Written examination; Report + presentation of the case study project	
Remarks	TEACHER WILL BE ANNOUNCED IN SEPTEMBER	
Department	3mE Department Process & Energy	

ME45135	Process and Power Plant Design	5
Responsible Instructor	Prof.dr.ir. W. de Jong	
Responsible Instructor	Dr.ir. R.M. Stikkelman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Design of equipment for mass and heat transfer Thermodynamics of mixtures and phases separation	
Course Contents	This course provides the fundamentals for conceptually designing processes of chemical and energy conversion plants, where basic unit operations such as reactions, energy conversion, heat and mass transfer, separations, etc. are designed, modeled and optimized. The conceptual process design involves the design of the flow sheet, creating the topology of the process by the optimizing the sequence of reactors, separation units and energy conversion systems to achieve the required product specification in a sustainable and economic feasible way. In the course the students learn how to select, sequence and model the different unit operations as well as the selection and rough sizing of the required equipment. In addition the students will learn how to integrate the different units into a process flow sheet, and how to model, optimize and rank of the different design alternatives. During the course the students get a hands-on experience with modern process design tools, which will be used in assignments in which conceptual designs of sustainable, economic feasible processes will be made The course includes the following elements: - Intro; context and scope of conceptual process design; - Modelling the thermodynamics of mixtures - Modelling of unit operations - Hierarchical approach to conceptual process design - o reaction and separation systems - o homogeneous systems - o heterogeneous systems - Simulation and optimization - Superstructures - Costing and project evaluation - Flow-sheet design using Aspen - Presentation of the case studies	
Study Goals	The student can conceptually design and optimize chemical process plants. More specifically, the student must be able to: 1. describe the context and scope of a conceptual design of a chemical plant, 2. model and design the most important unit operations of chemical process plants, 3. make a conceptual process design using the hierarchical Douglas approach for process design 4. select, sequence and model the different unit operations in the flowsheet of a chemical process plant and can determine the dimensions of the required equipment, 5. apply process flow-sheeting tools for the design and simulation of process flowsheets for a chemical plant including the recycle streams 6. Use the process simulation tools to evaluate, optimize and rank the different design alternatives for a process.	
Education Method	Lectures, assignments, presentations and oral examination	
Assessment	The examination will be as follows: Oral exam [50%, must be 6 or higher] + two sets of (report + recorded presentation) [10+40%]. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: exactly the same. The assessment is Covid-19 independent.	
Department	3mE Department Process & Energy	

ME45155	Modelling of Thermo- & Hydrodynamic Systems	5
Responsible Instructor	Dr.ir. M.J.B.M. Pourquie	
Instructor	Dr. R. Pecnik	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	WB2542 (thermofluids), elementary thermodynamics (energy equation), ME45042 or ME45043 (advanced fluid dynamics), WI3197MINOR or WBMT2049 T2 or similar (very basic numerical analysis), some elementary programming skill (matlab or any other)	
Course Contents	This is a basic course on the modeling of physical systems related to process and energy systems. Some very elementary knowledge of numerical methods (discretisation of a derivative, numerical solution of ordinary differential equations) is assumed. See expected prior knowledge. In general the course will cover: (1) general conservation equations for mass, heat and momentum transport (Navier-Stokes equations) (2) examples of specific problems in heat- and mass transfer for the process industry (3) reduction of 3D, time-dependent problems to 1D, lumped models, leading to systems of ODE's combined with algebraic equations (4) numerical solution of reduced problems (5) numerical solution of multi-dimensional, stationary and time-dependent problems as encountered in CFD (Computational Fluid Dynamics) (5.1) numerical schemes (5.2) boundary conditions (5.3) turbulence models, how to use eddy viscosity models in CFD (practical guidelines) (6) wave equations (7) introduction to compressible methods	
Study Goals	After learning the content of the course the student will have the following capabilities: (1) Describe the role of models in Process and Energy Systems Engineering, and describe examples of systems, processes, modeling paradigms, applications, software tools, methods. (2) Represent a process with process flow diagrams, and define and use on-design and off design steady state models, "open loop" dynamic models and their applications to design, operation and control. (3) Describe the two most popular methods in commercial CFD, finite differences and finite volumes (4) solve simple demonstrative problems in fluid flow and heat transfer by programming them in Matlab, using finite differences and finite volumes (5) recognize the effects of numerical methods on the solution, such as numerical diffusion and numerical dispersion and to explain how to make these effects smaller (6) recognize numerical instability, to list several ways to avoid it and to analyze stability of simple methods analytically (7) solve fluid flow and heat transfer problems with the commercial CFD package Fluent, which includes the following: set up a calculation, generate a simple grid, set appropriate boundary conditions, choose suitable discretisation, interpret and validate the results. Alternatively, in some cases, instead of using a commercial package, students can decide to write and validate their own code which they will use for calculating a fluid mechanics problem.	
Education Method	Lectures, practical exercises (2x2 hours per week)	
Literature and Study Materials	Course material: (1) Sheets/handouts (2) J.H. Ferziger and M. Peric, Computational methods for Fluid Dynamics, Springer Verlag. (3) P. Moin, Fundamentals of Engineering Numerical Analysis, Cambridge University Press, 2001. References from literature: (1) C. Hirsch, Numerical computation of internal and external flows, Volume I Fundamentals of numerical discretization, Volume II Computational methods for inviscid and viscous flows, Chicester, Wiley & Sons, 1988, 1990 (2) C.A.J. Fletcher, Computational techniques for Fluid Dynamics, Volume I Fundamental and general techniques, Volume II Specific techniques for different flow categories, Berlin, Springer, 2-nd ed. 1991. (3) H.K. Versteegh, W. Malalasekara, An introduction of computational fluid dynamics. The finite volume method. Second edition. Pearson Education.	
Assessment	Written + Report for assignments; written exam may be online type depending on circumstances	
Department	3mE Department Process & Energy	

ME45160	Advanced Applied Thermodynamics	5
Responsible Instructor	Dr. B.P. Tighe	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	EFPT track	
Expected prior knowledge	Thermodynamics at the level of a BSc in Mechanical Engineering	
Course Contents	- Review of fundamentals of thermodynamics, including thermodynamic potentials, Maxwell relations, and the 1st and 2nd Law - Equilibrium in one-component systems: equations of state, chemical potential, fugacity, Clausius-Clapeyron, etc. - Mixtures: ideal gas mixtures, partial molar quantities, Gibbs-Duhem, vapor- liquid equilibrium, activity coefficient models, and osmosis - Model equations of state - First introduction to statistical mechanics	
Study Goals	After this course, you will be able to calculate phase and material properties in thermodynamic equilibrium for multi-phase and/or multi-component mixtures under ideal and non-ideal process conditions. You will be able to use equations of state and activity coefficient models, and to calibrate and validate those models using thermodynamic data.	
Education Method	Lectures, optional homework, review session	
Computer Use	homework assignments require scientific computing tools such as python, Matlab (or comparable) and basic programming methods such as for loops	
Books	Holyst Poniewierski	
Assessment	Written final exam; open book	
Department	3mE Department Process & Energy	

ME45165	Equipment for Heat & Mass Transfer	5
Responsible Instructor	Dr. H.B. Eral	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Advanced reaction & separation systems Process plant design	
Expected prior knowledge	ME4500 (Advanced heat transfer)	
Summary	Heat transfer by conduction, convection, radiation, condensation and boiling. Principles and application of heat integration, exergy analysis and pinch technology. Design procedure for heat transfer equipment. Shell and tube heat exchangers, plate heat exchangers, air cooled heat exchangers, and furnaces. Condensers and boilers. Effectiveness and NTU methods in heat exchanger design. Selection of heat transfer equipment. Basic principles and methods for the process and equipment design of thermal separation processes. Theory on thermodynamics of mixtures, mass transfer by convection and diffusion across interfaces. Characteristics and design methods for vapour liquid, liquid-liquid, solid liquid separation processes or in processes in which mass separation agents and physical barriers are applied. Equipment design for the mentioned separation processes.	
Course Contents	Heat transfer by conduction, convection and radiation in process equipment; estimation of heat transfer rates during condensation and boiling, estimation of the overall heat transfer coefficients. Principles of heat integration, exergy analysis and pinch technology for heat exchanger network. Design and analysis of shell-and-tube heat exchangers, plate heat exchangers, air cooled heat exchangers, and furnaces. Design aspects of condensers and evaporators. Effectiveness and NTU methods in heat exchanger design. Selection of heat transfer equipment. Basic principles and methods for the process and equipment design of thermal separation processes Principles and theory of thermodynamics of mixtures. Phase equilibria, mass transfer by convection and diffusion across interfaces. Characteristics and design methods for the most important thermal separation processes such as distillation, absorption, tripping, crystallisation, adsorption, extraction and membrane separations. classification of the various separation processes the separation efficiency in single and multi-stage processes and the influence of the process conditions and contacting modes on the separation. design procedures of the different separation processes based on limitations of the thermodynamic equilibria. quantification of the influence of process condition on the design procedures of the different separation processes based on limitations of the thermodynamic equilibria. characteristics for the equipment for the various separation processes the role of mass transfer in the sizing and design of the separation equipment methods for the design and dimensioning of separation equipment. the use of computer tools for the design of separation processes	
Study Goals	The student is able to select and make an hydraulic and heat technical design of heat transfer equipment. More specifically, the student: 1. Can quantify heat transfer by conduction, convection & radiation in process equipment; 2. Can estimate the heat transfer by condensation and boiling; 3. Can calculate overall heat transfer coefficients; 4. Can select heat transfer equipment for given application; 5. Can apply effectiveness and Number of Transfer Units method in heat exchanger design; 6. Can design & analyse shell-and-tube heat exchangers (with / without phase change); 7. Can design & analyse plate heat exchangers (with / without phase change) 8. Can design & analyse finned-tube heat exchangers (with / without phase change); 9. Can design & analyse furnaces; 10. Can apply pinch technology / heat integration / exergy analysis for heat exchanger network. Furthermore, the student is able to make a design of thermal separation processes, including the dimensioning of the equipment. More specifically, the student: 11. Can describe the characteristics and principles of the most relevant thermal separation processes, 12. Can describe and formulate thermodynamics equations and models for the selection and design of thermal separation processes, 13. Can quantify the mass transfer rate over an interface by diffusion and convection, 14. Knows the role of mass transfer in the dimensioning of equipment for separation processes 15. Can determine the separation efficiency for single and multi-stage separations and can quantify the influence of the process conditions and the contacting modes on the separation. 16. Can describe principles and design methods for processes and equipment used in equilibrium stage separation processes, such as distillation, absorption, stripping (desorption), extraction, crystallization and membrane separations 17. Can use computer based process simulation tools to analyse and design separation equipment 18. Can design and analyse thermal separation process including dimensioning of the equipment.	
Education Method	Lectures + assignments	
Computer Use	Simulation of performance of distillation trays Development of homework assignments	
Literature and Study Materials	Course material: 1: H.B.Eral, A.De Haan, B.Schuur, "Industrial Separation Processes" Second Edition, De Guyter 2020 ISBN-13: 978-3110654738 2: Henley, E.J., Seader, J.D., and Roper, D.K. Separation Process Principles, third edition (or 4th Edition), International student version, J. Willey & Sons, 2011. 3: VDI Heat Atlas (2010) ISBN 9783540778776, Springer. 4: Course sheets / relevant papers Further reading	

Assessment	Incropera, Dewitt, Fundamentals of heat and mass transfer, 1985, John Wiley and Son Hewitt, G.F., Shriners, G.L. and Bott, T.R. Process heat transfer, 1994, CRC Press Coulson, J.M., Richardson, J.F., Sinnott; Chemical Engineering Vol. 6, Ch. 7,112: Homework assignments Written exam The homework assignments are distributed during the course and concern: For heat transfer three homework assignments: 1. Design & analyse shell-and-tube heat exchangers (with / without phase change) 2. Design & analyse plate heat exchangers (with / without phase change) 3. Design & analyse finned-tube heat exchangers (with / without phase change) For mass transfer three homework assignments will be provided from the following list: 5. Thermodynamics and mass transfer 6: Distillation 7: Crystallisation 8: Extraction 9: Adsorption 10: Membrane separation To get access to the exam the homework assignments must be delivered before the end of the education period. The final mark is the result of homework assignments + exam. Each assignment counts for 7.5%. Exam accounts for 55%. At least a grade of 5 is required from both heat and mass transfer sections of the exams.
Permitted Materials during Tests	Course books and course handouts
Percentage of Design	80
Design Content	Hydraulic and heat technical design of equipment of heat transfer; Basic principles of dimensioning of equipment used in the above mentioned processes
Department	3mE Department Process & Energy

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Power 2022

ET4103	High Voltage Technology	4
Responsible Instructor	Prof.ir. P.T.M. Vaessen	
Responsible Instructor	Dr. M. Ghaffarian Niasar	
Practical Coordinator	Ing. G.J. Kamphuis-de Rooij	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course presents the basic principles of High Voltage technology and practical application of this knowledge in the power industry. Proper design of High Voltage components requires knowledge and insight of electric fields and breakdown mechanisms.	
Study Goals	By the end of the course students will be able to: - Demonstrate insight in the relation between charge, electric field and voltage - Identify and calculate capacitive voltage distributions - Calculate and analyze electric fields using analytical and numerical methods - Describe the breakdown mechanisms in different insulation materials and the phenomena leading to this - Explain methods of field grading and field control - Describe impulse breakdown phenomena (for example lightning) - Evaluate the effect of combinations of materials and interfaces on electric fields - Remember maximum permissible field strengths in different materials and configurations	
Education Method	Note:the information provided in this page may change depending on the developments around the coronavirus Lectures, demonstrations	
Literature and Study Materials	Kreuger, F.H., Industrial High Voltage. Vol.1, 2, 3, ISBN 90-6275-561-5 (available for purchase at the course), hand-outs (available at the course)	
Prerequisites	Electromagnetics (EE3P11) or equivalent courses	
Assessment	Written exam (70%) and project (30%)	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

OE44076	Offshore Renewable Energy	6
Responsible Instructor	Dr.ir. A. Jarquin Laguna	
Instructor	Ir. A.C.M. van der Stap	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course will introduce the students to the technical aspects and detailed understanding of ocean energy technologies (wave, tidal current and ocean thermal energy OTEC). The course combines lectures on offshore renewable energy technologies, CO2 and energy storage. The students will participate in a realistic project teamwork with young industry coaches and working on actual cases with stakeholder and risk management assessment, modelling and energy extraction analysis from the offshore environment, including an overview of the project development of safe and reliable offshore energy technologies.	
Study Goals	by successfully completing the course you should be able to: Explain and describe the physical principles of energy conversion for wave energy, tidal stream and OTEC. Compare different marine renewable energy sources based on calculations of their energy content. A few examples of such sources include ocean waves, tidal currents and temperature difference of the ocean. Identify and evaluate the design considerations and technical challenges of wave and tidal energy technologies. Generate future-proof solutions for offshore energy projects with a focus on minimization of the CO2 footprint. Define technical, political, contractual, economic and operational constraints imposed on industrial projects throughout its entire life cycle. Explain the crucial importance of uncertainties and economics in the decision process in the course of developing a project. Apply basic project analysis and management techniques. Accommodate conflicting requirements in an offshore design environment.	
Education Method	Lectures and project teamwork with industry coaches.	
Assessment	Grading based on a group assignment.	
Department	3mE Department Maritime & Transport Technology	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Economics & Society 2022

SEN1211	Agent-based Modelling	5
Module Manager	Dr.ir. I. Nikolic	
Instructor	Dr.ir. I. Nikolic	
Instructor	Dr. C.N. van der Wal	
Responsible for assignments	Dr.ir. I. Nikolic	
Co-responsible for assignments	Dr.ir. E.J.L. Chappin	
Co-responsible for assignments	Dr. C.N. van der Wal	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Our human society consists of many intertwined Large Scale Socio-Technical Systems (LSSTS), such as infrastructures, industrial networks, the financial and legal systems etc. Environmental pressures created by these systems on Earth's carrying capacity are leading to exhaustion of natural resources, loss of habitats and biodiversity, and are causing a resource and climate crisis. To avoid this sustainability crisis, we urgently need to transform our production and consumption patterns. Given that we, as inhabitants of this planet, are part of a complex and integrated global system where and how should we begin this transformation? And how can we also ensure that our transformation efforts will lead to a sustainable world? Society, with its industrial, social, economic en institutional systems, embedded in the planetary ecosystems are Complex Adaptive Systems (CAS). According to John Holland CAS are "...a dynamic network of many agents (which may represent cells, species, individuals, firms, nations) acting in parallel, constantly acting and reacting to what the other agents are doing. The control of a CAS tends to be highly dispersed and decentralized. If there is to be any coherent behavior in the system, it will have to arise from competition and cooperation among the agents themselves. The overall behavior of the system is the result of a huge number of decisions made every moment" by many individual agents. Understanding Complex Adaptive Systems requires tools that themselves are complex to create and understand. Shalizi defines Agent Based Modeling as "An agent is a persistent thing which has some state we find worth representing, and which interacts with other agents, mutually modifying each others states. The components of an agent-based model are a collection of agents and their states, the rules governing the interactions of the agents and the environment within which they live." This course has two focus points: 1-Complex Adaptive Systems Theory and 2- Agent Based Modelling. We will explore Complex Adaptive Systems theory, across many different types of systems. Lectures will be quite philosophical in nature, and are meant to provide a coherent, high level understanding of systems, their properties and their interactions. Lecture materials provide further reading to broaden your view and deepen your understanding. Second focus point is developing Agent Based Modelling skills. This course is not meant as an introduction to basic programming, or a thorough introduction to NetLogo environment, but will rather focus on understand the concepts of ABM and how model design choices affect the model created and how it can be used. Additional reading materials are provided for self-study in getting up to speed on basic programming and NetLogo.	
Study Goals	The two goal of the course are, first, to understand Complex Adaptive Systems theory and its relation to the socio-technical systems around us. Second goal is for the student to learn about the the basics of Agent Based Modeling. More formally, there are 5 course goals: 1. Students should know the definitions of CAS properties, as listed in the course topics 2. Students should be able to identify relevant properties of a system and determine whether or not the system can be classified as CAS. 3. Students should be able to create a coherent description of a system from both top down and bottom up 4. Students should be able to understand, modify and create Agent Based Simulations in NetLogo. 5. Students should be able to reflect on the traditional engineering systems thinking from the CAS perspective and understand the implications of changing the traditional perspective.	
Education Method	Course consists of a series of high-level lectures, modelling practicals and a final modeling project. Lectures Lectures and practicals are intertwined, building on top of each other. Where possible, the theory discussed in the lectures will be explored in a practical. Lectures will explore the interrelated properties of complex systems. Practicals The practicals are meant to introduce you to Agent Based Modelling concepts. The practicals build up in complexity, exploring the various components of ABMs. Limited attention is payed to the NetLogo language, as sufficient resources are available for self study of the programing language. Modeling project The second part of the course is dedicated to creating a model in small teams, applying the skills learned during the practicals. The project deliverable consists of the simulation code and report documenting the problem statement, model design, performed experiments and data analysis. This is a compulsory project, contributing 50% to the final grade.	
Computer Use	The course involves developing and experiment with simulations in NetLogo.	
Assessment	The final grade of the course is based 50% on the written theory exam and 50% on the modeling project (simulation + report). Both parts must be completed with a passing grade (at least 5.75) in order to pass the course.	

SEN1531	Design of Integrated Energy Systems	5
Module Manager	Dr.ir. E.J.L. Chappin	
Instructor	Dr.ir. P.W. Heijnen	
Instructor	Dr.ir. G. Korevaar	
Responsible for assignments	Dr.ir. E.J.L. Chappin	
Co-responsible for assignments	Dr.ir. G. Korevaar	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The SEN Energy Track modules taught in Q1 and Q2.	
Course Contents	Students are challenged to tackle an integrated energy system design problem in the form of a modelling project. Students are expected to come up with a creative (set of) design(s). Afterwards, they evaluate the expected performance of one or more of these designs as a future system or improvement to existing energy systems. The design challenge is complex and can only be met: 1) by taking into account dynamics on the short and long term, 2) by tapping into relevant knowledge on technological developments, 3) by using various quantitative analyses, 4) by considering various parts of the value chain, in terms of energy, mass, financial and information flows and 6) by relating and addressing the multi-actor and multi-level complexities. The quantitative analyses are to be executed with a set of quantitative tools, some used in earlier courses in the Energy Track, some developed by research institutes and companies, and others used in ongoing academic research at the faculty, made available for this course.	
Study Goals	Understand and execute the main steps in the design of integrated energy systems Is able to provide an overview of and explore the design space of integrated energy systems, consisting of multiple carriers, multiple relevant levels, multiple time-scales, multiple sectors Is able to identify and conceptualize core processes that affect the performance of an integrated energy system that play out on short, medium and long term Is able to select, develop and use quantitative simulation models for designing integrated energy systems Is able to reflect on aspects that are placed outside the design space, i.e. the relationship to food, water, built environment, etc.	
Education Method	The core task within this module is to carry out a group project in which an integrated energy system is designed and evaluated by using various models. Groups of students use various models and interact with model experts: teachers, researchers but also various companies are involved. There are lectures and visiting hours for debating modelling choices and details, but also for reflection and debate on modelling and simulation. Students do peer review and present their results for each other and for model experts. This enables academic reflection as well as direct interaction with the potential work field.	
Literature and Study Materials	A reader is provided with the approach applied and descriptions of all the models available. Various energy transition models and simulations are made available for use in the projects. Some were introduced in earlier courses, some are extracted from ongoing energy research at the TU Delft, and others come from various companies in the field that do their own energy system modelling and analysis.	
Assessment	An individual grade will be assigned on the basis of the following inputs: - The group project report that describes the integrated energy systems design including an assessment of its expected performance - An individual appendix with a personal reflection and description of contribution to the project - The contribution to peers, through peer review and discussions	

TPM001A	Sociotechnology of Future Energy Systems	4
Module Manager	Dr.ir. I. Bouwmans	
Instructor	Dr.ir. I. Bouwmans	
Instructor	Dr. A.F. Correlje	
Responsible for assignments	Dr.ir. I. Bouwmans	
Co-responsible for assignments	Dr. A.F. Correlje	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students need to have sufficient prior knowledge in the fields of energy technology, economics and policy.	
Course Contents	In this module the changing technological and societal characteristics of future energy systems are dealt with. Contemporary energy systems are confronted by technical developments and socio-economic concerns, such as maturing renewable technologies and ICT based control innovations or energy scarcity and environmental concerns. These changes will reflect on operational roles and responsibilities, market structures and actor interests, and business models for energy services. Moreover, they may necessitate new forms of governance to ensure a reliable operation that meets socio-economic performance criteria. In this light, the course discusses the technical and socio-economic aspects of future, mostly sustainable, energy systems, touching upon issues like the integration of new forms of distributive solar PV or large-scale off-shore wind into existing networks, various forms of new motor fuels, other more experimental renewable energy sources and storage methods, and the dynamics of transition processes.	
Study Goals	At the end of this module the students know: - Which technical and societal developments can be expected in the energy sector. - How new energy technologies and societal developments may reshape operational roles and responsibilities, energy markets and business models, and governance arrangements. - How to compare and analyze future energy systems, develop plausible transition paths towards them, and estimate the potential of new energy technologies within various scenarios. At the end of this module the students can: - Formulate descriptions of the technical and societal characteristics of future energy systems. - Make a conceptual design of an energy system, given certain demands. - Formulate a balanced advice on issues in the energy sector.	
Education Method	8 interactive sessions, self-study, Brightspace & class exercises, class discussion. Reading material: scientific papers.	
Assessment	Written exam (on the computer); active participation in the Brightspace exercises is a prerequisite for taking the exam. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: timed take-home exam (open book). This contains several open-ended essay questions in a restricted time period, comparable to the on-campus exam. A working internet connection is the responsibility of the students.	
Elective	Yes	
Category	MSc level	

TPM023A	Cost-Benefit Analysis: Theory and Application	4
Module Manager	Dr.ir. Z. Roosenboom-Kwee	
Instructor	N. Mouter	
Responsible for assignments	Dr.ir. Z. Roosenboom-Kwee	
Co-responsible for assignments	N. Mouter	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Summary	The course TPM023a focuses not only on private (e.g. project, firm, industry) costs and benefits but also on social costs and benefits. While the private CBA only considers financial aspects, the social CBA considers both financial and non-financial aspects. For instance, in the construction of new highway, besides the financial aspects such as toll revenues or construction costs, we also assess the non-financial aspects such as safety benefits (lives saved, accidents avoided). Hence, the course actually has a much broader view than the private CBA which offers students a higher level of practical applications.	
Course Contents	Cost-benefit analysis (CBA) has become a widely used (often mandatory) tool for public policy decision making on infrastructural projects or environmental and climate policies. Through CBA, the potential current and future gains and losses of a proposed policy are identified, and then converted into monetary units in order to make them comparable with policy alternatives; certain decision rules help determine whether the project is desirable from society's standpoint. Because CBA evaluates policy and/or project proposals from the point of view of the public interest, the market prices used in CBA are corrected for possible market distortions or failure (for example, un-priced environmental damage). Nevertheless, the application of CBA is fraught with difficulties and limitations that need to be taken into account. What rates of interest are appropriate for determination of present and future costs and benefits of a policy proposal? Why and how do the private benefits and costs of a policy proposal differ from the social benefits and costs? How can we express e.g. environmental damage, or human health and life risks, in monetary units? How can we adjust market prices to account for external effects? How can we use alternative measures of welfare to identify the benefits and costs of a policy proposal? How can we ensure that politicians who aspire to use the results of CBA in their decisions can actually do so?	
Study Goals	At the end of this course, students will be able to: * understand why CBA can be used to appraise projects with an objective to improve social welfare * gain an understanding of the fundamentals of CBA * perform steps or approaches to evaluate the financial and economic values of a project * understand the strength and limitations of CBA in project appraisal. * interpret the results of a cost-benefit analysis * criticize and defend the CBA methodology from an ethical perspective * explain the advantages and the disadvantages of applying a CBA methodology in a political process with multi-stakeholders	
Education Method	The lectures are on campus. However, depending on the COVID-19 measures, the lectures may be given through a hybrid form or online and remote form.	
Literature and Study Materials	* Textbook: Anthony E. Boardman, David H. Greenberg, Aidan R. Vining & David L. Weimer: Cost-Benefit Analysis: Concepts and Practice, 4th International Edition, Pearson, 2013. * Selected articles (available in Brightspace)	
Assessment	The course assessment is an individual assignment: writing a policy note on CBA. The retake of the assignment is only for students who submitted their assignment in the first attempt and NOT for new or existing students who did not previously submit their assignment in the first attempt.	
Special Information	In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: take-home individual assignment.	
Category	MSc level	

TPM404B	Technology Entrepreneurship and Internationalization	5
Module Manager	H. Khodaei	
Instructor	H. Khodaei	
Instructor	F. Delgado Medina	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course offers theoretical fundamentals of international entrepreneurship. You create a case study of an existing start-up (born global) analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an international start-up to academic research on the topics of International entrepreneurship. Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, internationalization marketing and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to: LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Identify and analysis international opportunities across different markets. LO 4 Describe and apply the born-global start-up growth process. LO 5 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 6 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 7 Describe and analysis the barriers to internationalization and evaluate the niche strategies. LO 8 Analyse an existing born-global start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations will be facilitated in an online mode (for example through zoom or Skype). Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (20%) - Individual: Open questions (30%) and theory quizzes (20%) All marks will be expressed on a scale of 1-10. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations and individual assessment will be facilitated in an online mode (for example through zoom or Skype)	

TPM411A	Idea to Start-up IT & AI	5
Module Manager	Dr. T.L. Dolkens	
Instructor	Dr. T.L. Dolkens	
Instructor	E. Çelik	
Responsible for assignments	Dr. T.L. Dolkens	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/2/2/0	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3 4	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	This course used to carry the name MOT9612 - Business Development Lab. In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity using computer sciences. Applications can range from mobility, cyber security to predictive maintenance etc. You will test its validity, and build a viable business model a roadmap from an idea to a sustainable innovation-driven company. On this journey, you will be helped by guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology startup, whether you want to build one or join one.	
Course Contents	You will focus on information and computing technologies (e.g AI, big data, blockchain) that are transforming traditional industries and societies at large. To analyse the opportunities for new startups we focus on entrepreneurial tools such as the opportunity canvas; value proposition development and customer empathy mapping, market segmentation and value chain analyses and the initial operations to develop a minimum viable product.	
Study Goals	Develop a business Idea by identifying a relevant economic problem or unmet need that is addressed by digital solutions, and advanced information and computing technologies. Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment Analyse the competitive landscape and devise your products competitive advantage, given specific challenges of protecting IP in the digital world. Create a blueprint of your product. Identify relevant stakeholders in your startups ecosystem, outline the go-to-market strategy, and assess profitability. Understand how these strategies adjust when trying to implement your solution in well-established systems of traditional industries. Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence. Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by improving the business model. Evaluate your entrepreneurial skills and motivate constructive feedback through peer reviewing	
Education Method	You will work in teams which are formed in the first sessions of the course. Topics are discussed using case studies. You will present your work, provide peer-feedback as well and receive advice from various coaches. The course ends with a jury pitch and a two-pager of the business proposition. Team coaching with industry experts will help you validate the idea and we provide individual coaching for personal entrepreneurial development.	
Books	The primary textbook for this course is an organic handbook by Bill Aulet: Disciplined Entrepreneurship: 24 Steps to a Successful Startup. Wiley, 2013.	
Assessment	The assessment of this course is Covid-19 proof. It includes class assignments and participation (40%), final business model report (30%), online pitch presentation (15%), and peer evaluation (15%). Participation and Peer Evaluation grades pertain to each student. Class assignments, final business report, and pitch presentations are evaluated on a group level.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM413A	Idea to Start-up Energy & Sustainability	5
Module Manager	Dr. L. Hartmann	
Responsible for assignments	Dr. L. Hartmann	
Co-responsible for assignments	Dr. A. Giga	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Summary	----- This course is NOT connected to the CleanTech Challenge. (A previous course Sustainable Business Study was linked to that competition, but this course is not in any way connected.) -----	
Course Contents	A sustainable business idea is a product, a process or a service that enables sustainability, where sustainability is defined as the capacity for the biosphere and human civilization to co-exist indefinitely. The course stimulates and assists students in creating a sustainable business idea and assessing the feasibility and the market potential thereof by testing it against prospective customers and key players in the specific industry of application. Students work together in teams which will be formed by the instructor with the aim to create optimal diversity of scientific background, gender and cultural background of the team members. Each team needs to come up with an original sustainable business idea, or a sustainable business idea connected to one of the sustainability challenges defined for the TU Delft Campus. Each sustainable business idea must be technology-based we do not accept apps and definitely no consulting of any kind. The course consists of two interconnected parts: 1. Students attend lectures and workshops. 2. Teams carry out a feasibility study for their sustainable business idea.	
Study Goals	- Students learn how to assess the market potential of a sustainable product, process, or service. - Students gain basic knowledge on the commercialization of a sustainable business idea. - Students learn what the specific challenges are for a technology-based start-up company. - Students learn about different business models and gain insight in how to choose a market opportunity in the value chain. - Students gain practical knowledge about starting up a business. - Students experience real business practice by feedback from both lecturers, external advisers and experienced entrepreneurs in particular. - Students learn how to give a convincing business presentation. - Students learn to be critical about their own business case and those of their fellow students. - Students learn how to make a proper evaluation of the feasibility of their idea and business plan.	
Education Method	We offer a mix of lectures, case studies, and coaching sessions. The class is fast-paced and interactive. Most of the classes are in the form of online coaching sessions and team exercises, therefore class attendance is mandatory. Online content and online lectures will provide students with general guidelines for each business model milestone. Depending on availability, we will bring guest speakers, usually entrepreneurs, with experience in the energy and sustainability industry. Team coaching will help students to validate their ideas and develop their entrepreneurial skills.	
Literature and Study Materials	Mandatory books: ----- Aulet Disciplined Entrepreneurship: 24 Steps to a Successful Startup MacKay Sustainable Energy - Without the Hot Air Recommended books: ----- Scott Berkun The Myths of Innovation Goldstein Planetary Improvement More literature and reading suggestions will be provided during the course.	
Prerequisites	A BSc is required. Admission to Master students only.	
Assessment	Group report and presentation	
Enrolment / Application	Enrol for the course on Brightspace	
Elective	Yes	
Category	MSc level	

TPM414A	Idea to Start-up Deep Tech	5
Module Manager	Dr. A. Giga	
Responsible for assignments	Dr. A. Giga	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Course Contents	In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity, test its validity, and build a viable business model a roadmap from an idea to a sustainable innovation-driven company. On this journey, you will be helped by guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology venture, whether you want to build one or join one. While the starting point is tackling the societal and market problems and unmet needs, the solution ideas must be based on substantial scientific advances and cutting-edge engineering that can be used to build competitive advantage in the market (via patent for example). Feel free to bring your own ideas and technologies, but you do not need one to enjoy this course. All you need is curiosity about building a business on solid technological foundations. Fast-paced and interactive, the class consists of mostly coaching sessions and team exercises, therefore class attendance is very important. Online content and short in-class lectures provide you with general guidelines for each business model milestone. There will be weekly assignments, building blocks to a full-fledged business model. Final deliverables include: a logbook of primary market research (yes, youll do interviews!), concise business model report, and a pitch presentation in front of a jury of startup professionals. You start the course with forming teams around a market problem or unmet need. Then, you gradually build a business model, mainly by answering questions such as: Who are my customers and what value do I create for them? How big is this market? How should my minimum viable product look like? How do I build a competitive advantage in the market (e.g. by patenting)? How will I market and deliver this product to the customer? Will this be a profitable business? You seek answers through customer, stakeholder, and expert interviews, by developing step-by-step interview protocols and executing them.	
Study Goals	- Develop a business idea based on a relevant economic problem or unmet need. - Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment - Develop a blueprint of your product and map its position in a competitive landscape. - Identify relevant stakeholders in your startups ecosystem, outline the go-to-market strategy, and assess profitability. - Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence. - Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by improving the business model.	
Education Method	The course offers a mix of online lectures, case studies, and coaching sessions. Depending on availability, we will bring guest speakers with relevant industry and entrepreneurial experience. Team coaching with industry experts will help you validate the idea and we provide individual coaching for personal entrepreneurial development.	
Assessment	The assessment of this course is Covid-19 proof. It includes class assignments and participation, final business model report, online pitch presentation, and peer evaluation. Participation and Peer Evaluation grades pertain to each student. Class assignments, final business report, and pitch presentations are evaluated on a group level.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Solar 2022

EE4585	Semiconductor Device Physics	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/2/0/0 lectures; 0/2/0/0 instructions	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).	
Course Contents	In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission. The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed. Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential. Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling. Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).	
Study Goals	This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device. The above description leads to the following learning objectives (LO) for this course: LO1. Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices). LO2. Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design. LO3. Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand. LO4. Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.	
Education Method	Lectures and instructions.	
Books	Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.	
Assessment	The final mark for this course, F, will be based on the average results for the assignments, A, and the final examination, E. All marks will be expressed on a scale of 0 to 10. For calculating A, the results of assignments that are not handed in on time have a score of 0; the lowest mark of all assignments will not be taken into account in calculation of A. The assignment results only will be valid within the academic year in which the assignments have been made. Assignment results obtained in previous academic years will not be taken into account. The final examination will give the mark E. This final examination will be taken on campus if COVID-19 restrictions allow with a written test, or online via a proctored (Moebius) test. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4670	PV Materials Processing and Characterization	4
Responsible Instructor	Prof.dr. A.W. Weeber	
Instructor	Dr. L. Mazzarella	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3 Different, to be announced	
Course Language	English	
Course Contents	The course is a follow-up of PV Technologies. Details of processing and characterization techniques of PV absorber materials and the solar cells functional coating will be discussed. Examples of lecture topics are advanced light management structures, novel surface passivating layers and contacting schemes. Background of deposition technologies such as Chemical Vapor Deposition (CVD), Physical Vapor Deposition (PVD) and Atomic Layer Deposition (ALD) will be given. Furthermore, working principles of special coatings and its functionality when applied to different solar cell concepts, including tandem devices, will be presented. This course is an elective one following the solar profile. PV Basics and PV Technologies should have been concluded successfully.	
Study Goals	After this course students are able to understand processing mechanisms, material properties, and functionality and working principle of the novel functional coating applied to different solar cell concepts. Students are expected to be able to explain these working principles and have a solid background on different deposition technologies and characterization techniques, and will be able to describe the effect on the solar cells output. Students will be able to analyse and evaluate recent technology developments and the background of it, and summarize them during an oral presentation for the entire group. As we can see, these learning objectives are specific about the content, and they are phrased in terms of observable behaviour: understand, explain, evaluate, present and describe.	
Education Method	This course consists of eight 2-hour lectures. In the same week a 2 hour work session / instruction will be organized for a deeper understanding of the technologies discussed in the lectures. During the whole course the students will work on assignments which need to be submitted as reports. All assignments/reports should be assessed as sufficiently good (no specific mark). In week 9 (final schedule will be given at the beginning of the course) you will present results of a literature study on a special topic covered by the course for the entire group.	
Literature and Study Materials	Book (not obligatory, expensive): Photovoltaic Solar Energy: from fundamentals to applications by Angèle Reinders et al. Material from online course: PVCDROM, PV-manufacturing.org Slides during lectures Articles (referred to in lectures)	
Assessment	The final grade consists of the following parts: presentation (team work) (50% of final grade); written exam (individual) (50%); Both marks will be expressed on a scale of 1 to 10 and should be 5.8 or higher. Reports on the assignments shall be completed as well, and assessed as sufficiently good (no specific mark). In case of a re-sit, you need to make an appointment with the lecturers.	
maximum aantal deelnemers	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Expected number of student: Maximum of 30. Use Brightspace and inform our secretary Mark Vielvoije (m.vielvoije@tudelft.nl) for attending the course.	

EE4680	Photovoltaic Modelling	4
Responsible Instructor	Dr.ir. R. Santbergen	
Instructor	Prof.dr.ir. O. Isabella	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Modelling is an important tool for gaining deeper insight in the working principles and loss mechanisms of solar cells and photovoltaic systems. Researchers use this insight to develop more efficient solar cells and to maximize the energy yield of photovoltaic systems. In this course we will cover the entire photovoltaic modelling chain: modelling of photovoltaic materials, devices (solar cells), modules and systems. The focus will be on understanding the working principles of these models, and using them to analyse the effects of design parameters on solar cell and photovoltaic system performance. The online course material consists of two parts, each consisting of several phases: 1) PV materials and device modelling 1.1) PV materials and device characterization 1.2) optical modelling of PV devices 1.3) electrical modelling of PV devices 2) PV systems modelling 2.1) modelling of irradiance 2.2) modelling of energy yield In addition there will be software demos, given by expert users, allowing you to familiarize yourself with the advanced modelling software commonly used by PV researchers. The prior knowledge from the courses PV basics and PV systems is required. This course PV modelling will prepare you for an MSc thesis project that involves modelling of solar cells or photovoltaic systems, as well as a future career in photovoltaics research.	
Study Goals	The online course material consists of two parts: 1) PV materials and device modelling, 2) PV systems modelling. Each consist of several phases: At the end of phase 1.1 (characterization of PV materials and devices) you should: - Be familiar with the main techniques used in the characterization of PV materials and devices - Understand the importance of a correct characterization for the modelling of PV - Be able to extract useful and important information (trends and properties) from the characterization of both PV materials and devices At the end of phase 1.2 (optical modelling of PV devices) you will; - have a complete overview of the modelling approaches commonly used for photovoltaic optical modelling, and which are their strength and limitations - get a thorough understanding of different optical modelling methods and which structures they are best suited to model - improve and practice your skills on optical modelling, by carrying out activities with our own GenPro4 software At the end of phase 1.3 (electrical modelling of PV devices) you should be able to: - have a complete overview of the modelling approaches commonly for photovoltaics, both optical and electrical - get a thorough understanding of different electrical modelling methods - improve and practice your skills on optical and electrical modelling, by carrying out activities with our own advanced semiconductor analysis (ASA) software. At the end of phase 2.1 (modelling of irradiance) you should be able to: - Gather and analyse meteorological data, including the utilization of geographic information systems, - Obtain information on horizon and shading at a specific location, and derive the relevant location parameters, - Combine meteorological data and location properties to calculate the irradiance on the angle of incidence with different levels of detail, and - Optimize the orientation of a module to maximize the irradiance on the angle of incidence. At the end of phase 2.2 (modelling of the energy yield) you should be able to: - Model the operating temperature of modules, - Obtain information from module datasheet, - Combine module properties with temperature and irradiance modelling to calculate DC power (and yield), - Model the performance of inverters (DC to AC) and batteries, - Analyse load profiles, and use them to size a PV system (module and batteries), and - Calculate the AC power and energy yield of an entire system, and analyse its performance from an economic point of view. After the software demos you will be able to explain the input, working principles and output of the following commonly used software packages: - Complete EASE (extracting optical constants from spectroscopic ellipsometry measurement of PV materials) - HFSS (rigorous solving of Maxwell equation for EM-wave (light) propagation in PV devices) - Sentaurus (modelling charge transport in PV devices by solving semiconductor equations) - Radiance (PV module plane of array irradiance calculation using backward ray-tracing) - PV Syst (PV system energy yield calculation)	
Education Method	This course consists of recorded video lectures, interactive lectures and instructions. The video lectures provide you with the basic information. The interactive lectures give you the opportunity to ask questions to the teacher and discuss selected topics in depth. In addition you will work on modeling assignments. During the instruction sessions the modeling assignments will be explained and feedback will be given on your (preliminary) results.	
Assessment	Every week you make and submit modeling assignments, which are graded. The weighted average counts for 70% of the final grade. In addition, every week you write a short report about the software demo, which is graded as well. The average of this counts for the remaining 30% of the final grade.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The on-campus Q&A session is not mandatory. Because the course material (video lectures, live demos) is given online, this course can also be taken by students who cannot physically come to the campus. Because the assessment is done through weekly assignments, without an exam, the final grade (and credits) can be obtained already before the exam period (end of October). This can be relevant for those who have not yet obtained sufficient credits to start their graduation project.	

ET4379	Photovoltaic Lab Course	4
Responsible Instructor	Prof.dr.ir. O. Isabella	
Instructor	Dr.ir. R. Vismara	
Contact Hours / Week x/x/x/x	4/0/0/0 Q1=Q5! Also possible online in Q6, Q7, Q8!	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The aim is to offer you an advanced practical course on solar cells and photovoltaic systems. The following areas of practical training will be part of the course: (1) Characteristics of artificial and natural light (2) Characterization of the performance of lab-scale solar cells (3) Behavior of photovoltaic modules in different temperature and shading conditions (4) Charge and discharge properties of batteries (5) Performance of power electronic components (charge controllers, inverters, maximum power point trackers) (6) Operation of entire photovoltaic systems	
Study Goals	(1) Experiment with photovoltaic systems and all of their components under lab and real-life conditions. (2) Describe the working principle behind each component of a photovoltaic system. (3) Analyze the operation of all the elements of photovoltaic systems and of the system as a whole. (4) Compare the performance of different types of components of photovoltaic systems.	
Education Method	Laboratory sessions	
Literature and Study Materials	Manuals and templates are available on BrightSpace.	
Prerequisites	This course is meant for students who have already passed at least the PV Basics course (ET4376) and preferably also PV Technologies (ET4377) and PV Systems (ET4378). This course is also meant for 1st-year masters students with a bachelor degree in electrical engineering.	
Assessment	The assessment is based on reporting results from the individual tasks, using an Excel template provided on Brightspace. Two-person teams will be formed at the plenary session, the attendance of which is compulsory. The final grade is per team. The final grade is the average of the six technical reports. A sufficient grade (greater than or equal to 5.75) in all six technical reports is required to complete the course successfully. Max 1 retry session and re-submissions are allowed. In case of failure of the technical report: - 1 technical report: use buffer sessions on week 1.7 - 2 or more technical reports: failure of the course --- A virtual version of this course is made available for students to attend in Q2, Q3, and Q4 of the academic year 2022-2023. Each of the six tasks of the on-campus course will run in the internet browser of your choice, in the form of a web application accessible via Brightspace. The assessment of the online version is entirely automatic, using numerical exercises and multiple-choice questions. A sufficient grade (> 57.5/100) in all six tasks is necessary to pass the course. The final grade is the average of the six grades. A maximum of one retry session is allowed. Attending one of these instances of the course, no teams will be formed and the final grade is per person.	
Enrolment / Application	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The on-campus course can accommodate up to 48 students. Anyone interested in attending this course must apply during Q4 of the immediately previous academic year. A selection will then take place and, before the beginning of the instance in Q5, the line-up of students will be announced. Student(s) who failed on-campus in Q5 can re-take the virtual version of the course. There is no limit to the number of students who can attend the course's virtual version.	
Remarks	This practical course will be given on campus only in Q1. Conversely, a virtual version will be available for students to attend in Q2, Q3, and Q4. The enrolment is open on Brightspace from Q1 of the academic year 2022-2023.	

ME46065	Thin Film Materials	3
Responsible Instructor	Dr. J.G. Buijnsters	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	This is a mixed lectures and project-based course. The lectures will cover the basics of thin films: deposition techniques, film growth, microstructure, stresses and failure. The project consists of carrying out in a group of several students a literature research on a specific thin film topic, such as: hard coatings, self-lubricating coatings, adhesive layers, thin films in MEMS and NEMS, thin-film based batteries, thin-film photovoltaics, coatings for implants, etc. The students will present their research in oral and written form. Progress presentations will be given to the class.	
Course Contents	The course consists of 7-8 plenary lectures and a student project. The topics of the lectures are 1. Introduction, liquid deposition techniques 2. Physical vapor deposition (evaporation & sputtering) 3. Chemical vapor deposition 4. Atomic layer deposition 5. Thin-film diamond 6. Nucleation & Growth modes The project consists in carrying out in a group of few students a literature research on a specific thin film topic, chosen out of the following: 1. hard coatings 2. self-lubricating coatings 3. adhesive layers 4. thin films in MEMS/NEMS 5. thin-film based batteries 6. thin-film photovoltaics 7. coatings for implants The students can also propose a topic of their own interest. The students will present their progress in several plenary presentations. The class will challenge them with critical questions. The students will also prepare a review paper on their topic that will be peer reviewed by a different group of students. At the end of the course a final presentation will be given by all groups.	
Study Goals	At the end of the course, the student is able to: 1. describe the various liquid and vacuum deposition technologies 2. discuss the pros and cons of each of these techniques 3. identify deposition-microstructure-properties relationships for a given deposition method 4. explain the main mechanisms of thin film growth 5. find, extract and order relevant information from open literature 6. explore new trends in thin films and their applications 7. professionally present information in written and oral form 8. critically assess the information found 9. critically assess the information presented by the other groups 10. cooperate with groupmates such that everybody is responsible for the content.	
Education Method	Lectures and assignments (literature study in group, presentations, discussions, paper, and peer review). Preference will be given to on-campus lectures and on-campus presentations.	
Literature and Study Materials	Course material will be shared by the teacher and posted on Brightspace.	
Assessment	The grade will be composed as follows: 1. Final paper 50% 2. Final presentation 30% 3. Mid-term presentations 20% A facultative, individual multiple-choice quiz is held towards the end of the lecture series. In addition a peer evaluation form will be used to assess whether all team members have equally contributed to the work.	
Department	3mE Department Precision & Microsystems Engineering	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Storage 2022

AP3311	Neutrons, X-Rays and Positrons for Studying Microscopic Structures and Dynamics	6
Responsible Instructor	Dr. W.G. Bouwman	
Instructor	Dr.ir. N.H. van Dijk	
Instructor	Dr. H. Schut	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge		
Course Contents	The microscopic structure and dynamics of condensed matter in physics, chemistry and biology is obtained by a wide variety of scattering techniques. Structures are determined on length scales ranging from 10⁻¹¹ to 10⁻⁵ m, dynamics on time scales from 10⁻¹⁴ to 10⁻⁶ s. In this course we discuss the different neutron, X-ray and positron techniques, illustrated by a broad range of scientific and technological applications, for which recent scientific articles will be studied. After a general introduction an overview is given of the scattering theory, both for neutrons and X-rays. Then the different techniques will be discussed in more detail. Diffraction (either by X-rays or neutrons) yields information about the structure of single crystals, powders, and liquids. Small-angle scattering (SAXS and SANS) reveals the shape and arrangement of colloidal systems, micelles or polymers in a solvent. Reflectometry is used to find the composition of layered structures like proteins adsorbed at surfaces or magnetic layers in recording materials. Polarized neutrons are used in two different ways. First, for determining magnetic structures and dynamics. Second, as a special technique (spin-echo) to study the larger length scales and time scales (polymer dynamics). Inelastic scattering measures phonons in crystals, diffusion in liquids and the vibrational spectrum of molecules. Neutron sources (fission and spallation), X-ray sources (from X-ray tube, via synchrotron to XFEL) and instrumentation will be an integral part of this course. An introduction to positron annihilation (PA) will be given. Positron annihilation lifetime, Doppler Broadening techniques are particularly suited for the detection and characterisation of (open volume) defects such as vacancies, vacancy clusters and voids in e.g. metals, alloys, semiconductors and polymers. The main application of the Angular Correlation of Annihilation Radiation (ACAR) technique is the study of the electronic structure of solids, and that of defects, precipitates and nanocrystals. A visit to the neutron and positron experimental facilities at the Reactor Institute Delft will be part of the course.	
Study Goals	Being able to explain the theoretical background of neutron and X-ray scattering and positron annihilation techniques; Being able to talk about the applications of these techniques; Being able to perform basic calculations and interpretations of the techniques; Being able to read and discuss recent articles on the application of the techniques; Being able to talk in depth about one of the topics to be chosen by the candidate in combination with a recent research article.	
Education Method	The course consists of 14 times 2 hours contact on the following chapters: I. Theoretical background neutron and X-ray scattering II. Neutron experimental techniques III. X-ray experimental techniques IV. Diffraction from crystalline materials, theoretical, experimental, application V. Diffraction from amorphous materials and liquids, theoretical, experimental, application VI. Small-angle scattering, theoretical, experimental, application VII. Reflectometry, theoretical, experimental, application VIII. Polarized neutrons, theoretical, experimental, application IX. Imaging and tomography with neutrons and X-rays, theoretical, experimental, application X. Positron annihilation, theoretical, experimental, application Each 2-hours-lecture is accompanied by a set of assignments from the website https://www.e-neutrons.org/. This has to be completed before the next class. In the class question about the previous lecture will be discussed as well as recent scientific articles. In turn the students have to give short presentations about the assignments and about the scientific articles. Those students who have attended at least 12 of the 14 lectures and sent in at least 11 of the 13 assignments are allowed to enter the oral exam.	
Literature and Study Materials	Lecture notes. The accompanying material with the website https://www.e-neutrons.org/. For mathematical background we recommend the book D.S. Sivia, Elementary Scattering Theory for X-ray and Neutron Users, Oxford University Press, ISBN 978-0-19-922868-3.	
Assessment	Having done at least 11 of the 13 times homework is a requirement for taking the oral exam. The oral exam of 45 minutes determines the final grade. The student proposes an article or scientific topic connected to the course for the first 15 minutes of the exam.	
Elective	Yes	

CH3513	Electrochemistry for Renewable Energy	4
Responsible Instructor	D.A. Vermaas	
Instructor	Dr. A.J. Houtepen	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Thermodynamics (e.g. 4051CHTHEY) and standard chemistry background (e.g., BSc education MST or similar)	
Course Contents	Electrochemistry provides an important platform to convert (renewable) electricity into valuable products, chemicals, fuels and coatings. This course provides an overview of several electrochemical processes of societal importance, linking fundamental understanding with applied utilization. The scope of the course will cover processes from the nanoscale to the meter-scale, and methods to understand fundamental phenomena in practical systems. This course will include: - Electrochemical potentials and definitions (electromotive force, formal potential, Donnan potential) - Types of electrochemical reactions (reversibility, multi-step reactions) - Electrochemistry at liquid-solid, liquid-liquid, solid-solid and gas-solid interfaces - Electrode reaction kinetics in steady and unsteady state (Cottrell equation) - Microscopic theories of charge transfer and (Marcus model, energy states, tunneling) - Ion transport through diffusion and migration - Electrochemical characterization techniques (cyclic voltammetry, chronopotentiometry, impedance, etc.) - Electric double layers (Helmholtz model, Gouy-Chapman model) - Examples of water electrolysis, CO₂ reduction - Industrial electrochemical systems (large scale electrolyzers, batteries, etc.) - Electrochemical equipment (potentiostats, reference electrodes, electrodes, electrolytes, membranes) - Experimental setup (including practicum)	
Study Goals	After this course, the student is able to: - Explain electrochemical responses in complex electrochemical cells (steady and non-steady state, and mono- or multi-ionic systems) to peers; - Demonstrate how theories for electrochemistry thermodynamics, kinetics (Marcus model, Tafel model) and electrolyte processes are applied to basic redox systems - Break down limiting factors (overpotentials or limiting currents) in given practical electrochemical conversions - Set up and perform suitable electrochemical tests for characterizing redox couples in practical electrochemical conversions	
Education Method	This course consists of: Instruction videos, Lectures and exercise sessions (combined), Practicum, Guest lectures	
Literature and Study Materials	Bard & Faulkner, Electrochemical methods fundamentals and applications	
Assessment	Grading is based on: - Homework assignments (0%, mandatory) - Practicum report (30%) - Exam at the end of the course (70%)	
Permitted Materials during Tests	Calculator (no graphical) Formula sheet	
Elective	Yes	
Continuing Courses	Possible follow-up courses are: CH3222: Energy Storage in Batteries (Q3) ME45203: Electrochemical Energy Storage 2: Analytical Modelling (Q4) MSc thesis	

CH3531	Functional Ceramics	3
Responsible Instructor	Dr. E.M. Kelder	
Instructor	Dr. P. Braga Groszewicz	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc: general and inorganic chemistry, basic quantum chemistry	
Course Contents	Ceramic materials show a wide variety in chemical, optical and electrical properties. They are used in many different applications, ranging from catalysts, fuel cells, batteries and sensors to solar cells, photo-catalysts and luminescent nanoparticles. This course will treat the basic properties of functional ceramic materials. The following topics will be included: - crystal structures - bonding in inorganic solids - basic electronic structure and properties of semiconductors - defect chemistry including point defects, non-stoichiometry and doping - ionic and electronic charge transport - semiconductor junctions and space charges.	
Study Goals	After this course, the student is able to: - Correlate crystal structure and band structure to optical and electrical properties of ceramic materials - Describe the operating principles of simple devices based on semiconductor junctions (solar cells, diodes, photo-electrodes) and ionic conduction (Nernst sensors, (Li-ion)-batteries, solid oxide fuel cells (SOFCs), and catalysts) - Identify and predict relationships between point defects, dopants, space charges, and ionic/electronic conduction in both semiconductors and ionic conductors, and perform simple calculations	
Education Method	Lectures, self study, instruction classes	
Books	- "Physical Ceramics" by Y-M Chiang, D. P. Birnie, and W.D. Kingery, John Wiley & Sons, ISBN: 0-471-59873-9	
Reader	- Several handouts and the lecture slides will be made available on Brightspace	
Assessment	It is foreseen that the exam will be a closed book exam. Examples have been posted on Brightspace. It is stressed that the situation with respect to COVID-19 may change, and an open book exam may thus be expected instead.	
Permitted Materials during Tests	- Calculator, closed book (overview of most important equations will be handed out during exam, the reference sheets)	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Wind 2022

AE4202	CFD for Aerospace Engineers	3
Responsible Instructor	Prof.dr.ir. S. Hickel	
Instructor	Dr.ir. A.H. van Zuijlen	
Responsible for assignments	Dr.ir. A.H. van Zuijlen	
Co-responsible for assignments	Dr. D. Modesti	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM.	
Study Goals	After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.	
Education Method	Lectures, hands-on tutorials, take-home assignments	
Books	van Dyke: An Album of Fluid Motion; The Parabolic Press Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann Pope: Turbulent Flows; Cambridge University Press Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press Wilcox: Turbulence Modeling for CFD; DCW Industries	
Reader	Detailed lecture slides and tutorials	
Assessment	Written (proctored online) exam and report on final assignment.	
Permitted Materials during Tests	none	

AE4205	MDO for Aerospace Applications	4
Responsible Instructor	Dr.ir. G. la Rocca	
Instructor	Dr.ir. M.F.M. Hoogreef	
Instructor	Dr. F. Oliviero	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Familiarity with MATLAB and basic aircraft design knowledge. MATLAB is used for tutorials and home assignment, thus students without MATLAB experience must gain this through self-study Students with no experience in aircraft design should consult textbooks like Torenbeek, Raymer and Roskam to familiarize with the fundamentals of wing sizing process and general concepts such as wing loading, aerodynamic coefficients, fuel fraction method, drag and its components, etc.	
Course Contents	The course will introduce students to multidisciplinary design optimization (MDO), as a modern methodology to support complex product development. Main topics: - What and why MDO - Parameterization techniques - Numerical optimisation methods (focus on gradient based methods) - Decomposition and coordination of complex computational systems - the eXtended Design Structure Matrix (XDSM) - MDO architectures - Intro to surrogate modeling, discrete&mixed integer problems, genetic algorithms and Multi Objective Optimisation	
Study Goals	The main goals are the following: - understanding of Multidisciplinary Design Optimization (MDO) methods and architectures - hands-on experience in setting up and solving MDO problems	
Education Method	The course consists of classical lectures and computer tutorials using own laptop. During the tutorials, students are introduced to some of the tools discussed during lecture and can make hands-on experience during interactive sessions with the teachers. In particular, students will be introduced to the MATLAB optimization function fmincon and its use for gradient based optimization. A significant amount of time is left for self-study, where students will have to work in pairs on one take-home assignment.	
Computer Use	The use of a personal computer is required MATLAB (including its optimization toolbox) will be used for the MDO practical sessions and assignment. Student licenses are provided by the TUD	
Literature and Study Materials	- lecture sheets and some notes (only available digitally on Brightspace) - scientific papers - some software modules - Suggested (not compulsory) book: - Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszczanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3 The book is available in e-format from the TUD library (Chapter 9 is used in the course AE4204 on Knowledge Based Engineering) - Engineering Design Optimization, by J.R.R.A. Martins and A. Ning (free on line)	
Assessment	There is no conventional exam for this course. The assessment is based on the quality of one take-home assignment, which will be executed in teams of two students. The assignment is about setting up and solving an MDO problem using Matlab. The deliverables are the Matlab code and a report. Two different take-home assignments are defined for the two yearly examination sessions.	
Tags	Algorithmics Analysis Matlab Modelling Optimalisation Programming Project	

AE4T40	Airborne Wind Energy	3
Responsible Instructor	Dr.ing. R. Schmehl	
Instructor	Dr.ir. A.C. Viré	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	BSc. completed	
Parts	Lectures include latest material on the developments in the field of airborne wind energy, with a thematical focus on kite power generation.	
Course Contents	Lecture 1: Introduction to airborne wind energy Lecture 2: History of kite applications Lecture 3: Implemented AWE concepts Lecture 4: Physics of tethered flight Lecture 5: Theory of tethered flight in three dimensions Lecture 6: Flight navigation and maneuvering Lecture 7: Operation of AWE systems Lecture 8: Practical use of AWE systems Lecture 9: AWE in the future energy system Lecture 10: Design and flight dynamics of kites Lecture 11: Aerodynamics of kites Lecture 12: Aeroelasticity of kites Practical: AWE workshop with Jupyter Notebook	
Study Goals	Give an overview of developments in airborne wind energy, developng mathematical models for kites and designing kite power systems for industrial scale generation of energy.	
Education Method	Lecture & workshop	
Prerequisites	Any MSc student can take this course, given a good understanding of vector analysis and basic trigonometry.	
Assessment	Written exam	
Set-up	12 lectures of two lecture hours each in the first semester followed by a written examination.	

AE4W30	Wind Resource and Wind Farm Yield	4
Responsible Instructor	Prof.dr. S.J. Watson	
Instructor	Dr.ir. D.J.N. Allaerts	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	This course will cover how the output of wind farm arrays is predicted based on modelling of the wind resource and combined wake interactions. The first part of the course will cover the different wind resource models which are used ranging from simple mass consistent models, to linearised solvers, computational fluid dynamics models and mesoscale models. The use of models in both onshore and offshore environments will be covered including the importance of atmospheric stability. The second part of the course will consider the application of models to assess the yield from wind farms considering wakes effects. Wake models will be reviewed including simple semi-empirical models to kinematic models and the modelling of wakes in CFD models. Once again, the importance of atmospheric stratification will be considered and the potential impact on wind farms of certain mesoscale phenomena, e.g. gravity waves offshore, will be assessed and quantified. Large wind farm phenomena will be studied including the so-called deep array effect and wind farm blockage.	
Study Goals	Course Aim: to give students a firm foundation in how to predict wind farm yield from prevailing wind conditions. Course Objectives: 1) To understand what tools are available to predict wind resource over a wide area 2) To apply to such models to potential wind farm sites onshore and offshore 3) To quantify the importance of terrain effects and atmospheric stability on the wind resource 4) To understand what tools are available to predict wake effects in a large wind farm 5) To apply wake models in a realistic wind farm situation 6) To appreciate the importance to wind farm yield of phenomena such as terrain complexity (onshore), proximity to the coast (offshore), atmospheric stability, mesoscale effects (e.g. gravity waves), wind farm layout and wind farm scale	
Education Method	Students will learn through a combination of guided reading, group work, online resources (e.g. lectures, videos, quizzes), face-to-face lectures, coursework and exams.	
Assessment	The students will undertake two course assignments (resource assessment and wind farm yield prediction) worth 30% each, spread out during the course and an oral exam worth 40% at the end of the course. Assignment 1: application of a model to assess and optimise wind resource and energy yield for an onshore site with a report to be assessed. Assignment 2: application of a model to assess and optimise energy yield for an offshore site incorporating different wake models with a report to be assessed. Exam: oral test to assess understanding of the principles of wind resource assessment and wind farm yield prediction.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Electric Mobility 2022

EE4515	Advanced Power Electronics	4
Responsible Instructor	Dr. Z. Qin	
Instructor	Dr. T. Batista Soeiro	
Contact Hours / Week x/x/x/x	0/0/0/3 + lab	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	It is expected that the student has taken all the previous courses in the power electronics teaching line: ET4119, ET4116, ET4145	
Course Contents	The focus of this course is on providing the theoretical background of various design related aspects of power converters not covered in the previous courses. The topics covered include: switching behavior and loss modelling, soft switching vs hard switching, passive components, thermal management and control.	
	The lab projects integrate the knowledge from this course and the preceding courses in a design assignment of designing, building and testing an efficient and small power converter.	
Study Goals	After this course, the students will be able to: - design a transformer/inductor - design the control - design the cooling - analyse and improve the efficiency - design, build, and test a converter with small size and high efficiency	
Education Method	Lectures and lab project	
Literature and Study Materials	Lecture handouts N. Mohan, T.M. Undeland, W.P. Robbins, Power Electronics - Converters Applications and Design, Wiley, 2nd or 3rd edition.	
Assessment	projects (80%) and oral exam (20%)	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4C04	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation Dynamic response, relation to modes, the matrix exponential Realization of transfer function models by state space descriptions, coordinate changes, canonical forms Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback Application of LQ regulator Observability, detectability, unobservable modes, state-estimation observer design Output feedback synthesis and separation principle Reference signal modeling, integral action for zero steady-state error. Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design and - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. Due to COVID-19 the exam may be an online exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4116	Power Electronics	4
Responsible Instructor	Dr.ir. P. Bauer	
Responsible Instructor	Dr. Z. Qin	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	ET4119 ET4145	
Course Contents	Resonant converters, snubbers, Space Vector theory, Vector and scalar PWM modulation, Simulation and modelling of power electronics, High Power Converters, Applications of power electronics for: HVDC transmission, DC Grids, Electric Utility (Facts and Custom Power), Power Conditioners, UPS systems etc.	
Study Goals	Learn objectives per topic 1. Resonant Converters - to know why resonant converters are applied, - advantages/disadvantages - which categories of resonant converters there are - how they are constructed and how they work - when thyristors and when transistors are applied, influence on the switches, - to dimension the switches for a resonant converters, - to understand how to dimension the resonant tank 2. PWM modulation - to understand the different modulation techniques for single phase and three phase converters - to know the main feature of PWM modulation for different types - to understand space vector (phasor) - to know the Clark, Park transformation and to apply it for three phase converters - to know the following principles: - carrier modulation-scalar - carrier modulation-vector - over modulation - preprogrammed modulation, harmonic elimination - number of pulses for harmonic elimination, advantages, disadvantages - hysteresis modulation - carrier modulation with feedback - scalar and vector modulation with real time optimisation, predictive current - control method 3. Applications Understand the following: - HVDC Transmission	
Education Method	Lectures	
Literature and Study Materials	Mohan, N., T.M. Undeland and W.P. Robbins, 'Power Electronics', Wiley, 2nd ed. 1995 (to obtain by the ETV).	
Assessment	Written and oral disclaimer: information may change depending on the developments around the coronavirus. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Consultation by appointment, actual course information available on brightspace.	

ET4121	A.C. Machines	4
Responsible Instructor	J. Dong	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	Control of Electrical Drives (ET4291)	
Expected prior knowledge	Electrical machines and drives (ET4117)	
Course Contents	The following subjects are discussed: - winding function and inductances; - time and spatial harmonics; - skin effect in electrical machines; - phasor, space vector and transformation in electrical machines; - the salient pole synchronous machine; - modeling and simulation of the dynamic performance of DC and AC machines.	
Study Goals	After following this course the students should be able to - describe and calculate the steady-state performance of AC machines; - calculate spatial harmonics and inductances of given winding distributions; - derive the dynamic models of DC and AC machines using space vectors; - model and simulate the dynamic performance of DC and AC machines; - evaluate the limitations of the dynamic models.	
Education Method	Lecture / Computer Simulation / Discussion	
Literature and Study Materials	- A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill. - lecture notes on Brightspace	
Assessment	Oral exam	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) - Presence at the computer simulation session is obliged. - Submission of assignments on Brightspace is obliged.	

ME41106	Intelligent Vehicles 3ME	5
Responsible Instructor	Prof.dr. D. Gavrila	
Practical Coordinator	R.M. Ensing	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	ME41106 Intelligent Vehicles is the introduction course on the (on-board) AI technology of automated driving. Whereas the application setting is that of vehicles, most concepts apply to the broader mobile robotics setting. Students seeking a one-course overview of mobile robotics are thus encouraged to attend, also from outside 3ME (e.g. EEMCS, Civil Engineering and Aerospace Engineering faculties). Introduction Course organization Intelligent Vehicles (IV) domain Motivation for automated driving SAE levels of automation Main components of IV technology Remaining challenges Overview sensors (camera, radar, LiDAR, tactile) 3D Machine vision Perspective camera model Extrinsic and intrinsic camera transformations Image matching Stereo vision Object Detection and Classification Detection vs. classification Object proposals Handcrafted features (e.g. HOG) & classification (e.g. linear SVM) End-to-end learning: Neural Networks Performance metrics: confusion matrices, precision vs. recall, ROC curves State Estimation Bayesian Filtering Kalman Filtering Particle Filtering Object Tracking Data Association Track Management Self-Localization & Sensor Fusion Absolute vs. relative localization Ego-motion compensation (e.g. odometry) Extrinsic sensor calibration Environment representations (grids, voxels) Motion Planning Planning as graph-search (the A* algorithm) Trajectory generation and optimization The Road Ahead IV technology currently on the market Future developments The course has a substantial practicum component (about 65% of course work load), where learned concepts are put into practice by means of programming assignments (Python, using Jupyter notebooks).	
Study Goals	At the end of the course students will understand the main methodical components of a intelligent vehicle (mobile robot). They will be able to program basic algorithms and test these on simulated and real-world data. Students will be able to express an educated opinion on the benefits and risks of automated driving, the current developments from driver assistance to driver-less cars, and the forces driving this transformation.	
Education Method	Lectures (2-4 hours per week) Practicum (2 hours per week) Formative feedback: Practica - Pairwise collaboration (alternating random groups) - Pass/Fail (counts for knock-out criteria) - No late submissions or resit (except due to motivated medical conditions) Students can in principle work on their own laptops. However, we provide only limited support for setting up the software environment (we use Anaconda) on their systems. If the software environment cannot be installed properly on a particular laptop (e.g. a few processors like M1 and ARM might not work), the student will need to revert to using the computers in the designated lab rooms.	
Course Relations	This course, ME41106 Intelligent Vehicles *cannot* be selected in combination with RO47004 Machine Perception, due to the content overlap between the courses. This means, for example, that this course ME41106 cannot be selected by students following the MS Robotics program (for which RO47004 is required). This course mainly addresses the on-board AI technology of an intelligent vehicle. Complementary to this course is CIE 5805 Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment which focuses more on automated vehicles as part of a larger intelligent transportation system (i.e. including smart infrastructure, connected/cooperative driving) and on associated issues at the macro level (e.g. traffic flow efficiency, fuel consumption, behavior adaptation).	
Literature and Study Materials	Slides Hand-outs recommended text book: Artificial Intelligence: A Modern Approach (Russel and Norvig) 3rd edition freely available as PDF Jupyter notebooks	
Prerequisites	Basic linear algebra and probability theory Intermediate Python programming skills, e.g. Numpy, object oriented programming (classes, inheritance), modules, namespaces and scope. Recommended: Pattern Recognition / Machine Learning (RO47002, CSE2510, or equivalent) It is possible to start the course with beginner Python skills if your aim is to take advantage of this course to also improve your	

Assessment	programming expertise. Be aware, however, that this catching up will likely result in an increased study load (e.g. 1-2 EC). Final practicum assignment (50%) - Individual assignment - Late submission: max. one day late for -1.0 grade point - Students having an initial grade of < 5.8 will be offered an opportunity for a remake to obtain a (maximum) grade of 5.8 for this component in the Q3 quarter. Final written exam (50%) - Written exam on campus. Knock-out criteria - At most one of the Pass/Fail practicum assignments is Fail - The minimum final practicum assignment grade is 5 - The minimum final written exam grade is 5 - The minimum final course grade is 6 The (final) practicum assignments will be evaluated on the computers in the designated lab rooms. If a student choose to work on an own laptop, the student needs to test the software on those computers before handing-in.
Special Information	More information about this course is available at http://intelligent-vehicles.org/teaching/courses/. It includes a link to the software environment installation guide.

RO47017	Vehicle Dynamics and Control	5
Responsible Instructor	B. Shyrokau	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course provides fundamental knowledge on vehicle motion, vehicle-road interaction and main technical principles and aspects of automotive control systems. This course discusses vehicle modelling, controller design and testing procedures to evaluate vehicle motion and automotive control systems. Week 1: Tire dynamics and modelling Wheel slip, slip angle, tire force / moments, influence of camber angle, pressure, temperature effect, combined slip, friction circle, transient behavior Tire modelling and wheel dynamics Week 2: Lateral motion and vehicle handling Kinematics, understeer gradient, bicycle model, steady-state handling Transient motion, frequency response, handling testing, vehicle stability Week 3: Antilock braking system (ABS) and vehicle dynamics control (VDC) General concepts of ABS and different control principals: rule-based, mixed slip-deceleration, advanced control techniques General concept of VDC, yaw rate based controller, body slip angle based controller, integrated vehicle control using control allocation Week 4: Planar motion Planar vehicle modelling Static equilibrium, lateral load transfer, roll center and axis, roll stiffness, anti-roll bars Longitudinal load transfer, pitch center, anti-dive, anti-lift Week 5: Steering & assist logic Wheel kinematics, steering system modelling, column- and rack-assist Steering assist logic, steering feel and assessment Week 6: Path-following and evasive control of automated vehicles Vehicle road model, error definition, overview of path-following controllers, PD controller Evasive maneuver, advanced control techniques, MPC-based evasive steering assist Week 7: Vertical motion and comfort Introduction to vibration, quarter car, 2DoF and 3DoF models, Ride comfort Suspension control: skyhook, groundhook, and mixed control Week 8: Vehicle state estimation General concept of VSE, vehicle velocity estimation, Kalman filtering (linear and extended), wheel load reconstruction methods Final presentations of group project	
Study Goals	After completing this course, students will be able to: - explain the tire-road interaction and be able to implement own tire model and compare with established one - describe the concept of vehicle handling and lateral stability and derive the equations of motion of a nonlinear vehicle model - interpret the load transfer effects and implement four-wheel planar vehicle model including the wheel kinematics, steering system and steering assist logic - explain the concepts of wheel slip control, vehicle stability control, automated lateral control and vehicle state estimation - implement control strategies for vehicle stability control (PID with gain scheduling or LQR) and automated lateral control (MPC) - derive the equations of vertical motion of a quarter-car vehicle model and explain the influence of vehicle and road parameters on vehicle vibration behaviour	
Education Method	Contact hours Lectures (4 hours per week) Office hours (2 hours per week)	
Prerequisites	Basic programming experience (e.g. Matlab / Simulink) at the BSc level Dynamics and control at the BSc level knowledge about Model Predictive Control is recommended	
Assessment	Homework assignments (25%) individual Group project (25%) collaboration in groups of 4-5 students Written exam (50%) individual The examination is a closed-book digital exam on campus including multi-select and open questions. Knock-out criteria - o Minimum average grade for homework assignments is a 5.0 - o Minimum average grade for group project is a 5.0 - o Minimum grade for the exam is a 5.0 - o Final grade minimum is a 6.0	
Department	3mE Department Cognitive Robotics	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Recommended electives 2022

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Language courses 2022

TPM301B	Spoken English for Academic Purposes Intermediate	2
Module Manager	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This speaking course focuses on the language and skills students need to participate successfully at the university. Students will engage in a number of activities both inside and out of the classroom - to practise and improve their English. The vocabulary focus will be on general academic language, so that after the course students can interact with more confidence in an academic setting.	
Study Goals	The aim of the course is to develop language skills needed for the university environment. You will learn phrases and words for a range of skills needed here, for example participating in discussions; rephrasing and explaining; giving an opinion, (dis)agreeing, making suggestions; describing and presenting a concept.	
Education Method	Group and pair work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to feel (more) confident when communicating with others at the university.	
Assessment	When you have taken our Placement Test you will be recommended a language level, either Intermediate or Advanced. With this recommendation you can sign up for both of the courses offered at that level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor.	
Enrolment / Application	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris.	
Remarks	Please note that you must also choose a group to reserve a place in the class. Type 'TPM301B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc students	

TPM302B	Spoken English for Academic Purposes Advanced	2
Module Manager	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This speaking course focuses on expanding and improving the language students need to participate at the university. A particular focus point of the course is the vocabulary needed at a higher level of English, both in an academic and a more general context. A range of sources for language input is offered, such as podcasts, videos, websites, or (academic) articles. The sessions revolve around student interaction.	
Study Goals	The first aim of the course is to improve and expand the language skills needed at a higher level of English. These include paraphrasing and sharing information with others, using polite language, giving feedback, and presenting information. Another study course goal is to expand and improve the students' general vocabulary.	
Education Method	Group and pair work, combined with (individual) assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English language skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for either of the courses offered at that level.	
Assessment	For this course you need a recommendation to take courses at the Advanced level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor.	
Enrolment / Application	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM302B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	The course will only be offered if there is a sufficient number of participants. Please note that it is not a presentation course.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc level	

TPM303A	Intermediate Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on the acquisition of necessary language skills in the areas of grammar and usage, as well as reading and writing at a level appropriate to university study. Students will develop skills that develop an idea with relevant support for that idea by structuring the text according to current writing conventions, using various transitions to link ideas and demonstrating a knowledge of grammar and punctuation. Students will edit their own texts and work with other students in peer-editing groups.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at the Intermediate level.	
Books	Advance in Academic Writing 1 - Steve Marshall This textbook is available in e-format (approx. 34) or a combined e-book and paper book (approx. 55). Details of how to obtain a copy will be given on Brightspace after enrolment.	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at intermediate level. After receiving your Placement Test results you can enrol via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q2 and Q4. However, the course will only be offered if there are sufficient participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM304A	Advanced Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on writing effective, well-structured and coherent academic texts in English. It looks at the writing process, including planning and structuring the text. Students will then develop paragraphs into longer academic texts and work on text coherence and cohesion. The course also pays attention to advanced academic vocabulary, sentence structure, punctuation and grammar.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who already have a good understanding of academic writing, but who wish to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at Advanced level.	
Books	Advance in Academic Writing 2 - Steve Marshall This textbook is available in e-format (approx. 34) or a combined e-book and paper book (approx. 55). Details of how to obtain a copy are given on Brightspace.	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at Advanced level. After receiving your Placement Test results you can enroll via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q1 and Q3. However, the course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure an essay, report or thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are expected to submit weekly writing assignments on a subject that is closely related to their field of study. An 85% attendance record is required. This means you may miss not more than one session (preferably none). If you fall behind with your coursework or cannot attend a minimum of six classes, you may be asked to leave the course. You will receive the course credits when all conditions are met.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

WM1116TU	Dutch Elementary 2	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 1 (course code WM1117TU)	
Expected prior knowledge	A completed course Elementary 1 (WM1115TU). It is also possible to enrol if you have a similar level: CEF A2. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Elementary 2 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Elementary 2 (WM1116TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1117TU	Dutch Intermediate 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 2. This is a paid course, also for students and will not give you ECTS credits.	
Expected prior knowledge	A completed course Elementary 2 (WM1116TU). It is also possible to enrol if you have a similar level: CEF A2+. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Intermediate 1 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Wesdijk en Blom Sciarone e.a., "Tweede Ronde, herziene editie. Nederlands voor buitenlanders. De Delftse Methode". Text book.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Elementary 2 (WM1116TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretaries office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1135TU	Advanced English for the University	3
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Responsible for assignments	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	This course is given at C1 CEFR (Common European Framework of Reference) level. Advanced English for the University is the highest level integrated skills course offered by the English Unit within the ITAV institute at TPM. It is designed for those who already have a thorough working knowledge of English grammar and vocabulary but who still wish to further perfect their proficiency in order to become more adept in both writing and speaking. Time will be devoted to: Academic texts and writing conventions Academic vocabulary and collocations Grammar and punctuation Presentations and discussions Language learning strategies Advanced English for the University is a 14-week course with 1 lesson a week (2 hours) plus approx. 4 hours of homework assignments per week. The groups tend to be small (max. 18 participants) and the sessions are highly interactive. Attendance is mandatory. Students can miss a maximum of two lessons. Courses begin in the first week of September and in the first or second week of February . Please see the website for Placement Test dates and for any further course details: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit	
Study Goals	By the end of the course students should be familiar with different types of academic texts academic writing conventions academic vocabulary with a specific focus on collocations common grammatical errors punctuation conventions structural devices to improve coherence various language resources such as online dictionaries	
Education Method	Students will do in-class exercises completed either in small groups or in pairs. They will also do a group PowerPoint presentation and a discussion led in pairs. The feedback on course components will be given both by the lecturer and by fellow course participants.	
Literature and Study Materials	Course book: Cambridge Academic English - An integrated skills course for EAP (Student's Book; C1 Advanced). Authors: Martin Hewings and Craig Thaine Publisher: Cambridge University Press, 2012 ISBN: 978-0-521-16521-1	
Prerequisites	To qualify for the course students need to have received an Advanced English for the University recommendation following the completion of the Placement Test. Please see in this connection: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test	
Assessment	There is no final exam. Assessment is based on the written assignments and on the in-class discussions and presentations. The six writing assignments each account for 10% of your final grade while your grade for the discussion and your grade for the presentation each accounts for 20% of your final grade.	
Enrolment / Application	After receiving their Placement Test results students can enrol via Brightspace by first registering for the course and then selecting the group of their choice.	
Remarks	More information can be found on Brightspace WM1135TU	
Elective	Yes	
Tags	English proficiency	
Targetgroup	BSc, MSc and PhD students are welcome	
Category	BSc and MSc level	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Sustainable Energy Technology (SET)

Entrepreneurship courses 2022

TPM406A	Corporate Entrepreneurship and Start-ups	6
Module Manager	Dr.ing. V.E. Scholten	
Responsible for assignments	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	This course examines how large corporations engage startups to tap into new markets, technologies and business models. You will develop knowledge and skills to analyse strategic situations of existing corporates and how they innovate through engaging startups. The course provides an eye on corporate startup engagement from both the corporate and startup perspective.	
Course Contents	Large international organizations often find themselves in a luxurious position compared to small start-ups. They have more financial capital, access to markets, and more technological knowledge. However, large international organizations are confronted with increasing external turbulence, technological advancement and globalization, which puts their luxurious position in jeopardy. At the same time the luxurious position constrains them to depart from present activity and focus on radical new innovations. Hence, corporates try to renew themselves through internal entrepreneurial behaviour, but also by initiating start-ups and acquiring start-ups to experiment with new technology. Many times, these start-ups originate in academic environments. Therefore we pay specific attention to the emergence of academic start-ups and how corporate organizations provide incubation facilities and venture capital support to engage startups to renew themselves. This course examines the various forms of corporate entrepreneurship and the relationship with corporate strategy in particular. You will develop knowledge and skills to analyze strategic situations and design the most appropriate solutions in terms of corporate entrepreneurship strategy and organization.	
Study Goals	Understand the concept of corporate entrepreneurship; the contexts, the forms, and the relationship with strategic management Analyze strategic situations and design appropriate corporate entrepreneurship strategies and organizations Analyse the contribution of internal and external ventures in the renewal of large corporates Understand the emergence of academic spin-offs and how they relate to the technology development of large established organisations Identify the various key actors that facilitate and support the emergence of new business ventures and academic spin-offs	
Education Method	In case of Covid-19 restrictions for on-campus education we will offer this course in 14 online sessions. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Each student will be expected to be present during lectures (two times a week), actively participate during the lectures, complete assigned readings, read posted lectures, listen to thought leader presentations and submit written assignments when due.	
Literature and Study Materials	Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Practical Guide	Given the interactive nature of this course, attendance at classes is mandatory. Only very serious grounds for absence (such as illness accompanied with doctors note) are acceptable. Holidays and (paid or voluntary) work obligations do not count as acceptable reasons for absence. This is a full-time course with scheduled holidays. Unjustified non-attendance will result in expulsion from this course.	
Assessment	The assessment is Covid-19 proof and includes three components. 1) The first component is a team level online presentation of a case analyses of an existing company. Each team will analyse a practical company case provided to you and based on literature of a specific lecture session you will present your findings online in class (40%) 2) The second component is a written report of a large firm in which you will analyse with your team the startup engagement activities this firm has put in place to renew itself. The report is an analyses of the company activities w.r.t. corporate venturing and an reflection of these activities by making use of literature. The report accounts for 40% of the final grade. 3) The third component is an individual analyses of a startup that is involved in the corporate startup engagement activities of your team case company. It reflects on the motivation for the startup engagement from both startup and corporate perspective. It accounts for 20% of the final grade.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. Brightspace enrollment will be activated by education support a couple of weeks before the course starts. No other means of registration, such as the app MyTuDelft are needed.	
Special Information	Each student team will present during the lectures, actively participate during the lectures, complete assigned readings, read posted lectures, listen to thought leader presentations and submit written assignments when due.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

TPM416A	Turning Technology into Business	6
Module Manager	Dr. L. Hartmann	
Responsible for assignments	Dr. L. Hartmann	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Turning Technology into Business (TTiB) aims to equip participants with a strong conceptual foundation to actively understand the dynamic process of technology-based entrepreneurship. Participants learn how business strategies are best formulated and how (through entrepreneurship) technology can create value. TTiB consists of seven lectures, introducing the theoretical backgrounds of technological, market and business analyses. A unique aspect of TTiB is that existing technologies (developed and patented by the TU Delft) are used as case subjects. The patent project is the focal point of the curriculum. Each group of 4-5 students will be assigned an original patent, and is expected to evaluate the commercial potential of this technology. This includes choosing promising applications (products) for the technology, and making recommendations for the most suitable business model to commercialize the technology. The patent project should provide a coherent and structured answer to the central question: which strategy is most likely to generate business from this patent? In addition to equipping participants with a thorough knowledge of the course subject, the section Technology, Strategy & Entrepreneurship (TSE) hopes that this course will be the starting point for a variety of university spin-offs in which the students will participate. To accommodate this, TSE offers the course TPM420A- Ready to Startup!	
Study Goals	This course aims to equip students with a strong conceptual foundation to an active understanding of two domains: (1) The dynamic process of technological innovation through concepts such as technology life-cycles, dominant design, disruptive technologies, Schumpeterian competition and the diffusion of innovations (2) How business strategies are formulated and, through entrepreneurship, technology can create value. This multi-faceted process of technology commercialization process is addressed in terms of assessing technology position, discovering market opportunities, competitive analysis, appropriability and the various modes of entrepreneurship.	
Education Method	7 interactive lectures, participant-centered case studies, homework assignments, classroom assignments, individual group coaching. Final Presentations in February.	
Literature and Study Materials	Will be provided during the course	
Assessment	Each group writes a report on their findings, analysis and recommendations for the technology (patent) they used as a case for this course. On the Final Day, the group has to give a 10 minute oral presentation. The report makes up 80% of the final grade. This team grade will be modified according to the individual student's class contributions and his/her performance within the group (peer review).	
Enrolment / Application	MAXIMUM CAPACITY is limited to 75 participants. Register on BrightSpace from October 1. Because of the huge popularity of this course, there is a selection procedure which consists of pre-course assignments. One of these assignments is a letter of motivation: why do you want to participate in this course? Based on the motivation and the other pre-course assignments, the final selection of 75 participants will be made.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Targetgroup	Advanced Master students, PDEng and PhD students, and staff researchers from all faculties within the TU Delft.	
Category	MSc level	

TPM420A	Ready to Start-up	6
Module Manager	Dr.ing. V.E. Scholten	
Module Manager	Ir. B.R. Joseph	
Responsible for assignments	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	This course is offered twice in a year; First starting and ending in Q2 and second starting mid Q3 until end Q4	
Course Contents	Ready to Startup (RtS) is an elective master course that has been the crown jewel of entrepreneurial education at TU Delft for over two decades. Bringing together business experts and students eager to start companies, it has one simple goal: turn engineers into successful founders. RtS focuses predominantly on Business Strategy: gaining market traction, prototyping, growing the team, devising a financial plan and fundraising strategy, protecting IP, and foundational activities (setting up a company, sharing ownership, etc.) The course has the prerequisite that teams should be formed and have business ideas that are analysed and validated. It is meant for students who plan to start a company. After RtS, teams must be at the level required for YES!Delft incubation and Delft Enterprises investments.	
Study Goals	At the end of the course, you will be able to - Reflect on your current business model and identify further necessary validation activities. Analyse market entry strategies and consider alternative models. - Learn how to negotiate, attract first users, and seek endorsement of stakeholders (such as investors and advisors). - Plan your core operational activities such as company foundation, IP protection, product development strategy and team development - Devise financial plans and appropriate fundraising strategies - Convey a convincing narrative of your business and be able to provide constructive feedback to other startup plans. - Demonstrate effective entrepreneurial behaviour by making swift decisions based on newly gathered information and adjusting your model accordingly.	
Education Method	The course is spread out over 8 sessions (online in case of persistent Covid-19 restrictions for on campus education) in the afternoon (13h30-17h30) where the core elements of a startup plan are presented. You work on assignments that are aimed at specific elements of the core startup plan, and you will by apply them to your startup project and present your findings during the course. In this way you make evident progress with your startup. The course combines state of the art business methods, translated to the needs of starting entrepreneurs, with consults from practitioners. Topics are discussed using case studies. You will present your progress, provide peer-feedback as well as receive advice from mentor experts. The course ends with a jury pitch and an executive summary.	
Literature and Study Materials	Handouts and literature are available trough in Brightspace.	
Prerequisites	Enthusiasm, entrepreneurial spirit and a good business idea.	
Assessment	The course has 3 grading components that are Covid-19 proof. - Individual: Active online participation and constructive peer review feedback on other business plans (30%) - Team: class-based assignments (20%) - Team: final startup plan report (40%), supported by a jury pitch (10%)	
Enrolment / Application	Application process you need to register in advance to the course. Teams of students with clear ideas about a business they want to start supported with evidence that it addresses a need, will have priority to enter. All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education .	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

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